

AUTOREGRESSIVE MOVING AVERAGE MODELS: ARMA(P,Q)



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The Wold Decomposition Theorem

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- The **complex power spectral density (CPSD)** of a discrete-time wide-sense stationary (WSS) process $X[n]$ can always be expressed as the sum of two terms:

$$S_X(z) = S_X^c(z) + \sum_{i=1}^L 2\pi P_i \delta(z - e^{j2\pi f_i}), \quad r_1 < |z| < \frac{1}{r_1}$$

- The first term represent the continuous part of the spectrum, the second term is the discrete part, i.e. the Dirac's delta functions present in the spectrum.
- The continuous part of the spectrum corresponds to the so called **regular** component of the process $X[n]$, that cannot be predicted with zero error. Whereas, the discrete part corresponds to the **predictable** part of the process, that can be predicted with zero error, and it is formed by the periodic components present in $X[n]$.

- The **Wold decomposition theorem** states that any WSS process $X[n]$ can always be decomposed in the sum of two orthogonal (uncorrelated) processes, one of them, $X_r[n]$, is the so-called **regular** component and the other, $X_p[n]$, is the **predictable** component, i.e. it can be predicted with zero mean square error (MSE) by means of a proper linear predictor (that is a LMMSE estimator):

$$X[n] = X_r[n] + X_p[n]$$

- The **regular** component $X_r[n]$ corresponds to the continuous part of the PSD, whereas the **predictable** component $X_p[n]$ corresponds to the discrete part of the PSD:

$$CPSD\{X_r[n]\} = S_X^c(z), \quad CPSD\{X_p[n]\} = \sum_{i=1}^L 2\pi P_i \delta(z - e^{j2\pi f_i})$$

The Wold Decomposition Theorem

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- The **power spectral density (PSD)** is obtained from the complex PSD by calculating it on the unit circle, i.e. for $z=\exp(j2\pi f)$, under the assumption that the unit circle belongs to the **Region of Convergence (ROC)** of the bilateral z-transform:

$$S_X(e^{j2\pi f}) = S_X(z) \Big|_{z=e^{j2\pi f}} = S_X^c(e^{j2\pi f}) + \sum_{i=1}^L 2\pi P_i \delta(e^{j2\pi f} - e^{j2\pi f_i})$$

$$X[n] = X_r[n] + X_p[n]$$

$$S_X(e^{j2\pi f}) = \underbrace{S_X^c(e^{j2\pi f})}_{PSD\{X_r[n]\}} + \underbrace{\sum_{i=1}^L 2\pi P_i \delta(e^{j2\pi f} - e^{j2\pi f_i})}_{PSD\{X_p[n]\}}$$

- The lines in the spectrum are generated by the Z-transform of the periodic components (sinusoids, complex exponentials) of the autocorrelation function (ACF), which in turn are generated by the periodic components that are present in the realizations of the random process:

$$X_p[n] = \sum_{i=1}^L A_i e^{j(2\pi f_i n + \theta_i)}, \quad \{A_i\}_{i=1}^L \text{ and } \{\theta_i\}_{i=1}^L \text{ mutually independent r.v.'s}$$

$$\{\theta_i\}_{i=1}^L \in \mathcal{U}(-\pi, \pi)$$

$$\text{If } X_p[n] \text{ is real} \rightarrow X_p[n] = \sum_{i=1}^{L/2} A_i \cos(2\pi f_i n + \theta_i), \quad 0 < f_i < \frac{1}{2} \quad \forall i$$

- The expression above represents a **parametric model** for the **predictable** component $X_p[n]$ of a process $X[n]$. It depends on $3L$ parameters, if the process is complex-valued, or $3L/2$ parameters, if it is real-valued.

The Wold Decomposition Theorem

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$$\begin{aligned}
 R_{X_p}[m] &= E\{X_p[n]X_p^*[n-m]\} = E\left\{\sum_{i=1}^L A_i e^{j(2\pi f_i n + \theta_i)} \sum_{k=1}^L A_k e^{-j(2\pi f_k (n-m) + \theta_k)}\right\} \\
 &= \sum_{i=1}^L \sum_{k=1}^L E\{A_i A_k e^{j(\theta_i - \theta_k)}\} e^{j2\pi(f_i n - f_k (n-m))} = \sum_{i=1}^L \sum_{k=1}^L E\{A_i A_k\} E\{e^{j(\theta_i - \theta_k)}\} e^{j2\pi(f_i n - f_k (n-m))} \\
 &= \sum_{i=1}^L \sum_{k=1}^L E\{A_i A_k\} \delta[i-k] e^{j2\pi(f_i n - f_k (n-m))}, \quad \text{where} \quad E\{e^{j(\theta_i - \theta_k)}\} = \delta[i-k] \\
 &= \sum_{i=1}^L E\{A_i^2\} e^{j2\pi f_i m} = \sum_{i=1}^L P_i e^{j2\pi f_i m}, \quad \text{where} \quad P_i \triangleq E\{A_i^2\}
 \end{aligned}$$

$$i \neq k \Rightarrow E\{e^{j(\theta_i - \theta_k)}\} = E\{e^{j\theta_i} e^{-j\theta_k}\} = E\{e^{j\theta_i}\} E\{e^{-j\theta_k}\} = 0$$

Wold decomposition theorem

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$$\begin{aligned} S_X(z) &= \sum_{m=-\infty}^{+\infty} R_X[m]z^{-m} = \sum_{m=-\infty}^{+\infty} \left(R_X^c[m] + \sum_{i=1}^L P_i e^{j2\pi f_i m} \right) z^{-m} \\ &= \sum_{m=-\infty}^{+\infty} R_X^c[m]z^{-m} + \sum_{m=-\infty}^{+\infty} \sum_{i=1}^L P_i e^{j2\pi f_i m} z^{-m} \\ &= \sum_{m=-\infty}^{+\infty} R_X^c[m]z^{-m} + \sum_{i=1}^L P_i \sum_{m=-\infty}^{+\infty} e^{j2\pi f_i m} z^{-m} \\ &= S_X^c(z) + \sum_{i=1}^L 2\pi P_i \delta(z - e^{j2\pi f_i}) \end{aligned}$$

- Hence, the lines in the spectrum are generated by the periodic components in the ACF, which in turn are generated by the periodic components in the realizations of the random process.

■ **Example:** cosinusoidal process in AWGN:

$$X[n] = S[n] + W[n] = A \cos(2\pi f_0 n + \theta_0) + W[n], \quad 0 < f_0 < \frac{1}{2}$$

■ The amplitude A is modelled as a Rayleigh r.v., the phase θ_0 is modelled as a r.v. uniformly distributed in $[-\pi, \pi)$.

$$A \in \mathcal{R}(\sigma_A^2) \quad \theta_0 \in \mathcal{U}(-\pi, \pi) \quad A \text{ and } \theta_0 \text{ independent}$$


$$f_A(a) = \frac{a}{\sigma_A^2} e^{-\frac{a^2}{2\sigma_A^2}} u(a), \quad E\{A^2\} = 2\sigma_A^2$$


$$R_X[m] = E\{X[n]X[n+m]\} = R_S[m] + R_W[m] = \sigma_A^2 \cos(2\pi f_0 m) + \sigma_W^2 \delta[m]$$

Wold decomposition theorem

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$$S_X(e^{j2\pi f}) = 2\pi \frac{\sigma_A^2}{2} \delta(e^{j2\pi f} - e^{j2\pi f_0}) + 2\pi \frac{\sigma_A^2}{2} \delta(e^{j2\pi f} - e^{-j2\pi f_0}) + \sigma_W^2$$

$$S_X(z) = \underbrace{2\pi \frac{\sigma_A^2}{2} \delta(z - e^{j2\pi f_0}) + 2\pi \frac{\sigma_A^2}{2} \delta(z - e^{-j2\pi f_0})}_{\text{Discrete part}} + \underbrace{\sigma_W^2}_{\text{Continuous part}}$$

Discrete part

Continuous part

■ Hence, $W[n]$ is the regular part, whereas $S[n]$ is the predictable part of $X[n]$.

$$\begin{aligned} E\{S^2[n]\} &= R_S[0] = \int_{-1/2}^{1/2} \left[2\pi \frac{\sigma_A^2}{2} \delta(e^{j2\pi f} - e^{j2\pi f_0}) + 2\pi \frac{\sigma_A^2}{2} \delta(e^{j2\pi f} - e^{-j2\pi f_0}) \right] df \\ &= 2 \int_0^{1/2} 2\pi \frac{\sigma_A^2}{2} \delta(e^{j2\pi f} - e^{j2\pi f_0}) df = \sigma_A^2 \int_0^\pi \delta(e^{j\omega} - e^{j\omega_0}) d\omega = \sigma_A^2 \end{aligned}$$

$$X[n] = X_r[n] + X_p[n] = X_r[n] + \sum_{i=1}^L A_i e^{j(2\pi f_i n + \theta_i)}$$

$$R_X[m] = \underbrace{R_X^c[m]}_{ACF\{X_r[n]\}} + \underbrace{\sum_{i=1}^L P_i e^{j2\pi f_i m}}_{ACF\{X_p[n]\}}$$

Parametric model for the predictable component $X_p[n]$

$$S_X(e^{j2\pi f}) = \underbrace{S_X^c(e^{j2\pi f})}_{PSD\{X_r[n]\}} + \underbrace{\sum_{i=1}^L 2\pi P_i \delta(e^{j2\pi f} - e^{j2\pi f_i})}_{PSD\{X_p[n]\}}$$

- The **predictable** component $X_p[n]$ corresponds to the discrete part of the PSD.
- The **regular** component $X_r[n]$ corresponds to the continuous part of the PSD.

- A **white process** is a **regular** process, its PSD is continuous and constant:

$$S_X^c(e^{j2\pi f}) = \sigma^2$$

- It is a **parametric model** that depends on only 1 parameter, that is the power of the process σ^2 . Once we know (estimate) σ^2 we know the PSD:

$$\hat{\sigma}^2 = \frac{1}{N} \sum_{n=0}^{N-1} X^2[n] \Rightarrow \hat{S}_X^c(e^{j2\pi f}) = \hat{\sigma}^2$$

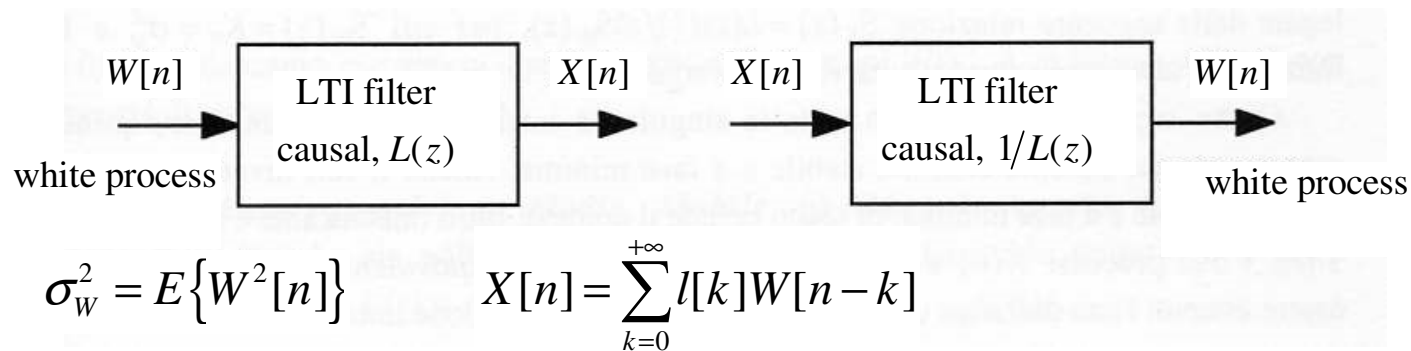
- **Problem**: How to model parametrically a non-white **regular process**?

- **Problem**: How to model parametrically a non-white **regular process**?
- If the continuous part of the PSD satisfies the **Paley-Wiener condition** for discrete-time random processes:

$$\int_{-1/2}^{1/2} \left| \ln S_X^c(e^{j2\pi f}) \right| df < +\infty$$

- then, the **regular** part of $X[n]$ can be represented as the output of a linear time-invariant (LTI) filter with system function $L(z)$, causal and stable, called the **innovation filter**, with inverse $I(z)=1/L(z)$, that is also causal and stable, called the **whitening filter**, having at the input a white process $W[n]$, called the **innovation process**.

- A **regular** process $X[n]$ whose PSD satisfies the Paley-Wiener condition can be represented as follows:



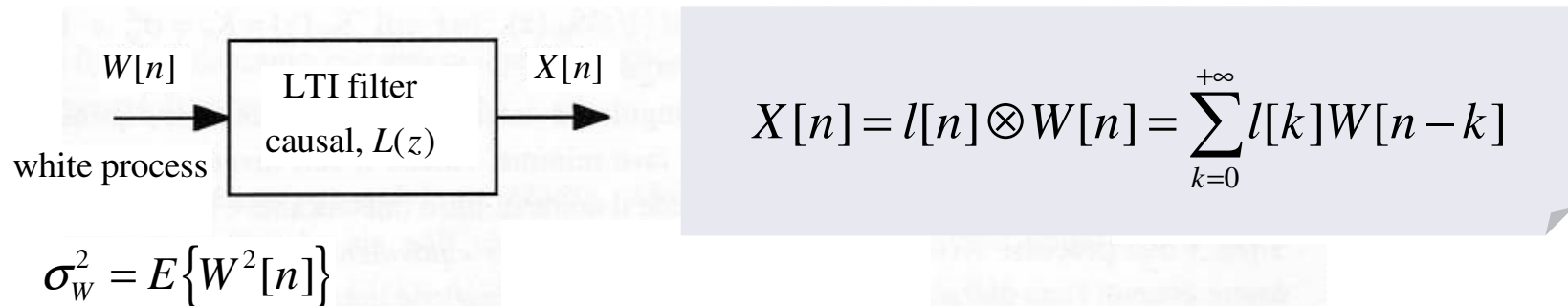
■ Innovation Filter

■ Whitening Filter

$l[n]$ impulse response of the innovation filter, $l[0] = \lim_{z \rightarrow \infty} L(z) = 1$

$L(z) = Z\{l[n]\}$ system function, $L(e^{j2\pi f}) = FT\{l[n]\}$ frequency response

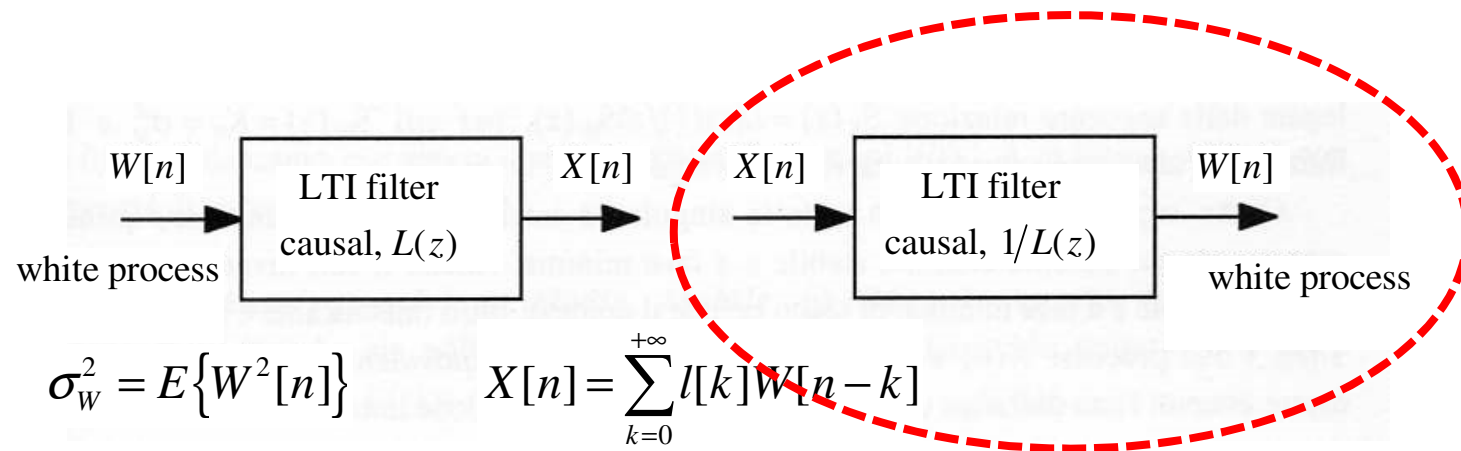
$$L(z) = \sum_{n=0}^{+\infty} l[n]z^{-n} \Rightarrow L(e^{j2\pi f}) = L(z)|_{z=e^{j2\pi f}} = \sum_{n=0}^{+\infty} l[n]e^{-j2\pi fn}$$



$$\left. \begin{aligned} R_W[m] &= E\{W[n]W[n+m]\} = \sigma_W^2 \delta[m] \\ S_W(e^{j2\pi f}) &= \sum_{m=-\infty}^{+\infty} R_W[m]e^{-j2\pi fm} = \sigma_W^2 \end{aligned} \right\} W[n] \text{ innovation process}$$

$$S_X^c(z) = S_W(z)L(z)L(1/z) = \sigma_W^2 L(z)L(1/z)$$

$$S_X^c(e^{j2\pi f}) = \sigma_W^2 L(e^{j2\pi f})L(e^{-j2\pi f}) = \sigma_W^2 L(e^{j2\pi f})L^*(e^{j2\pi f}) = \sigma_W^2 |L(e^{j2\pi f})|^2$$



$$\Gamma(z) = \frac{1}{L(z)} \quad \text{system function of the whitening filter}$$

$\gamma[n]$ impulse response of the whitening filter

$$S_X^c(z) = S_W(z) L(z) L(1/z) = \sigma_W^2 L(z) L(1/z)$$

$\Downarrow$

$$S_W(z) = \Gamma(z) \Gamma(1/z) S_X^c(z) = \frac{1}{L(z)} \cdot \frac{1}{L(1/z)} S_X^c(z) = \sigma_W^2 \quad \text{white process}$$

- Signals with rational spectra (ratio of polynomials in z)
- According to the **Weierstrass Theorem**, the complex PSD of a WSS **regular** process can be always represented as the ratio of polynomials of proper orders (as a limit the order can be infinity):

$$S_X(z) = \sigma_w^2 L(z)L(1/z) = \sigma_w^2 \frac{B(z)B(1/z)}{A(z)A(1/z)}, \quad r_1 < |z| < 1/r_1$$

- where $A(z)$ and $B(z)$ are polynomials with all roots internal to the unit circle.
- The ROC is annular and it is determined by the dominant pole, i.e. the pole closest to the unit circle: if the dominant pole is $p_1 = r_1 \exp(j\phi_1)$ and r_1 is the modulo of this pole, the ROC is as above.
- A PSD of this kind always satisfies the Paley-Wiener condition.

- **Innovation filter:** causal, stable, minimum phase, with inverse (the **whitening filter**) causal, stable, and minimum phase.

$$L(z) = \frac{B(z)}{A(z)} = \frac{\sum_{k=0}^Q b_k z^{-k}}{1 - \sum_{k=1}^P a_k z^{-k}}, \quad r_1 < |z|$$

$$\Gamma(z) = \frac{1}{L(z)} = \frac{A(z)}{B(z)}$$

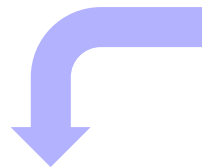
$$X[n] = l[n] \otimes W[n] = \sum_{k=0}^{+\infty} l[k] W[n-k],$$

$$W[n] = \gamma[n] \otimes X[n] = \sum_{k=0}^{+\infty} \gamma[k] X[n-k],$$

$$l[0] = \lim_{z \rightarrow \infty} L(z) = b_0 = 1$$

$$\gamma[0] = \lim_{z \rightarrow \infty} \frac{1}{L(z)} = 1$$

- The input-output relationship of the **innovation filter** in the time-domain can be obtained as follows:


$$X(z) = L(z)W(z) = \frac{B(z)}{A(z)}W(z) = \frac{\sum_{k=0}^Q b_k z^{-k}}{1 - \sum_{k=1}^P a_k z^{-k}} W(z)$$

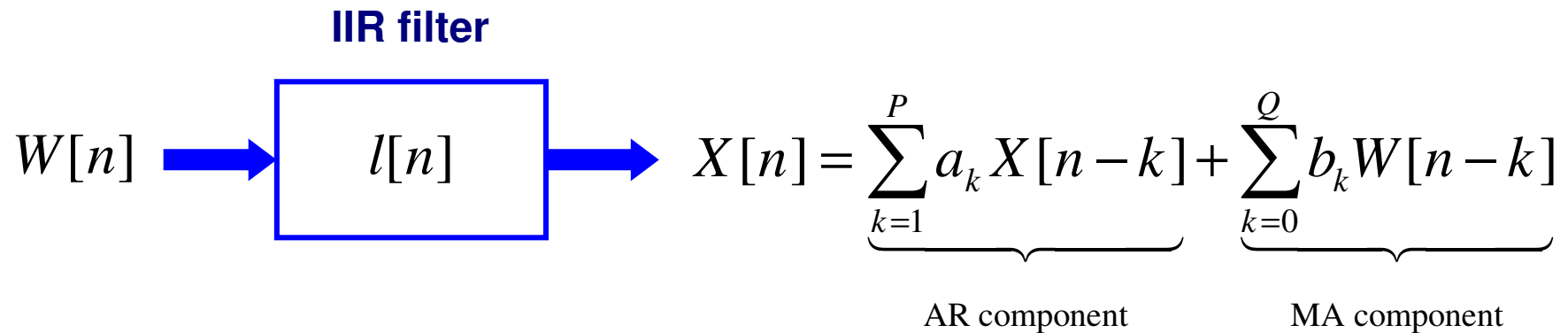
$$\left(1 - \sum_{k=1}^P a_k z^{-k}\right) X(z) = \left(\sum_{k=0}^Q b_k z^{-k}\right) W(z) \Rightarrow X(z) = \sum_{k=1}^P a_k z^{-k} X(z) + \sum_{k=0}^Q b_k z^{-k} W(z)$$

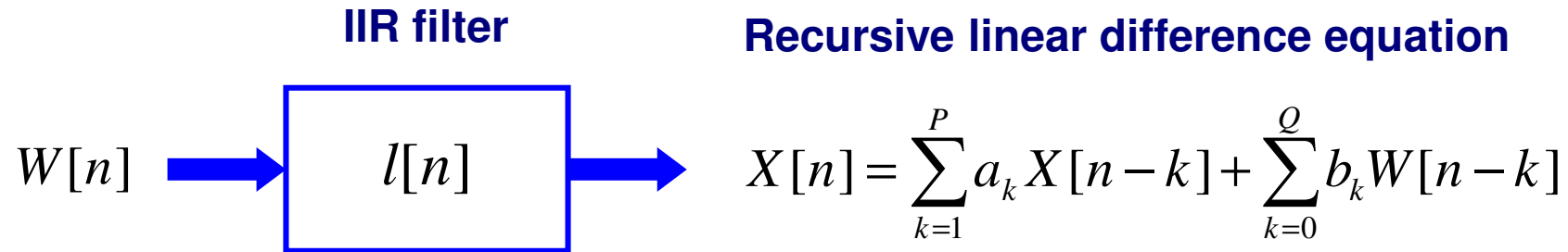
- Calculating the inverse Z-transform, we get the *input-output* relationship in the time-domain:

$$X[n] = \sum_{k=1}^P a_k X[n-k] + \sum_{k=0}^Q b_k W[n-k]$$

- Processes of this kind, whose *input-output* relationship is described by a recursive linear difference equation with constant coefficients, are called **Autoregressive Moving Average** processes of order (P,Q): **ARMA(P,Q)**

$$X[n] = \sum_{k=1}^P a_k X[n-k] + \sum_{k=0}^Q b_k W[n-k]$$





e.g. if $X[n]$ is Gaussian, $W[n]$ is white Gaussian $\mathbf{w} = \text{randn}(N,1)$.

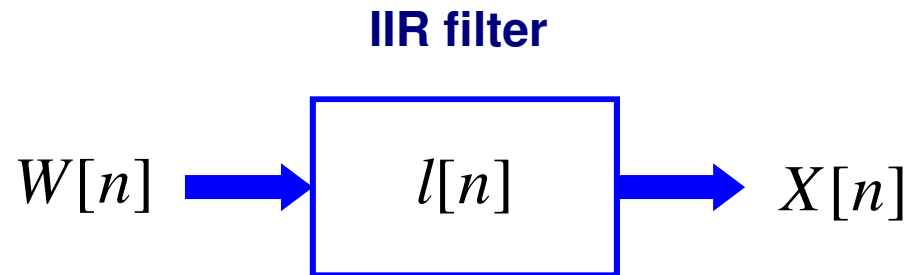
MATLAB:
 $\mathbf{x} = \text{filter}(\mathbf{b}, [1 \ -\mathbf{a}^T]^T, \mathbf{w})$

$$\mathbf{a} \triangleq [a_1 \quad a_2 \quad \cdots \quad a_P]^T$$

$$\mathbf{b} \triangleq [b_0 \quad b_1 \quad \cdots \quad b_Q]^T$$

$$X[n] - \sum_{k=1}^P a_k X[n-k] = \sum_{k=0}^Q b_k W[n-k]$$

■ **Important:** Any regular process that satisfies the Paley-Wiener condition can be modelled as an **ARMA(P,Q)** process, with orders P and/or Q possibly infinite.



System function:
$$L(z) = \frac{B(z)}{A(z)} = \frac{\sum_{k=0}^Q b_k z^{-k}}{1 - \sum_{k=1}^P a_k z^{-k}}$$

Frequency response:
$$L(e^{j2\pi f}) = L(z) \Big|_{z=e^{j2\pi f}} = \frac{B(e^{j2\pi f})}{A(e^{j2\pi f})} = \frac{\sum_{k=0}^Q b_k e^{-j2\pi kf}}{1 - \sum_{k=1}^P a_k e^{-j2\pi kf}}$$

$$\text{Complex PSD: } S_X(z) = \sigma_W^2 L(z) L(1/z) = \sigma_W^2 \left(\frac{\sum_{k=0}^Q b_k z^{-k}}{1 - \sum_{k=1}^P a_k z^{-k}} \right) \left(\frac{\sum_{k=0}^Q b_k z^k}{1 - \sum_{k=1}^P a_k z^k} \right)$$

$$\text{PSD: } S_X(e^{j2\pi f}) = \sigma_W^2 \left| L(e^{j2\pi f}) \right|^2 = \sigma_W^2 \left| \frac{\sum_{k=0}^Q b_k e^{-j2\pi k f}}{1 - \sum_{k=1}^P a_k e^{-j2\pi k f}} \right|^2$$

$$S_X(e^{j2\pi f}) \Leftrightarrow \{\sigma_W^2, \mathbf{a}, \mathbf{b}\} \rightarrow P+Q+1 \text{ parameters (if } b_0 = 1)$$

$$\mathbf{a} \triangleq [a_1 \quad a_2 \quad \cdots \quad a_P]^T, \quad \mathbf{b} \triangleq [b_1 \quad b_2 \quad \cdots \quad b_Q]^T$$

$$\begin{aligned}
 L(z) &= \frac{\sum_{k=0}^Q b_k z^{-k}}{1 - \sum_{k=1}^P a_k z^{-k}} = \frac{z^{-Q} \sum_{k=0}^Q b_k z^{Q-k}}{z^{-P} \left[z^P - \sum_{k=1}^P a_k z^{P-k} \right]} \\
 &= z^{P-Q} \cdot \frac{\sum_{k=0}^Q b_k z^{Q-k}}{z^P - \sum_{k=1}^P a_k z^{P-k}} = z^{P-Q} \cdot \frac{b_0 z^Q + b_1 z^{Q-1} + \dots + b_{Q-1} z + b_Q}{z^P - a_1 z^{P-1} - \dots - a_{P-1} z - a_P} \\
 &= b_0 z^{P-Q} \cdot \frac{z^Q + \frac{b_1}{b_0} z^{Q-1} + \dots + \frac{b_{Q-1}}{b_0} z + \frac{b_Q}{b_0}}{z^P - a_1 z^{P-1} - \dots - a_{P-1} z - a_P} = b_0 z^{P-Q} \cdot \frac{\prod_{n=1}^Q (z - z_n)}{\prod_{n=1}^P (z - p_n)}
 \end{aligned}$$

Poles of the filter = roots of the characteristic equation

$$z^P - a_1 z^{P-1} - \dots - a_{P-1} z - a_P = 0$$

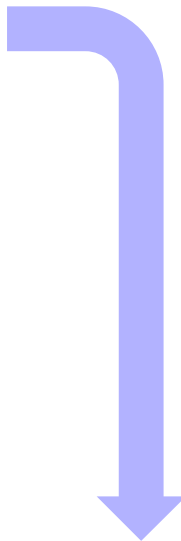
$$L(e^{j2\pi f}) = b_0 e^{j2\pi f(P-Q)} \cdot \frac{\prod_{n=1}^Q (e^{j2\pi f} - z_n)}{\prod_{n=1}^P (e^{j2\pi f} - p_n)}$$

- The behavior of the **frequency response** of the **innovation filter** is a function of the coefficients $\{a_k\}$ and $\{b_k\}$ of the recursive equation, i.e. it is a function of the location of the **poles** and **zeros** of the innovation filter.

MATLAB: $b=\text{poly}(z)$, $z=\text{roots}(b)$

$$S_X(e^{j2\pi f}) = \sigma_W^2 \left| L(e^{j2\pi f}) \right|^2 = \sigma_W^2 b_0^2 \cdot \frac{\prod_{n=1}^Q |e^{j2\pi f} - z_n|^2}{\prod_{n=1}^P |e^{j2\pi f} - p_n|^2}$$

where $\sigma_W^2 = E\{W^2[n]\}$

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- Hence, also the behavior of the PSD (and so of the ACF) depends on the coefficients $\{a_k\}$ and $\{b_k\}$, i.e. on the location of the poles and zeros of the innovation filter.
 - Usually, we set $l[0]=b_0=1$ to avoid ambiguity with the innovation power.

- The shape of the PSD (and so of the ACF) is uniquely a function of the locations of the poles and zeros of the innovation filter.
- The power of the innovation process is just a scale factor, it does not affect the PSD shape.

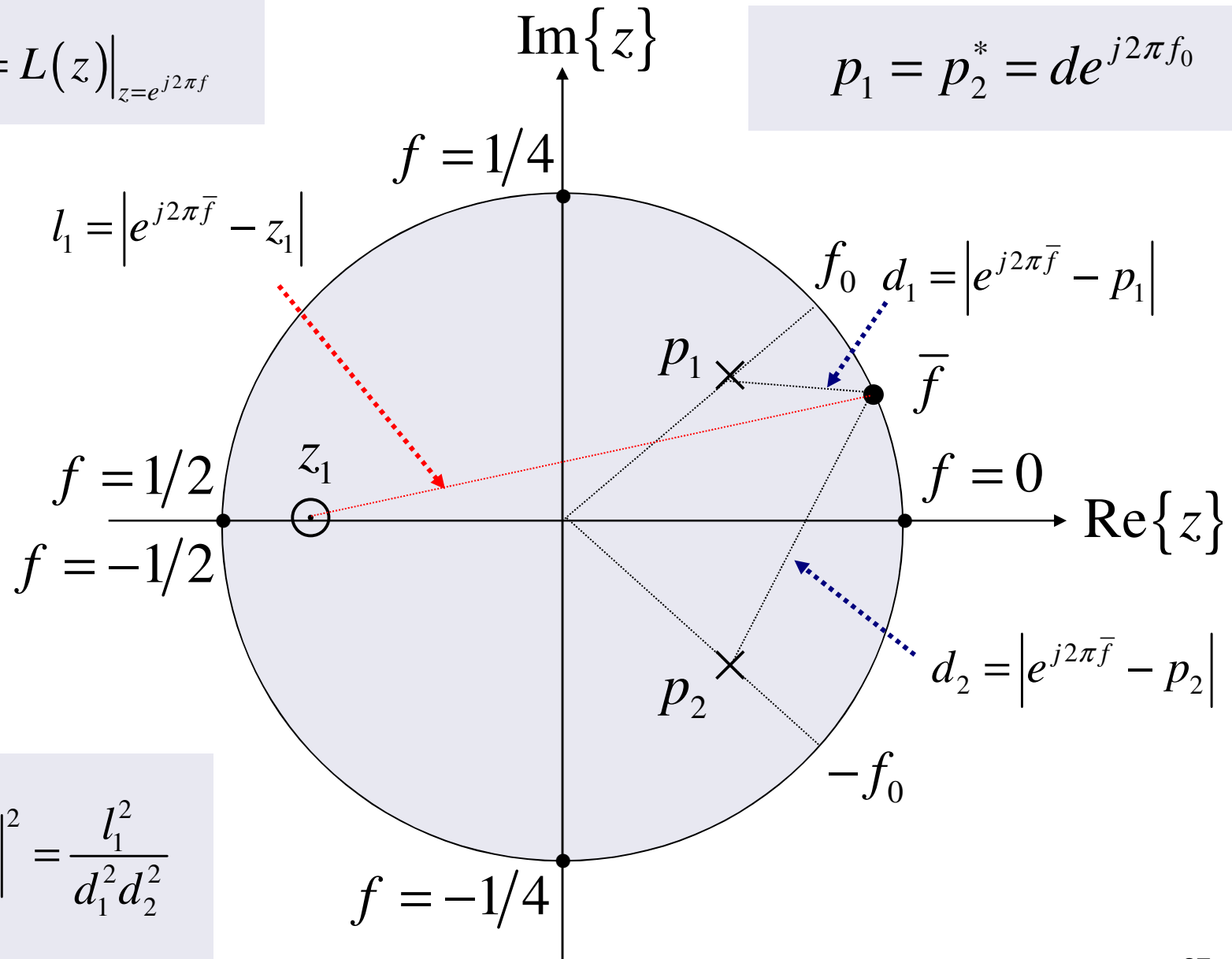
$$S_X(e^{j2\pi f}) \Leftrightarrow \{\sigma_w^2, \mathbf{a}, \mathbf{b}\} \Leftrightarrow \{\sigma_w^2 b_0^2, \mathbf{p}, \mathbf{z}\}$$

$$\mathbf{p} \triangleq [p_1 \quad p_2 \quad \cdots \quad p_P]^T, \quad \mathbf{z} \triangleq [z_1 \quad z_2 \quad \cdots \quad z_Q]^T$$

$$b_0 = 1 \Rightarrow \{\sigma_w^2, \mathbf{p}, \mathbf{z}\} \rightarrow P + Q + 1 \text{ parameters}$$

$$L(e^{j2\pi f}) = L(z) \Big|_{z=e^{j2\pi f}}$$

$$p_1 = p_2^* = de^{j2\pi f_0}$$

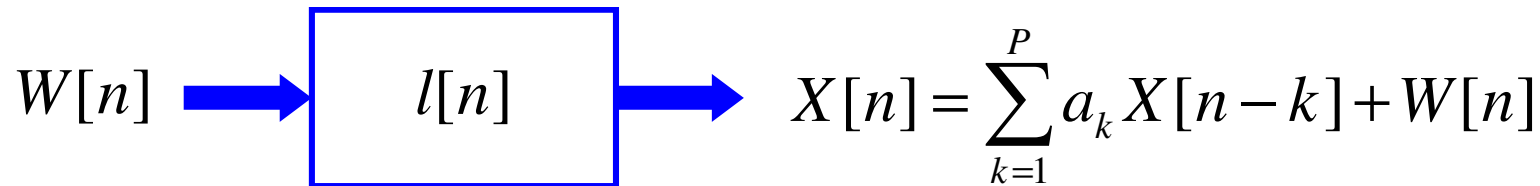


$$\left| L(e^{j2\pi \bar{f}}) \right|^2 = \frac{l_1^2}{d_1^2 d_2^2}$$

- **In summary:**
- The shape of the PSD, and so also of the ACF, is uniquely determined by the locations of the poles and zeros of the innovation filter $L(z)$.
- The phases of the poles determine the frequencies where the peaks in the PSD are located. How much a peak is pronounced is determined by the modulo of the pole: the closer the pole is to the unit circle and the narrower the peak is.
- The phases of the zeros determine the frequencies where the valleys in the PSD are located. How much a valley is narrow is determined by the modulo of the zero: the closer the zero is to the unit circle and the narrower the valley is.
- In most of the cases, we are more interested on identifying the locations of the peaks and their widths rather than those of the valleys.

IIR filter of order P

The difference equation is recursive



$$(b_0 = 1 \text{ and } b_k = 0 \text{ for } k > 0)$$

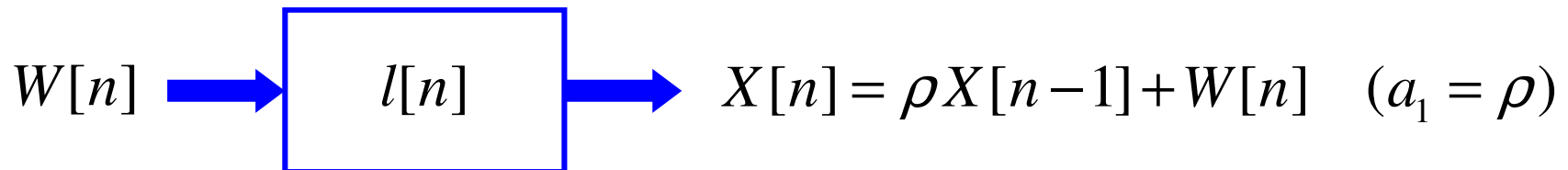
MATLAB:
`x=filter(1,[1 -a^T]^T,w)`

**Poles of the filter
 = roots of the
 characteristic equation**

$$z^P - \sum_{k=1}^P a_k z^{P-k} = 0$$

$$L(e^{j2\pi f}) = \frac{1}{1 - \sum_{k=1}^P a_k e^{-j2\pi kf}} = \frac{e^{j2\pi fP}}{\prod_{k=1}^P (e^{j2\pi f} - p_k)}$$

IIR filter of order P



MATLAB:
`x=filter(1,[1 -rho]^T,w)`

Characteristic equation: $z - \rho = 0$

**AR(1): the recursive equation is
of order 1 $\rightarrow$ 1 pole in $z=\rho$**

$$X(z) = \rho z^{-1} X(z) + W(z) \rightarrow (1 - \rho z^{-1}) X(z) = W(z)$$

$$L(z) = \frac{X(z)}{W(z)} = \frac{1}{1 - \rho z^{-1}} = \frac{z}{z - \rho}, \quad |\rho| < 1$$

$$L(e^{j2\pi f}) = L(z) \Big|_{z=e^{j2\pi f}} = \frac{1}{1 - \rho e^{-j2\pi f}}$$

- The impulse response of the innovation filter of an AR(1) process can be evaluated by solving the relevant linear difference equation with a Kronecker delta as input sequence:

$$l[n] = \rho l[n-1] + \delta[n], \quad \text{with } l[-1] = 0 \text{ and } |\rho| < 1$$

$$\Downarrow$$

$$\left\{ \begin{array}{l} l[0] = \rho l[-1] + \delta[0] = 1 \\ l[1] = \rho l[0] + \delta[1] = \rho \\ l[2] = \rho l[1] + \delta[2] = \rho^2 \\ \vdots \\ l[n] = \rho l[n-1] + \delta[n] = \rho^n \end{array} \right.$$



■ Then, finally: $l[n] = \rho^n u[n], \quad |\rho| < 1$

■ The transfer function $L(z)$ of the innovation filter can also be derived as the Z-transform of the impulse response:

$$L(z) = \sum_{n=0}^{+\infty} \rho^n z^{-n} = \sum_{n=0}^{+\infty} (\rho z^{-1})^n = \frac{1}{1 - \rho z^{-1}},$$

$$|\rho z^{-1}| < 1 \rightarrow |z| > |\rho|$$

□ **Autocorrelation function (ACF)** of an AR(1) process:

$$\begin{aligned} R_X[m] &= E\{X[n]X[n+m]\} \\ &= E\{X[n](\rho X[n+m-1] + W[n+m])\} \\ &= E\{\rho X[n]X[n+m-1] + X[n]W[n+m]\} \\ &= \rho R_X[m-1] \end{aligned}$$

in fact $E\{X[n]W[n+m]\} = 0$ if $m > 0$

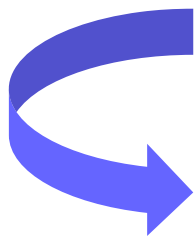

$$R_X[m] - \rho R_X[m-1] = 0, \quad m > 0$$

$$R_X[m] = R_X[0]\rho^m, \quad m > 0$$

■ To derive $R_X[0]$:

$$\begin{aligned}\sigma_X^2 &= R_X[0] = E\{X^2[n]\} = E\{(\rho X[n-1] + W[n])^2\} \\ &= \rho^2 E\{X^2[n-1]\} + 2\rho E\{X[n-1]W[n]\} + E\{W^2[n]\}\end{aligned}$$

$$= \rho^2 \sigma_X^2 + \sigma_W^2$$



$$\sigma_X^2 = \frac{\sigma_W^2}{1 - \rho^2}$$

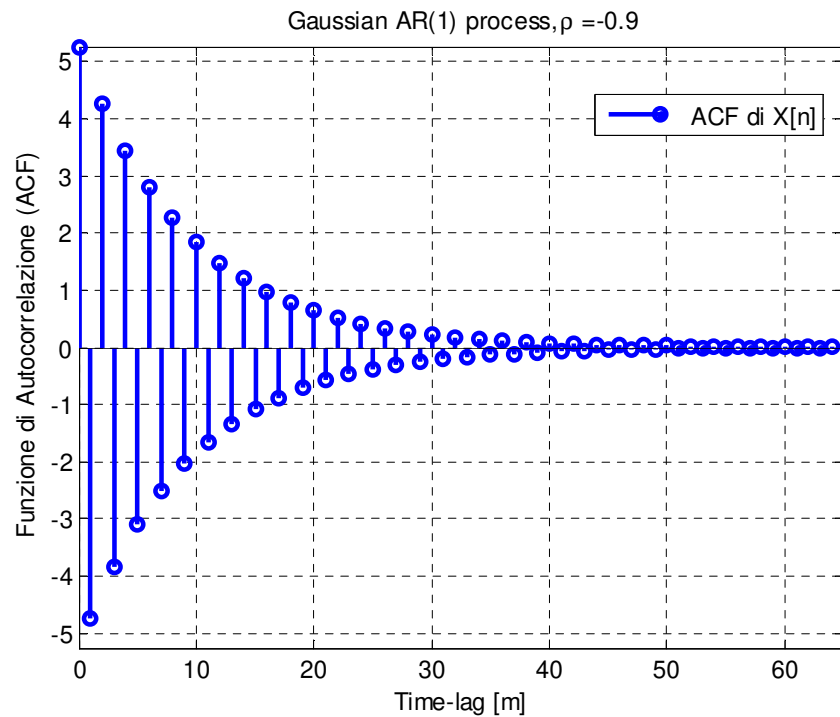
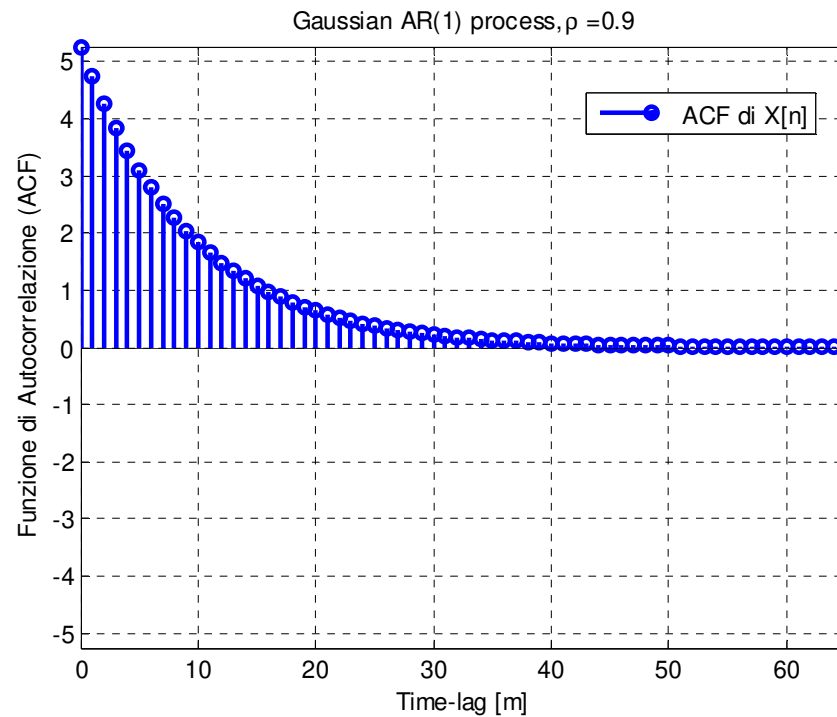
■ Taking into account that $R_X[-m] = R_X[m] \rightarrow R_X[m] = \frac{\sigma_W^2}{1 - \rho^2} \rho^{|m|}$

$$\rho_{X[n], X[n+m]} = \frac{\text{cov}\{X[n], X[n+m]\}}{\sqrt{\text{var}\{X[n]\} \cdot \text{var}\{X[n+m]\}}} = \frac{C_X[m]}{C_X[0]} = \frac{R_X[m]}{R_X[0]} = \rho^{|m|}$$

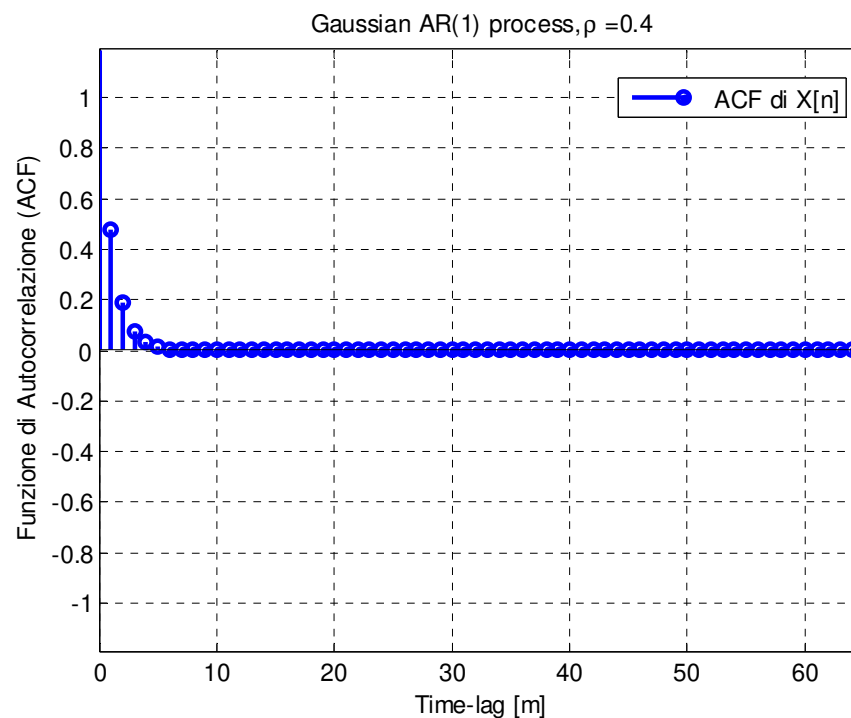
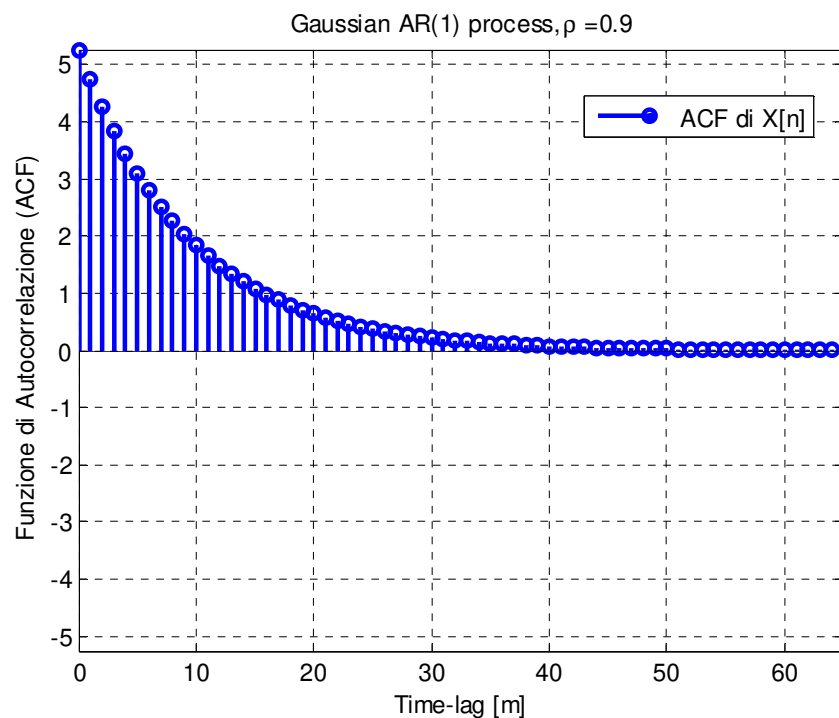
Markov process: AR(1)

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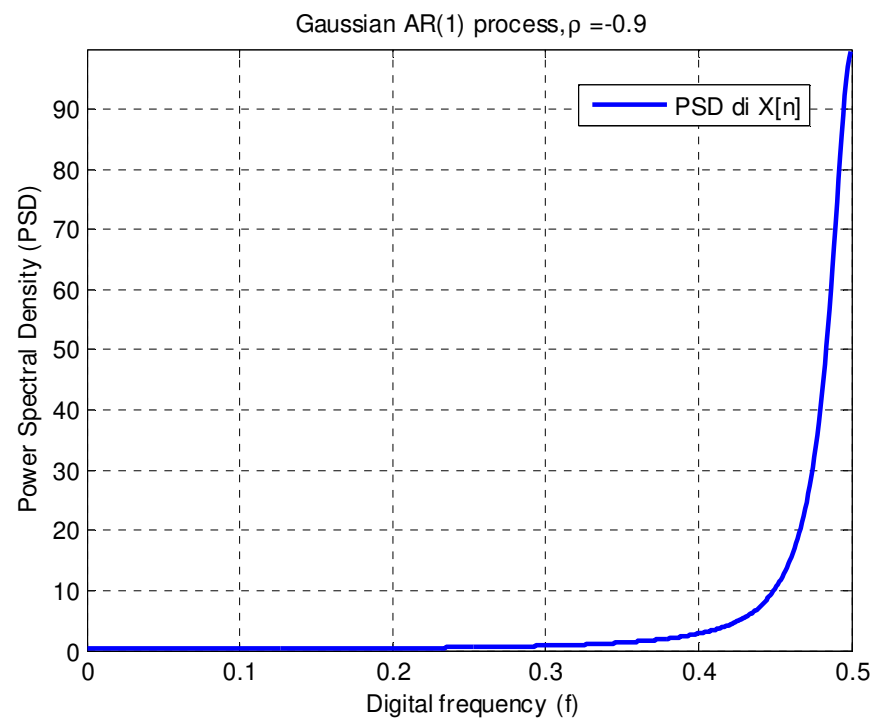
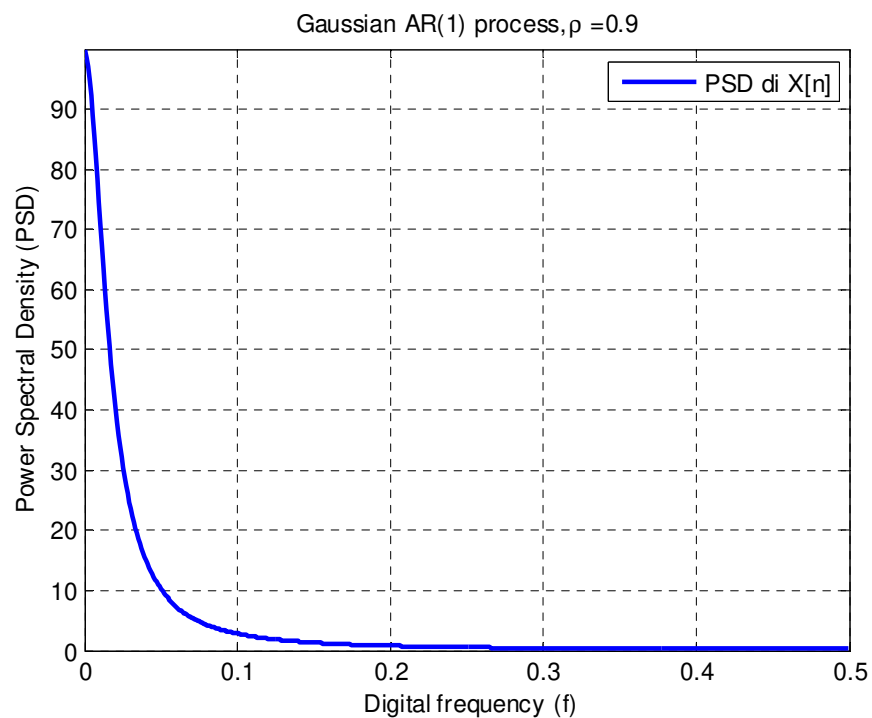
$$R_X[m] = \sigma_X^2 \rho^{|m|} = \frac{\sigma_W^2}{1 - \rho^2} \rho^{|m|}$$



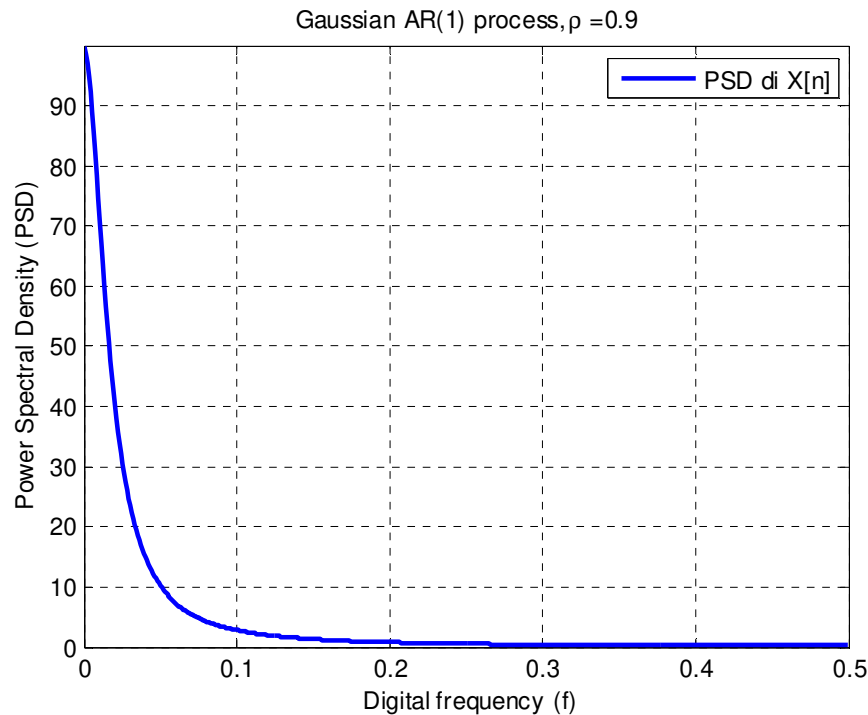
- ❑ The duration of the ACF (**correlation time** or **coherence time**) is larger when $|\rho|$ is closer to the unit circle.



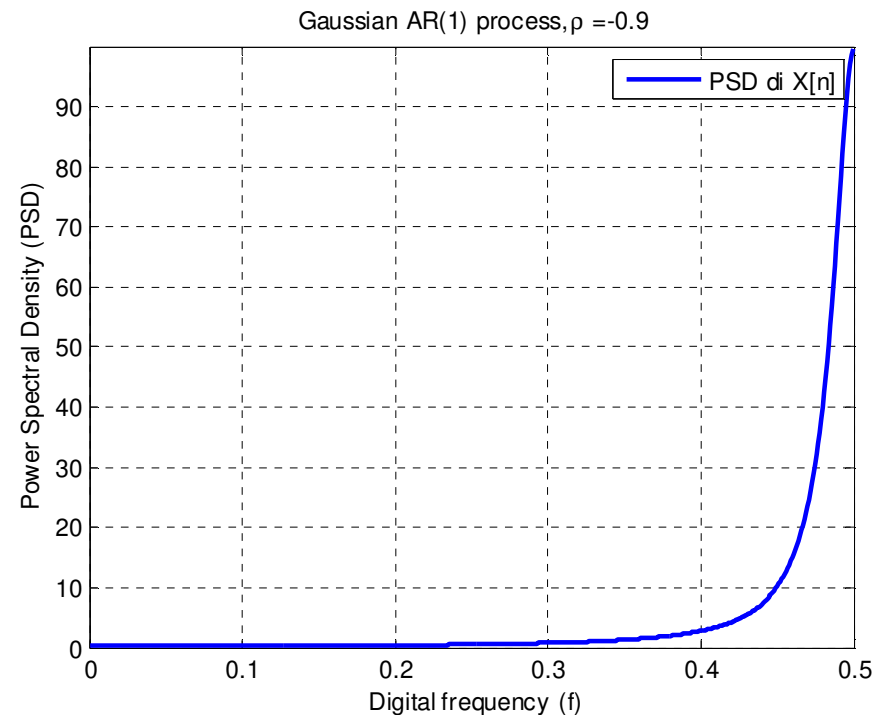
$$S_X(e^{j2\pi f}) = \sigma_W^2 \left| L(e^{j2\pi f}) \right|^2 = \sigma_W^2 \left| \frac{1}{1 - \rho e^{-j2\pi f}} \right|^2 = \frac{\sigma_W^2}{1 + \rho^2 - 2\rho \cos(2\pi f)}$$



$\rho > 0 \rightarrow$ low-pass process

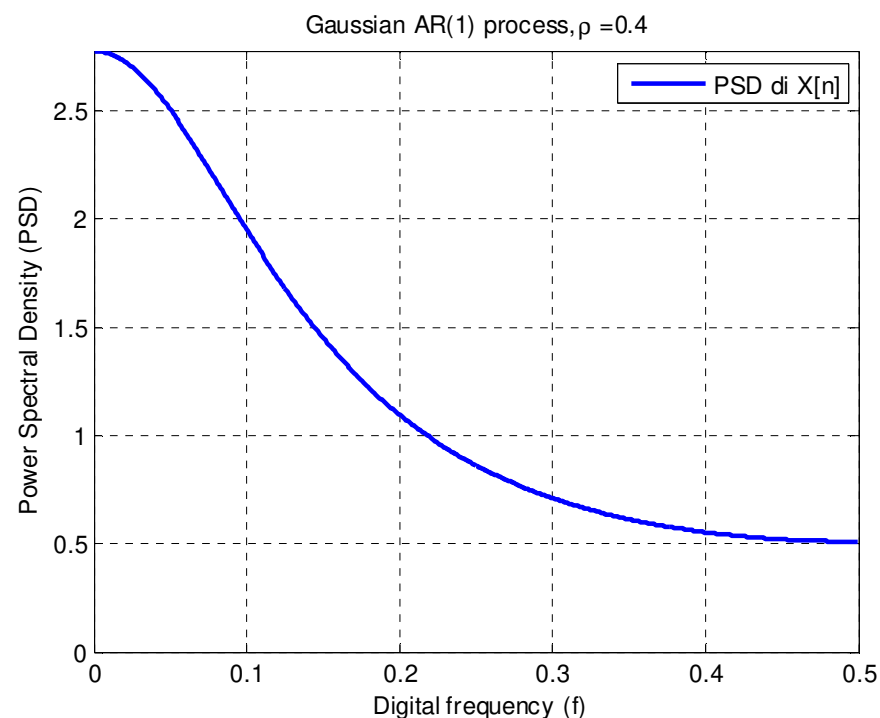
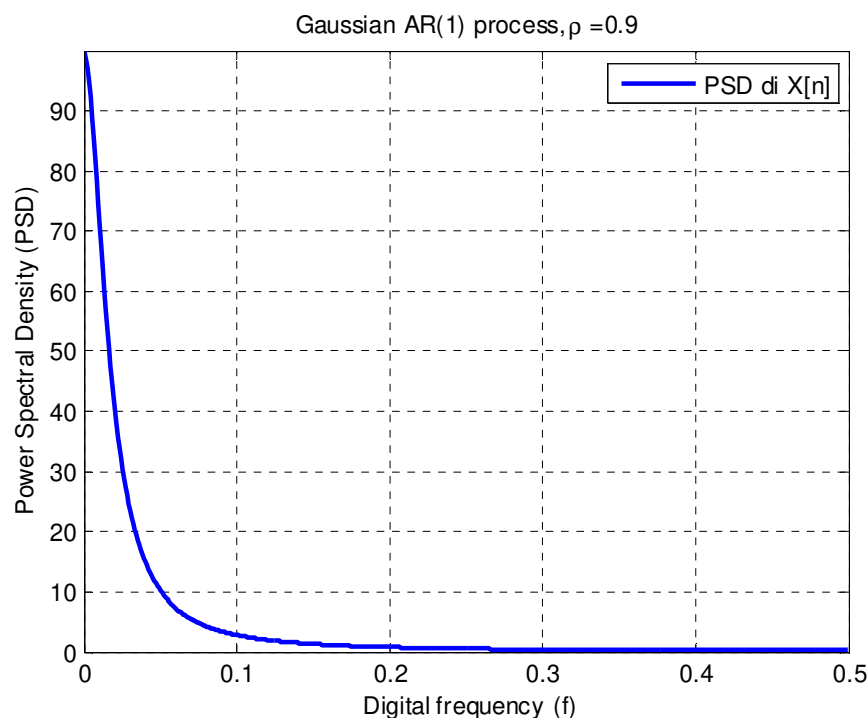


$\rho < 0 \rightarrow$ high-pass process



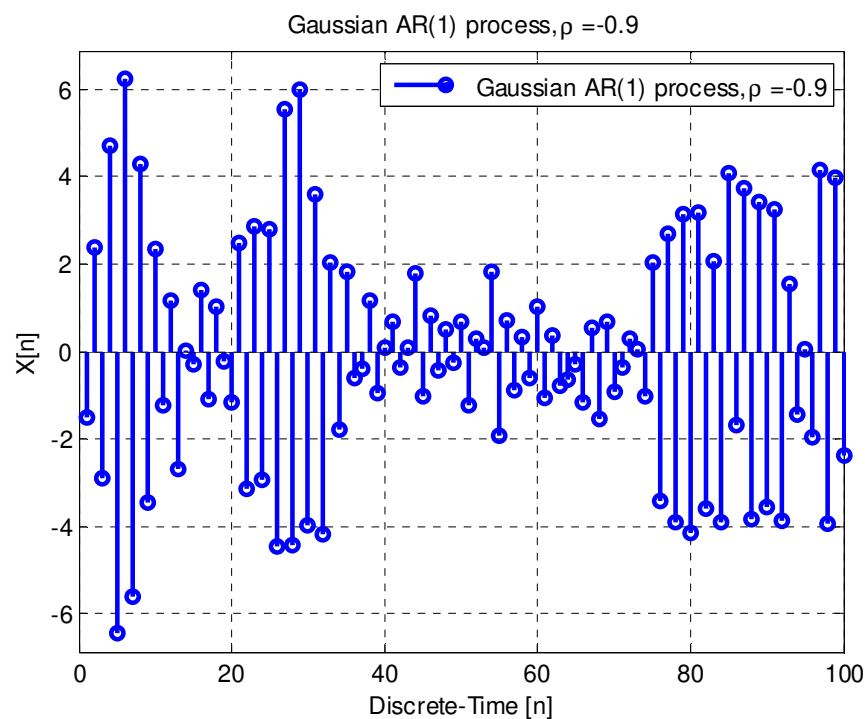
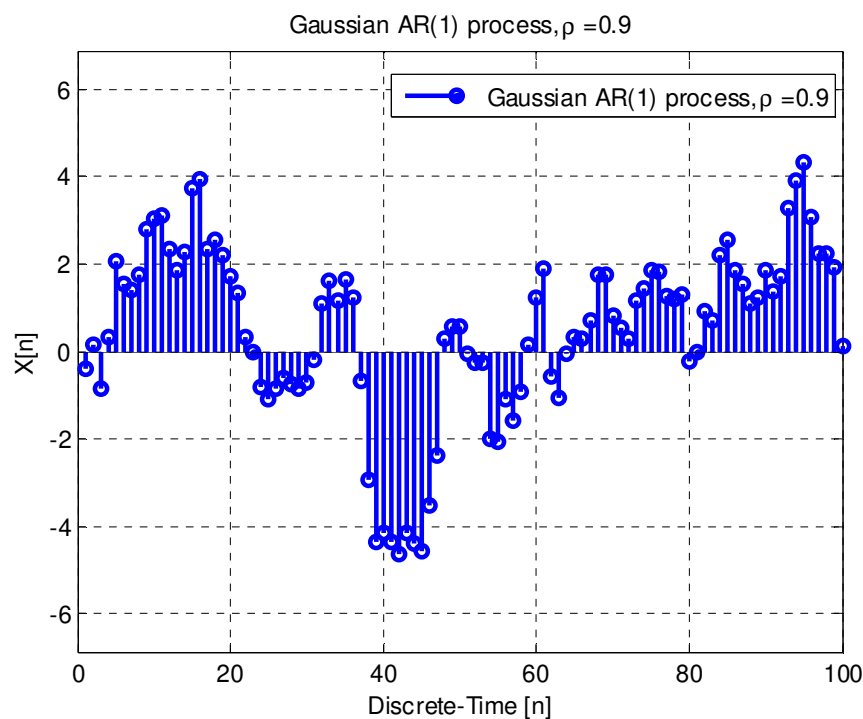
- ❑ With an AR(1) model we can obtain/represent only low-pass or high processes, but not pass-band processes.

- ❑ The bandwidth B of the process depends on how close the pole is to the unit circle: B gets lower when ρ gets closer to the unit circle.



$$\rho_{X[n],X[n+m]} = \rho^{|m|} \Rightarrow \rho_{X[n],X[n+1]} = \rho$$

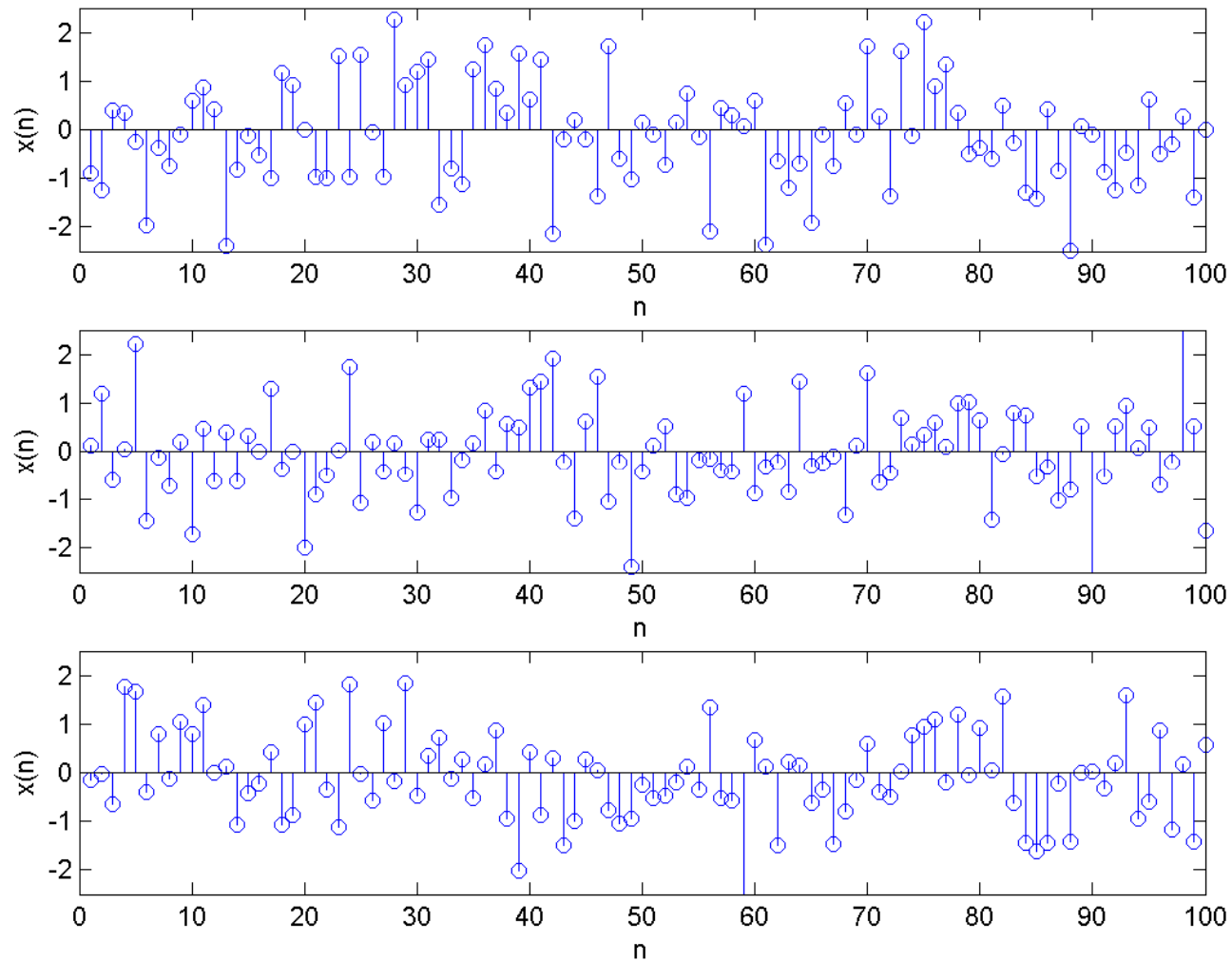
□ Parameter ρ represents the **one-lag correlation coefficient** of the process.



Markov process: AR(1)

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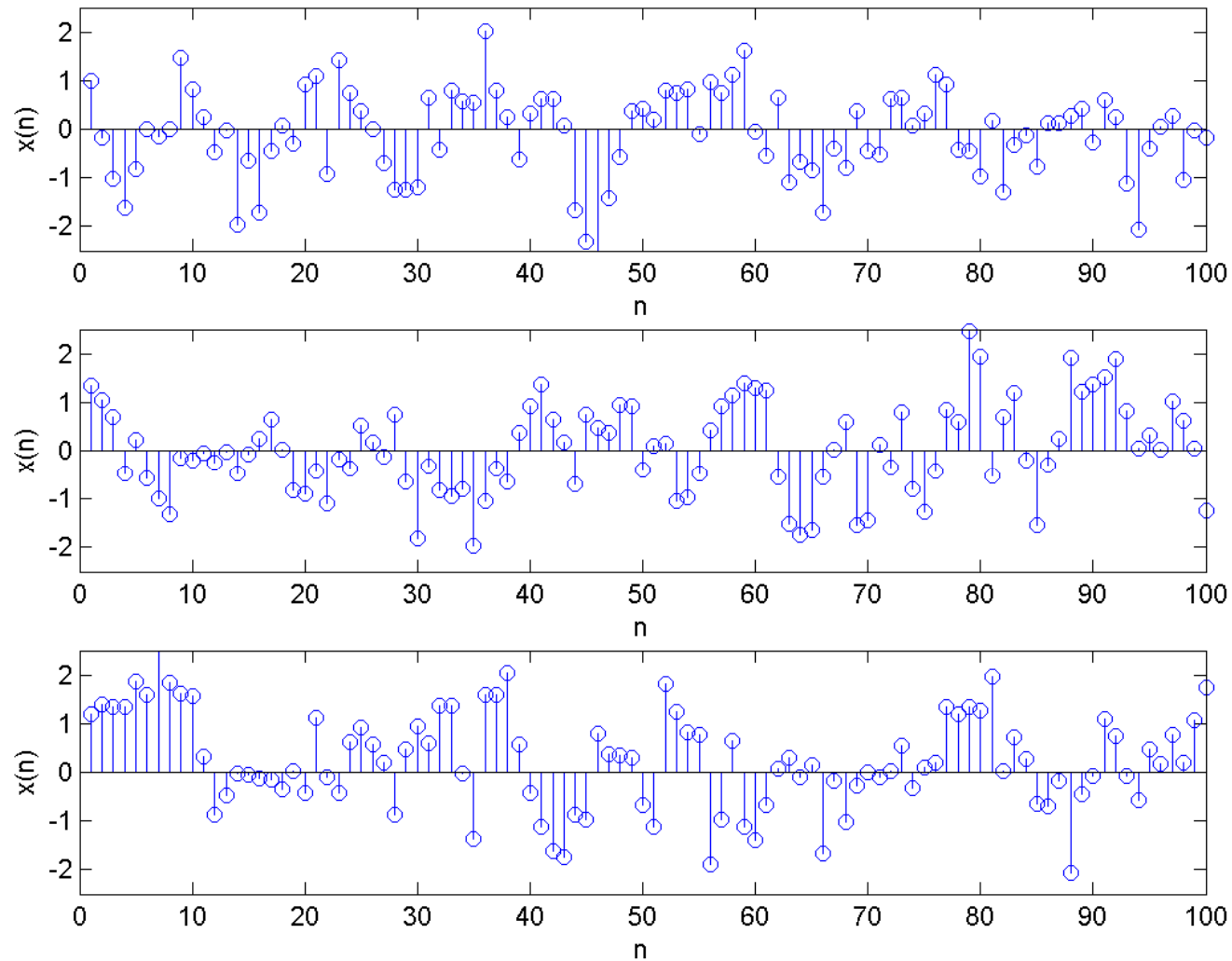
$\rho=0.1$



Markov process: AR(1)

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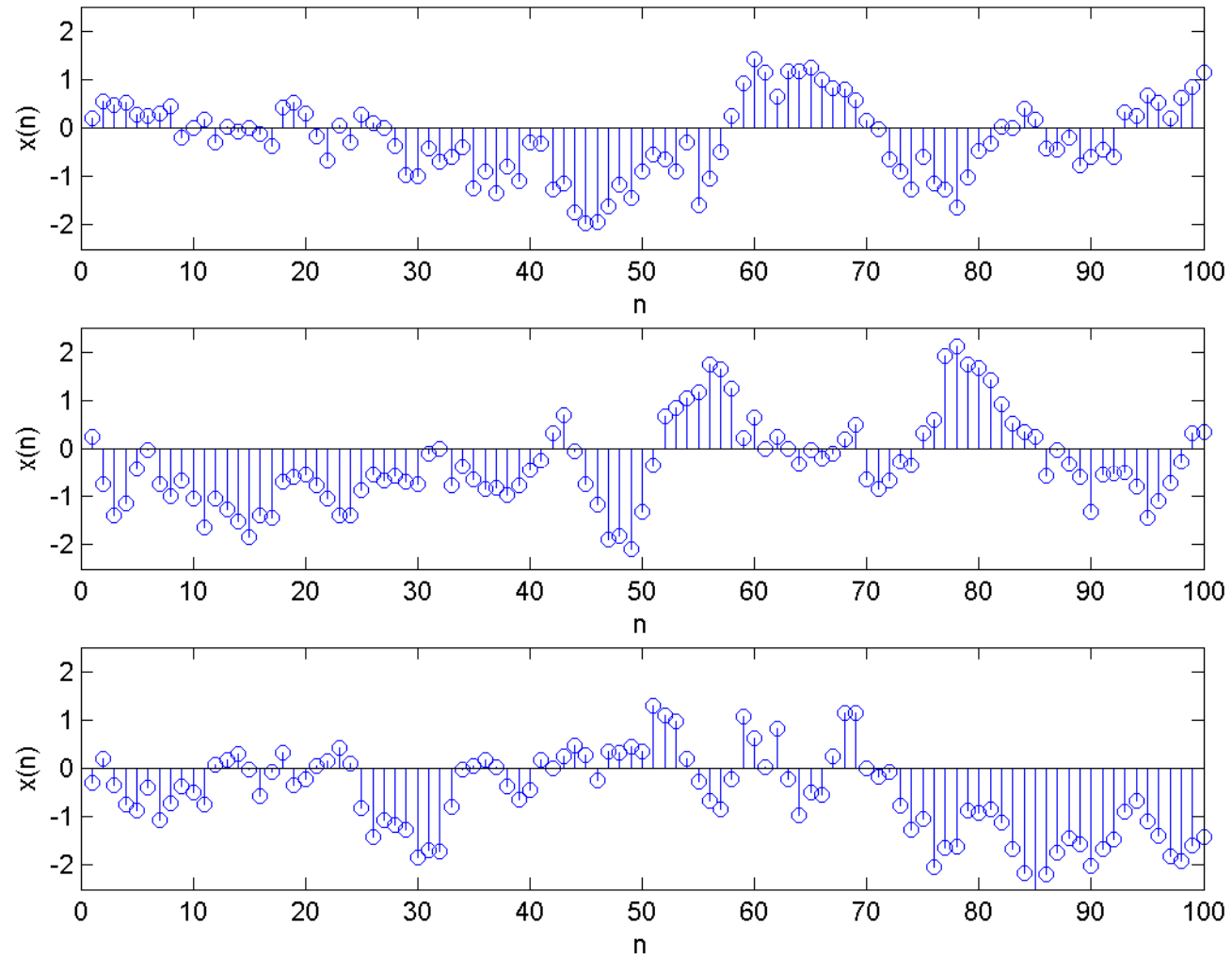
$\rho=0.5$



Markov process: AR(1)

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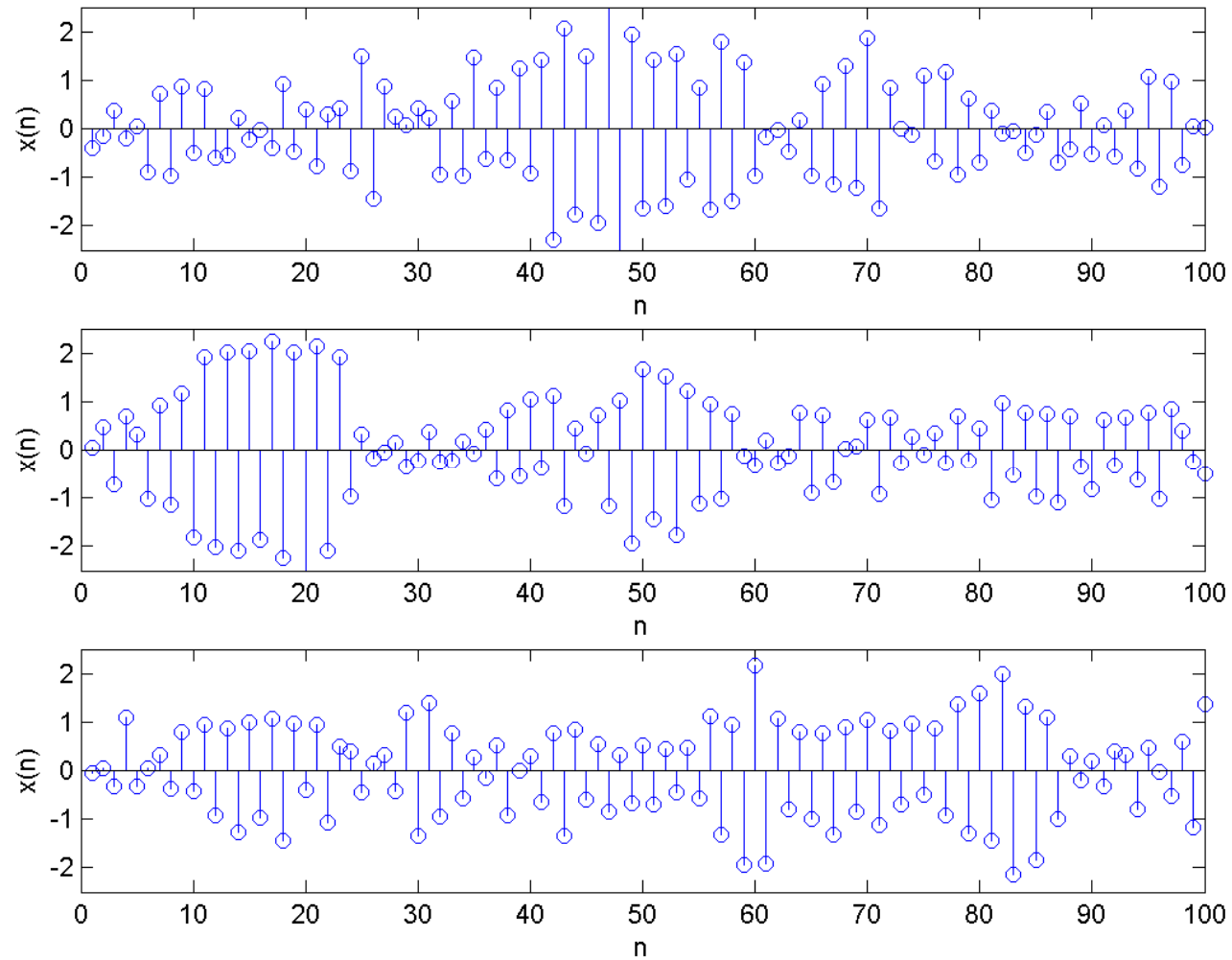
$\rho=0.9$



Markov process: AR(1)

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$\rho = -0.9$



- The PSD/ACF shape depends on the vectors of parameters **a** and **b**, or equivalently on the pole and zeros locations.
- We have analyzed this in detail for an **AR(1)** process. We have seen that varying the value of the parameter $a_1=\rho$ we can obtain a low-pass process ($\rho>0$) or an high-pass process ($\rho<0$), but we cannot obtain a pass-band process. The PSD bandwidth depends on $|\rho|$.
- When $\rho=0$ we have the degenerate case $X[n]=W[n]$, so process $X[n]$ is white.
- Let us consider now an **AR(2) process** and let see how to obtain a **pass-band process** by properly selecting the values of a_1 and a_2 , i.e. the locations of the two poles p_1 and p_2 .

$$X[n] = a_1 X[n-1] + a_2 X[n-2] + W[n]$$

- Let us derive first the **system function** and the **frequency response** of the **innovation filter**:

$$X[n] = a_1 X[n-1] + a_2 X[n-2] + W[n]$$


$$X(z) = a_1 z^{-1} X(z) + a_2 z^{-2} X(z) + W(z)$$


$$L(z) = \frac{X(z)}{W(z)} = \frac{1}{1 - a_1 z^{-1} - a_2 z^{-2}} = \frac{z^2}{z^2 - a_1 z - a_2}$$

$$L(e^{j2\pi f}) = \frac{1}{1 - a_1 e^{-j2\pi f} - a_2 e^{-j4\pi f}}$$

- The filter has 2 poles and 2 zeros at $z=0$.
- The poles are derived by solving the characteristic equation:

$$z^2 - a_1 z - a_2 = 0 \Rightarrow p_{1,2} = \frac{a_1 \pm \sqrt{a_1^2 + 4a_2}}{2}$$

- The complex PSD and the PSD can be immediately derived:

$$\begin{aligned} S_X(z) &= \sigma_w^2 L(z) L\left(\frac{1}{z}\right) = \frac{\sigma_w^2}{(1 - a_1 z^{-1} - a_2 z^{-2})(1 - a_1 z - a_2 z^2)} \\ &= \frac{\sigma_w^2}{(1 - p_1 z^{-1})(1 - p_2 z^{-1})(1 - p_1 z)(1 - p_2 z)}, \quad d < |z| < \frac{1}{d} \end{aligned}$$



$$d = \max\{|p_1|, |p_2|\}$$

- The PSD can be immediately derived:

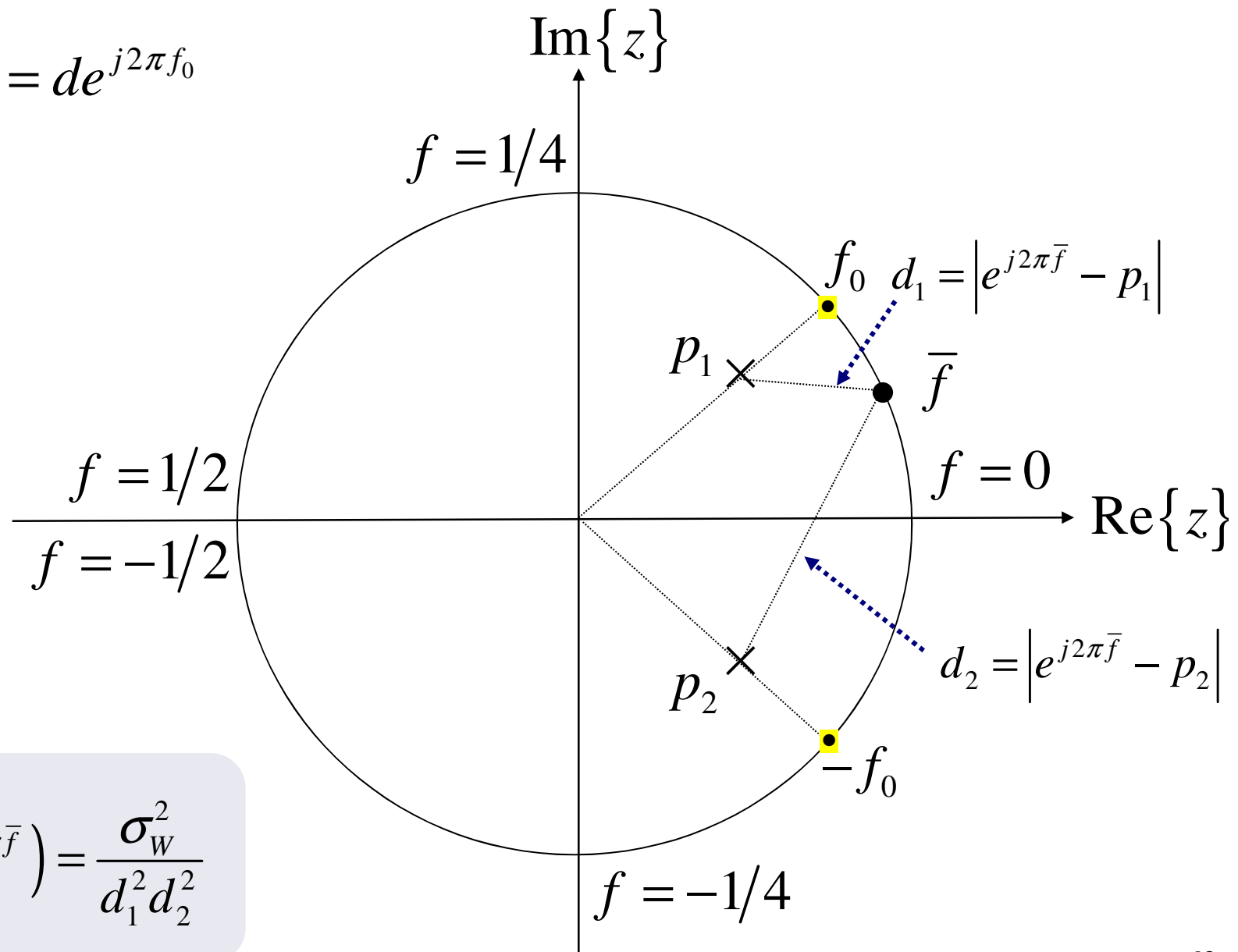
$$\begin{aligned} S_X(e^{j2\pi f}) &= \sigma_w^2 \left| L(e^{j2\pi f}) \right|^2 = \frac{\sigma_w^2}{\left| 1 - a_1 e^{-j2\pi f} - a_2 e^{-j4\pi f} \right|^2} \\ &= \frac{\sigma_w^2}{\left| e^{j2\pi f} - p_1 \right|^2 \left| e^{j2\pi f} - p_2 \right|^2} \end{aligned}$$

- The PSD of an AR(2) process with complex conjugate poles:

$$p_1 = p_2^* = d e^{j2\pi f_0}$$

is pass-band, with central frequency (close to) f_0 and bandwidth B which depends on d : the closer d is to 1 and the smaller B is.

$$p_1 = p_2^* = de^{j2\pi f_0}$$



$$S_X(e^{j2\pi \bar{f}}) = \frac{\sigma_W^2}{d_1^2 d_2^2}$$

- The **autocorrelation function (ACF)** can be obtained by the inverse FFT (IFFT) of the PSD:

$$R_X[m] = \text{IFFT} \left\{ S_X \left(e^{j2\pi f} \right) \right\}$$

- The ACF can also be obtained by solving the recursive equation derived for an AR(P) process in the following (see at p. 77):

$$R_X[m] = \sum_{k=1}^P a_k R_X[m-k] + \sigma_w^2 \delta[m], \quad m \geq 0$$

$P = 2$

$$R_X[m] = a_1 R_X[m-1] + a_2 R_X[m-2] + \sigma_w^2 \delta[m], \quad m \geq 0$$

- Calculating it for $m=0$ we get:

$$R_X[0] = a_1 R_X[-1] + a_2 R_X[-2] + \sigma_W^2 = a_1 R_X[1] + a_2 R_X[2] + \sigma_W^2$$

- If we now define the **normalized ACF**, which is the sequence of the correlation coefficients between $X[n]$ and $X[n+m]$ as a function of m :

$$\rho_X[m] \triangleq \frac{R_X[m]}{R_X[0]} = \frac{C_X[m]}{C_X[0]} = \frac{\text{cov}\{X[n], X[n+m]\}}{\sqrt{\text{var}\{X[n]\} \cdot \text{var}\{X[n+m]\}}} \Rightarrow \rho_X[0] = 1$$

$$R_X[0] = a_1 R_X[1] + a_2 R_X[2] + \sigma_W^2 = a_1 R_X[0] \rho_X[1] + a_2 R_X[0] \rho_X[2] + \sigma_W^2$$

- Thus, we get:
$$R_X[0] = \frac{\sigma_W^2}{1 - a_1 \rho_X[1] - a_2 \rho_X[2]}$$

- Calculating it for $m > 0$ we obtain the homogeneous difference equation:

$$R_X[m] - a_1 R_X[m-1] - a_2 R_X[m-2] = 0, \quad m > 0$$

$$\rho_X[m] - a_1 \rho_X[m-1] - a_2 \rho_X[m-2] = 0, \quad m > 0$$

- The solution of the difference equation for $m > 0$ is given by:

$$\rho_X[m] = \begin{cases} C_1 p_1^m + C_2 p_2^m, & m > 0, \quad \text{if the poles are different} \\ (C_1 + m C_2) p_1^m, & m > 0, \quad \text{if the poles are coincident} \end{cases}$$



$$\text{Coincident poles} \Leftrightarrow \Delta = a_1^2 + 4a_2 = 0$$

- Taking into account that the ACF is an even function, e.g. for the case of different poles, we have:

$$\rho_X[m] = C_1 p_1^{|m|} + C_2 p_2^{|m|}$$

- The two constants are obtained by imposing the initial conditions:

$$\begin{cases} \rho_X[0] = C_1 + C_2 = 1 \\ \rho_X[1] = C_1 p_1 + C_2 p_2 = \frac{a_1}{1 - a_2} \end{cases}$$

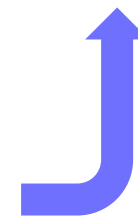


$$m=1 \Rightarrow \rho_X[1] - a_1 \rho_X[0] - a_2 \rho_X[-1] = 0 \Rightarrow \rho_X[1] = \frac{a_1}{1 - a_2}$$

- Solving the system of 2 linear equations we find the two constants:

$$\begin{cases} C_1 = -\frac{1}{p_2 - p_1} \left(\frac{a_1}{1 - a_2} - p_2 \right) = -\frac{1}{\sqrt{a_1^2 + 4a_2}} \left(\frac{a_1}{1 - a_2} - \frac{a_1 + \sqrt{a_1^2 + 4a_2}}{2} \right) \\ C_2 = \frac{1}{p_2 - p_1} \left(\frac{a_1}{1 - a_2} - p_1 \right) = \frac{1}{\sqrt{a_1^2 + 4a_2}} \left(\frac{a_1}{1 - a_2} - \frac{a_1 - \sqrt{a_1^2 + 4a_2}}{2} \right) \end{cases}$$

$$p_1 = \frac{a_1 - \sqrt{a_1^2 + 4a_2}}{2}, \quad p_2 = \frac{a_1 + \sqrt{a_1^2 + 4a_2}}{2}$$



$$\Rightarrow p_2 - p_1 = \sqrt{a_1^2 + 4a_2}$$

- Once we have C_1 , C_2 and $\rho_X[m]$, we can obtain immediately $R_X[0]$:

$$R_X[0] = \frac{\sigma_W^2}{1 - a_1\rho_X[1] - a_2\rho_X[2]}$$

- In the case of **complex poles**, since the coefficients of the characteristic equation a_1 and a_2 are real, the poles should be complex conjugated:

$$\text{Complex conjugate poles} \quad \Leftrightarrow \quad \Delta = a_1^2 + 4a_2 < 0$$

$$p_{1,2} = \frac{a_1 \pm j\sqrt{-(a_1^2 + 4a_2)}}{2} = de^{\pm j\omega_0} = de^{\pm j2\pi f_0}$$

- In the case of complex conjugate poles, also the two constants C_1 and C_2 are complex conjugated:

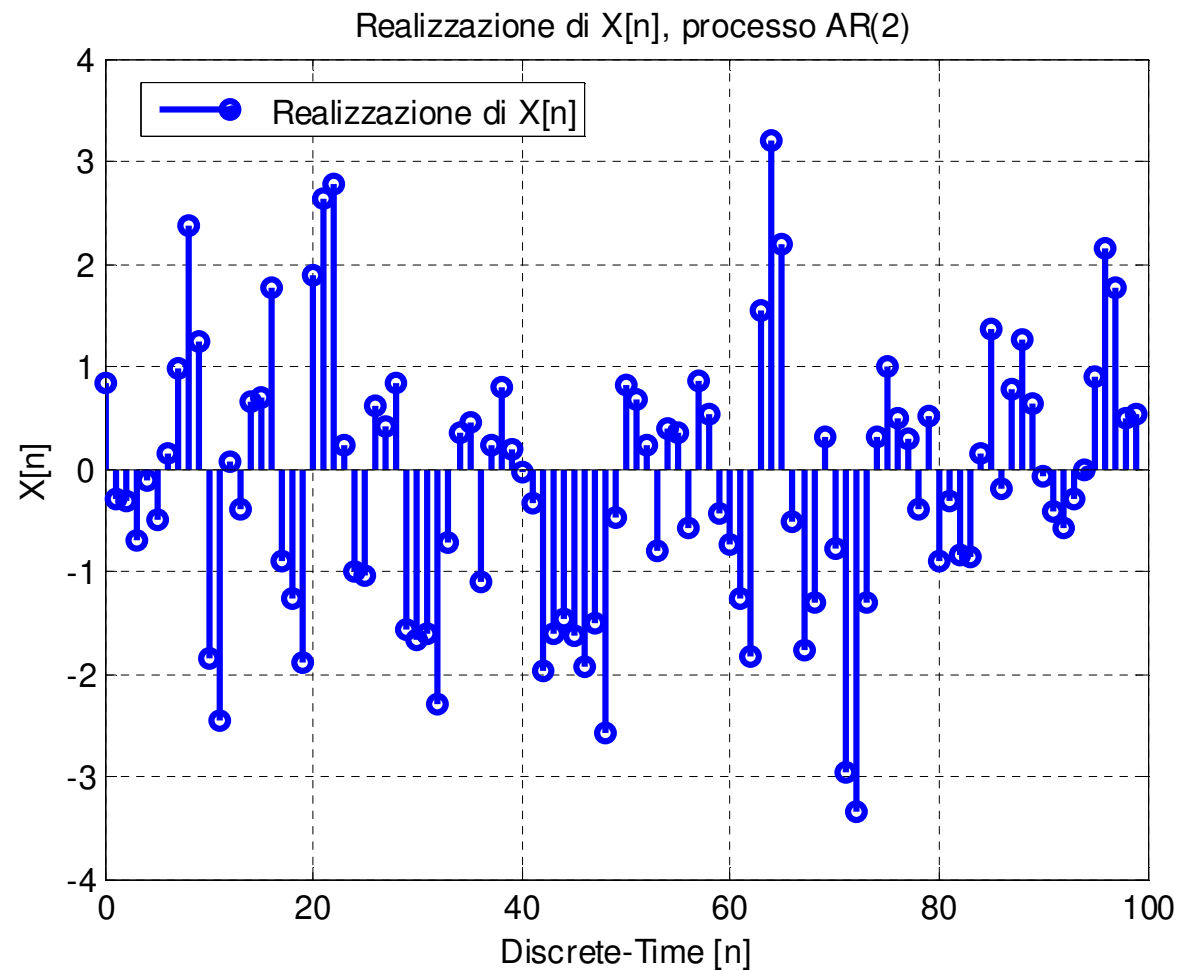
$$p_1 = de^{j2\pi f_0}, \quad p_2 = de^{-j2\pi f_0} \quad \Rightarrow \quad C_1 = Ce^{j\theta_0}, \quad C_2 = Ce^{-j\theta_0}$$



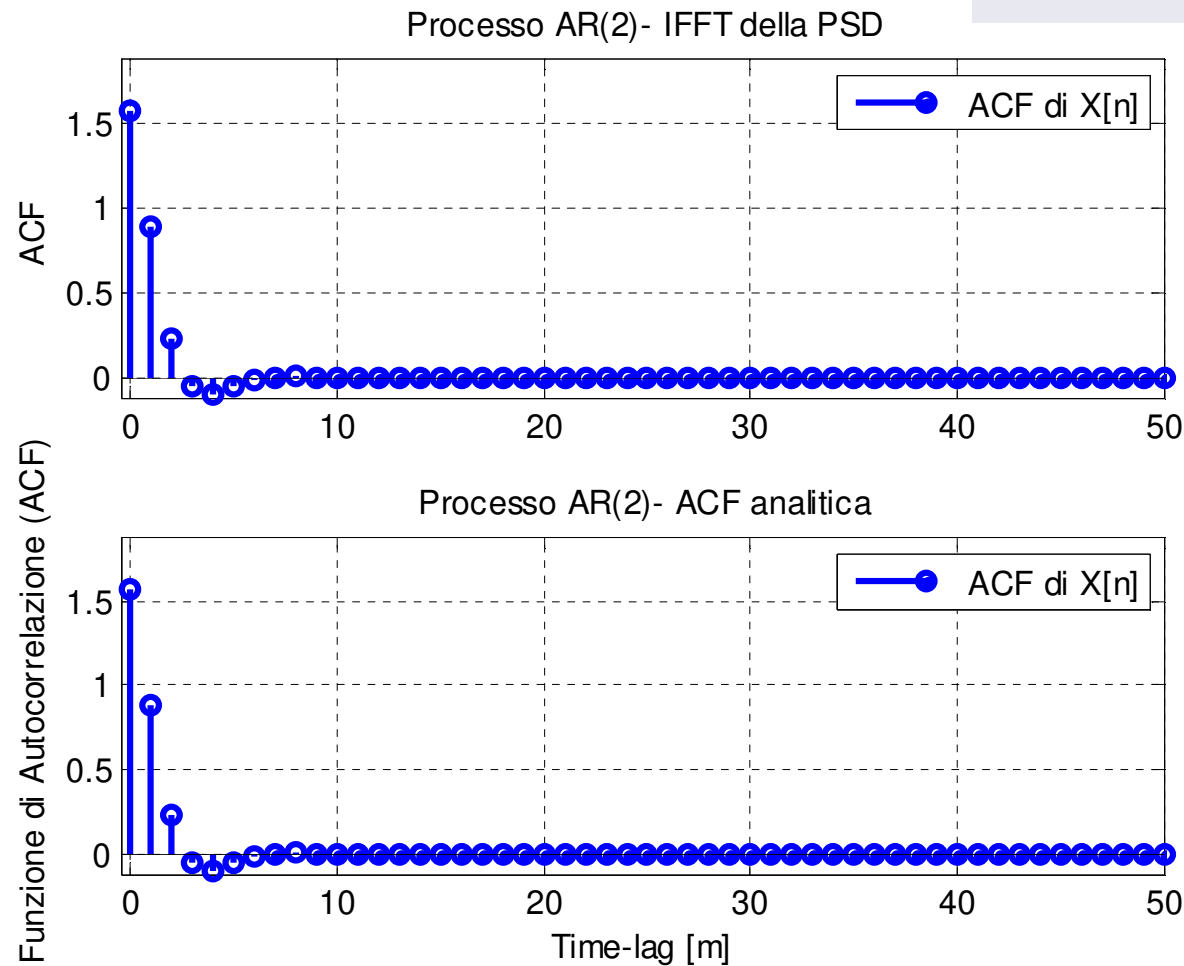
$$\begin{aligned} \rho_X[m] &= C_1 p_1^{|m|} + C_2 p_2^{|m|} = Ce^{j\theta_0} d^{|m|} e^{j2\pi f_0 |m|} + Ce^{-j\theta_0} d^{|m|} e^{-j2\pi f_0 |m|} \\ &= 2\Re\{Ce^{j\theta_0} d^{|m|} e^{j2\pi f_0 |m|}\} = 2Cd^{|m|} \cos(2\pi f_0 |m| + \theta_0) \end{aligned}$$

- The constants C and θ_0 are obtained as the modulo and the phase of C_1 .
- The ACF has an oscillatory behavior with period $T_0=1/f_0$ which depends on the phases of the poles $\pm 2\pi f_0$. Whereas the envelope of the ACF tends to 0 with a speed that depends on the modulo d of the poles.

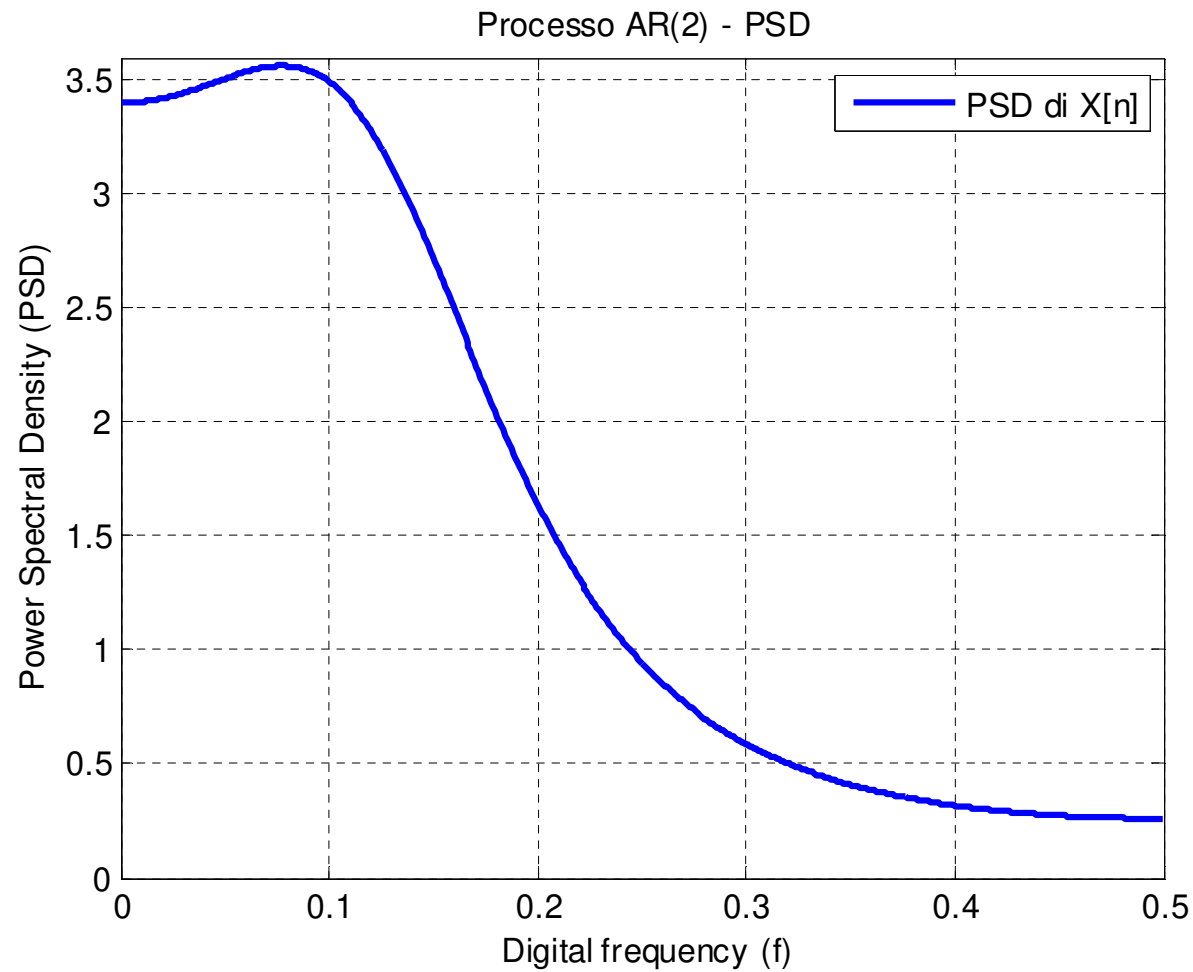
$$p_1 = p_2^* = de^{j2\pi f_0} = 0.5 \cdot e^{j2\pi \frac{1}{8}}$$



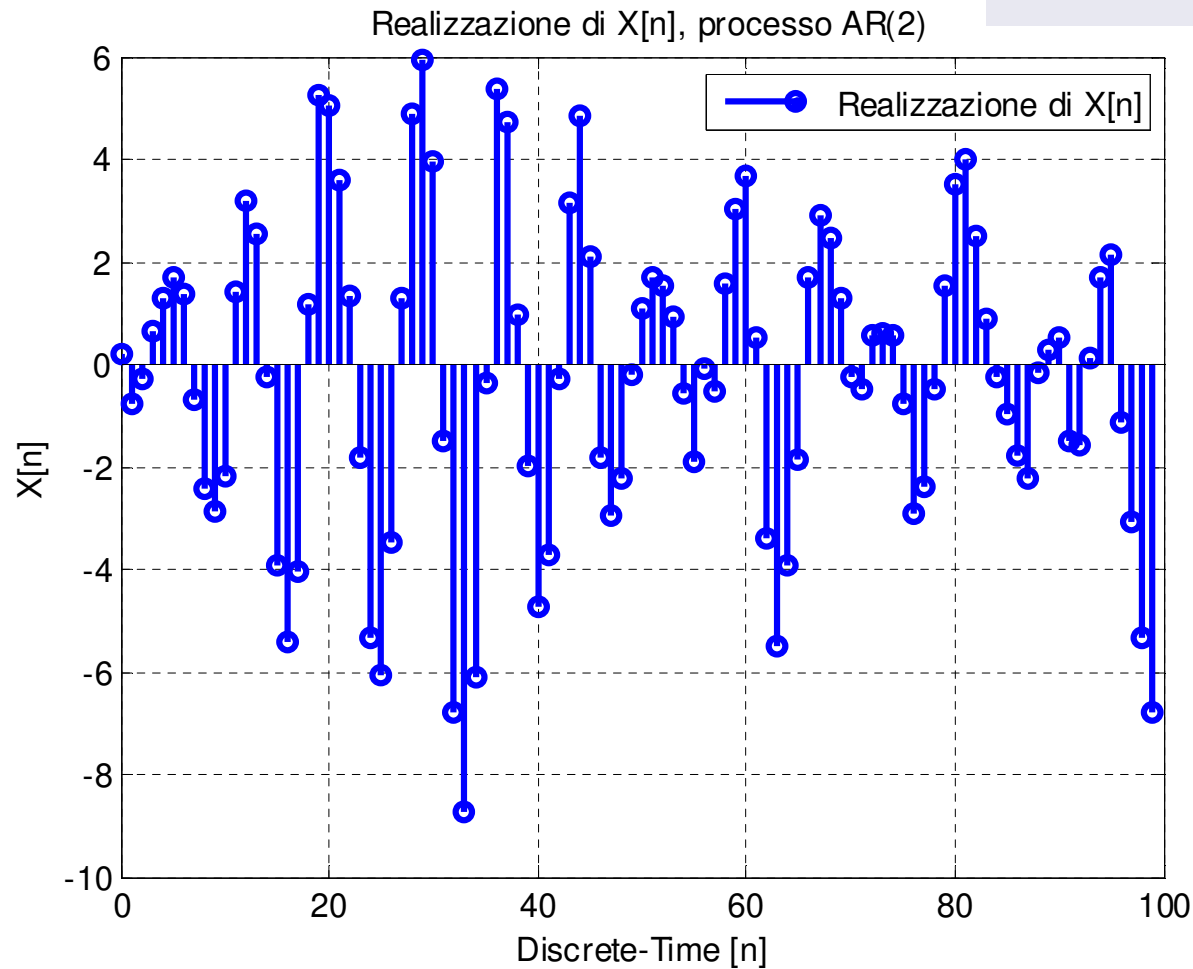
$$d = 0.5, \quad T_0 = \frac{1}{f_0} = 8$$



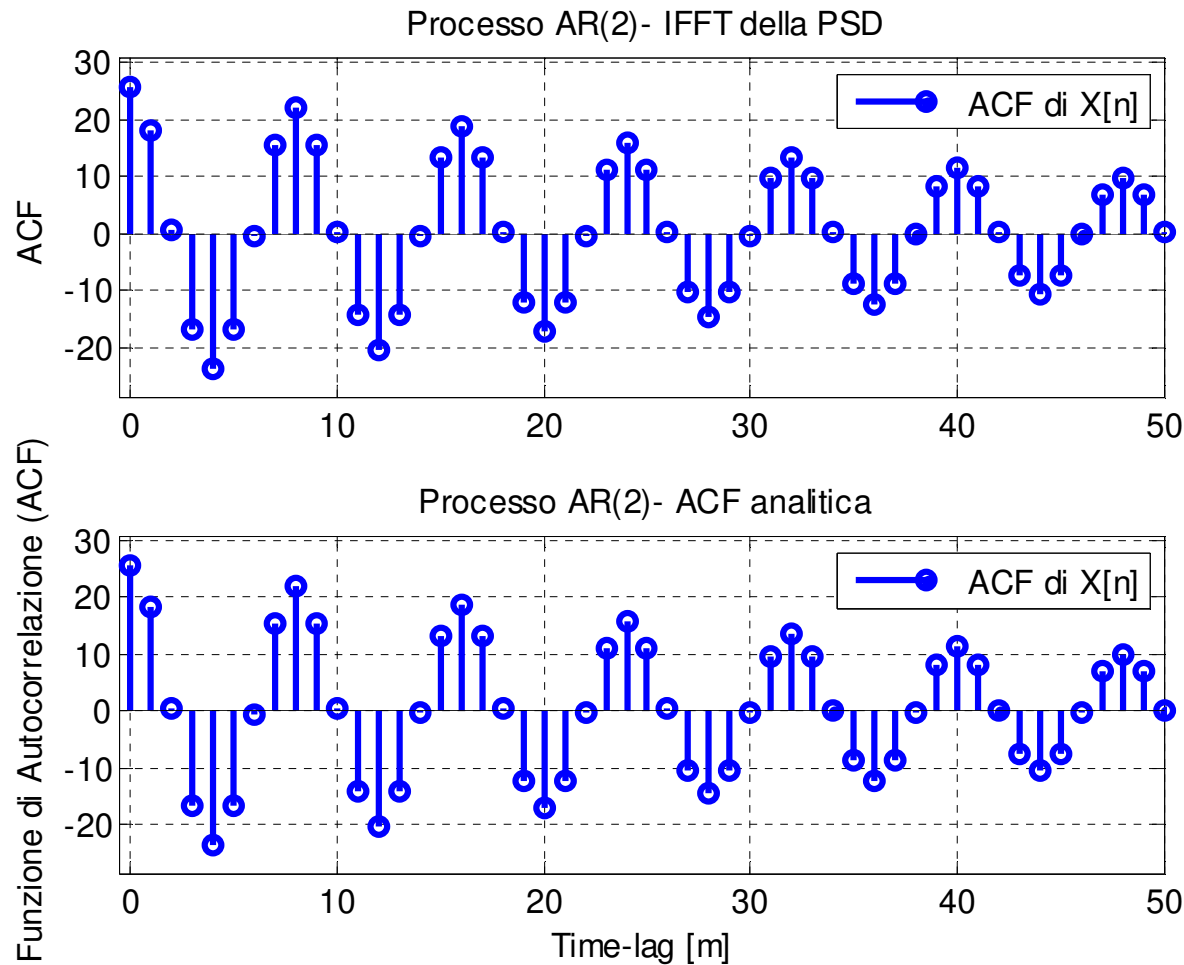
$$d = 0.5, \quad f_0 = \frac{1}{8} = 0.125$$



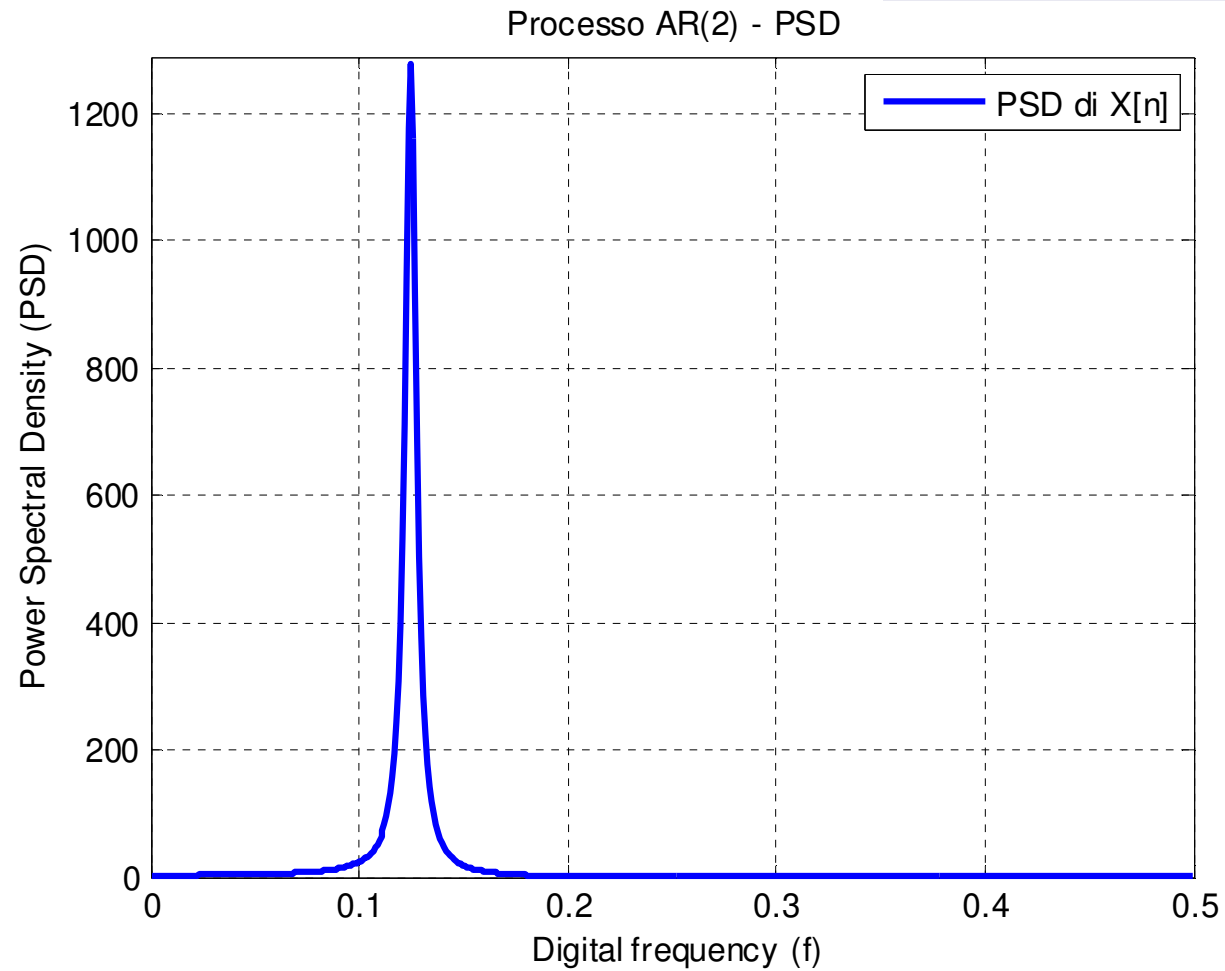
$$p_1 = p_2^* = 0.98 \cdot e^{j2\pi\frac{1}{8}}$$



$$d = 0.98, \quad T_0 = \frac{1}{f_0} = 8$$



$$d = 0.98, \quad f_0 = \frac{1}{8} = 0.125$$



FIR filter of order Q

The difference equation is not recursive

$$W[n] \rightarrow \boxed{l[n]} \rightarrow X[n] = \sum_{k=0}^Q b_k W[n-k]$$

($a_k = 0$ for $k > 0$)

Impulse response: $l[n] = \sum_{k=0}^Q b_k \delta[n-k]$

MATLAB:
`x=filter(b,1,w)`

Zeros of the filter
= roots of the equation: $\sum_{k=0}^Q b_k z^{Q-k} = 0$

$$L(e^{j2\pi f}) = \sum_{k=0}^Q b_k e^{-j2\pi kf} = b_0 e^{-j2\pi fQ} \prod_{k=1}^Q (e^{j2\pi f} - z_k)$$

- **Important:** for an MA(Q) process the **correlation time** is equal to the length of the impulse response of the innovation filter, i.e. $L_x=Q$.

$$l[n] = \sum_{k=0}^Q b_k \delta[n-k] = \begin{cases} b_n, & n = 0, 1, \dots, Q \\ 0, & \text{otherwise} \end{cases}$$

$$R_X[m] = \sigma_w^2 l[m] \otimes l[-m]$$

If $l[n] \neq 0$ for $0 \leq n \leq Q$ then $R_X[m] \neq 0$ for $|m| \leq Q$

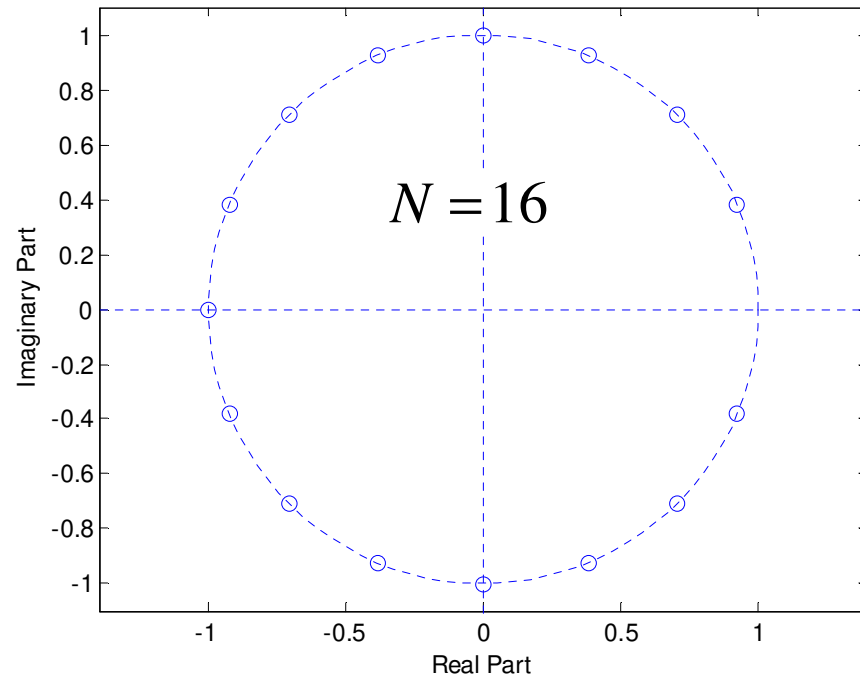
Frequency response:
$$L(e^{j2\pi f}) = \sum_{n=0}^Q b_n e^{-j2\pi f n}$$

Moving Average (MA) FIR filter

$$X[n] = \frac{1}{N} \sum_{k=0}^{N-1} W[n-k] \quad (Q = N-1)$$

$$l[n] = \sum_{k=0}^{N-1} \frac{1}{N} \delta[n-k]$$

$$= \begin{cases} \frac{1}{N}, & n = 0, 1, \dots, N-1 \\ 0, & \text{otherwise} \end{cases}$$



The MA FIR filter has $Q=N-1$ zeros equispaced on the unit circle:

**Zeros of the filter
= roots of the equation:** $\sum_{k=0}^{N-1} \frac{1}{N} z^{N-1-k} = 0$

$$z_k = e^{j2\pi \frac{k}{N}},$$

$$k = 1, 2, \dots, N-1$$

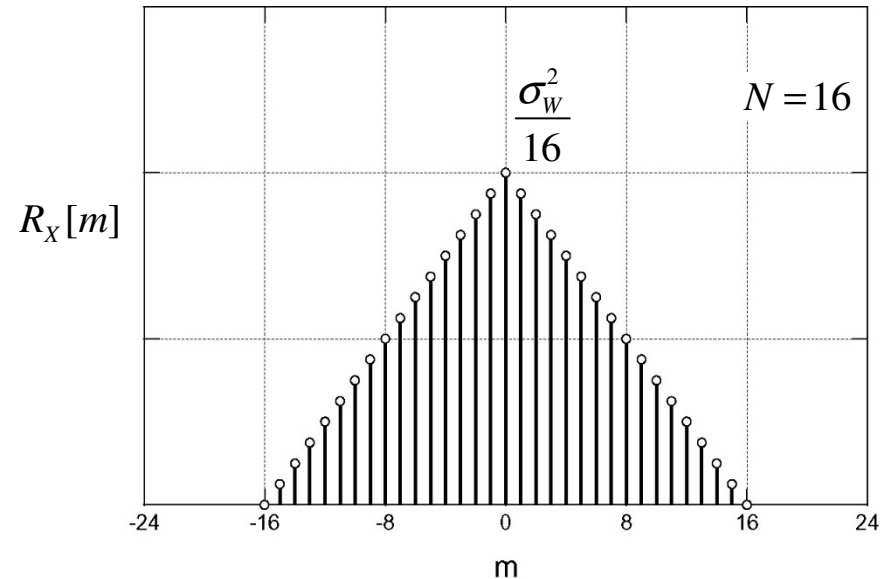
Moving Average (MA) FIR filter

$$Q = N - 1$$

$$X[n] = \frac{1}{N} \sum_{k=0}^{N-1} W[n-k]$$

$$l[n] = \begin{cases} \frac{1}{N}, & n = 0, 1, \dots, N-1 \\ 0, & \text{otherwise} \end{cases}$$

$$R_X[m] = \sigma_w^2 l[m] \otimes l[-m] = \frac{\sigma_w^2}{N} \left[1 - \frac{|m|}{N} \right], \quad m = 0, \pm 1, \dots, \pm N$$



Innovation filter = low-pass
→ $X[n]$ is a low-pass process



Moving Average processes: MA(Q)

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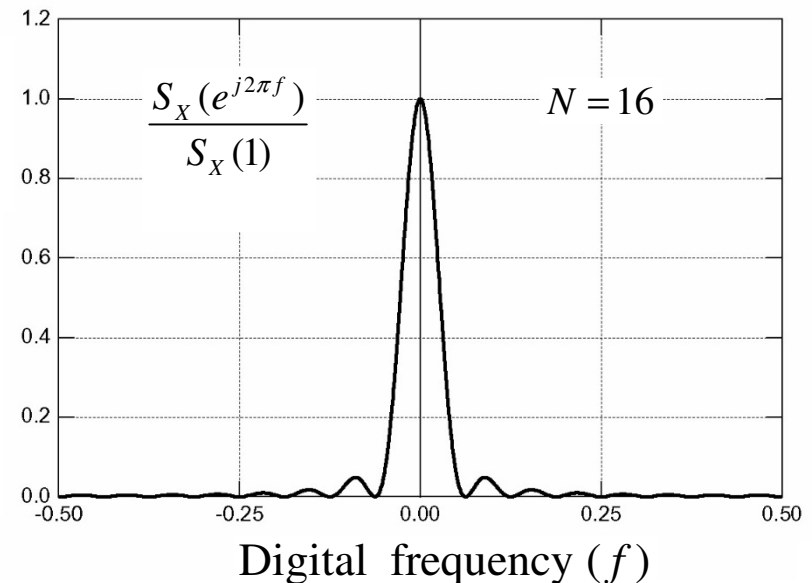
$$\begin{aligned} L(e^{j2\pi f}) &= \sum_{n=0}^{N-1} \frac{1}{N} e^{-j2\pi f n} = \frac{1}{N} \sum_{n=0}^{N-1} (e^{-j2\pi f})^n \\ &= \frac{1}{N} \cdot \frac{1 - e^{-j2\pi f N}}{1 - e^{-j2\pi f}} = \frac{1}{N} \cdot \frac{e^{j\pi f N} - e^{-j\pi f N}}{e^{j\pi f} - e^{-j\pi f}} \cdot \frac{e^{-j\pi f N}}{e^{-j\pi f}} \\ &= \frac{1}{N} \cdot \frac{e^{j\pi f N} - e^{-j\pi f N}}{e^{j\pi f} - e^{-j\pi f}} e^{-j\pi f (N-1)} \\ &= \frac{1}{N} \cdot \frac{e^{j\pi f N} - e^{-j\pi f N}}{2j} \cdot \frac{2j}{e^{j\pi f} - e^{-j\pi f}} e^{-j\pi f (N-1)} \\ &= \frac{1}{N} \cdot \frac{\sin(\pi f N)}{\sin(\pi f)} e^{-j\pi f (N-1)} = \frac{\pi f}{N\pi f} \cdot \frac{\sin(\pi f N)}{\sin(\pi f)} e^{-j\pi f (N-1)} \end{aligned}$$

$$L(e^{j2\pi f}) = \frac{\text{sinc}(Nf)}{\text{sinc}(f)} e^{-j\pi f(N-1)} \quad \text{"discrete sinc"}$$

$$S_X(e^{j2\pi f}) = \sigma_W^2 \left| L(e^{j2\pi f}) \right|^2 = \sigma_W^2 \cdot \frac{\text{sinc}^2(Nf)}{\text{sinc}^2(f)}$$

$$S_X(e^{j2\pi f}) \Big|_{f=0} = S_X(1) = \sigma_W^2$$

$$S_X(e^{j2\pi f}) = 0 \quad \text{for } f = \frac{k}{N}$$
$$k = 1, 2, \dots, N-1$$



- **Model identification problem:** given the observation of N samples of the process $X[n]$, $n=0,1,2,\dots,N-1$, we want to find the best model to represent this process.
- Identify the ARMA(P,Q) model means to find the parameters of the model that best represent these data:

$$\{\sigma_w^2, \mathbf{a}, \mathbf{b}\} \rightarrow P + Q + 1 \text{ unknown parameters}$$

- Actually, the orders P and Q of the AR and MA parts are also generally unknown. The problem of estimating P and Q from the data is called **Model Order Selection (MOS)**. We do not treat here this problem.
- In order to solve the model identification problem we apply the **Method of Moments** (Maximum Likelihood estimate is also possible, but the solution is much more difficult to find and it is not in closed form).

- The first step is to find a relationship between the unknown parameters and some moments of the process. We use to this purpose the ACF, that in the case of real-valued processes is defined as:

$$ACF : \quad R_X[m] = E\{X[n]X[n+m]\}$$

$$R_X[m] = R_X[-m] = E\{X[n]X[n-m]\}$$

- To find the relationship that we need, let us multiply all the terms of the ARMA recursive equation by $X[n-m]$, assuming that $m \geq 0$:


$$X[n] = \sum_{k=1}^P a_k X[n-k] + \sum_{k=0}^Q b_k W[n-k]$$
$$X[n]X[n-m] = \sum_{k=1}^P a_k X[n-k]X[n-m] + \sum_{k=0}^Q b_k W[n-k]X[n-m]$$

- Then, we calculate the statistical expectation of all terms:

$$E\{X[n]X[n-m]\} = \sum_{k=1}^P a_k E\{X[n-k]X[n-m]\} \\ + \sum_{k=0}^Q b_k E\{W[n-k]X[n-m]\}$$

$$R_X[m] = \sum_{k=1}^P a_k R_X[m-k] + \sum_{k=0}^Q b_k R_{XW}[m-k]$$

- The cross-correlation function between the innovation process $W[n]$ (input of the innovation filter) and the process $X[n]$ (output of the innovation filter) is defined as follows and it is not a symmetric function:

$$\begin{aligned} R_{XW}[m] &\triangleq E\{X[n]W[n+m]\} = E\{X[n-m]W[n]\} = E\{W[n]X[n-m]\} \\ &= R_{WX}[-m] \neq R_{XW}[-m] = E\{X[n]W[n-m]\} \end{aligned}$$

- To derive this cross-correlation function, remember that $X[n]$ is the output of a causal filter, the innovation filter, whose impulse response $l[n]$ is causal, when the input is the white process $W[n]$ (it is a sequence of uncorrelated random variables):

$$X[n] = l[n] \otimes W[n] = \sum_{k=0}^{+\infty} l[k]W[n-k]$$

$$\begin{aligned}
 R_{XW}[m] &= E\{X[n]W[n+m]\} = \sum_{k=0}^{+\infty} l[k]E\{W[n-k]W[n+m]\} \\
 &= \sum_{k=0}^{+\infty} l[k]R_W[m+k] = \sigma_W^2 \sum_{k=0}^{+\infty} l[k]\delta[m+k] = \sigma_W^2 l[-m]
 \end{aligned}$$

- Since $l[n]$ is the impulse response of a causal filter, it is a causal signal, i.e. $l[n]=0$ for $n<0$. Hence, we have:

$$R_{XW}[m] = \begin{cases} 0, & m > 0 \\ \sigma_W^2 l[-m], & m \leq 0 \end{cases}$$

$$\sum_{k=0}^Q b_k R_{XW}[m-k] = \sigma_W^2 \sum_{k=0}^Q b_k l[k-m] = \begin{cases} 0, & m > Q \\ \sigma_W^2 \sum_{k=m}^Q b_k l[k-m], & 0 \leq m \leq Q \end{cases}$$



$$\sum_{k=0}^Q b_k R_{xw}[m-k] = \begin{cases} 0, & m > Q \\ \sigma_w^2 \sum_{k=m}^Q b_k l[k-m], & 0 \leq m \leq Q \end{cases}$$

if we define $i \triangleq k - m$ we get:

$$\sum_{k=0}^Q b_k R_{xw}[m-k] = \begin{cases} 0, & m > Q \\ \sigma_w^2 \sum_{i=0}^{Q-m} l[i] b_{i+m}, & 0 \leq m \leq Q \end{cases}$$

$$R_X[m] = \sum_{k=1}^P a_k R_X[m-k] + \sum_{k=0}^Q b_k R_{XW}[m-k]$$
$$= \begin{cases} \sum_{k=1}^P a_k R_X[m-k], & m > Q \\ \sum_{k=1}^P a_k R_X[m-k] + \sigma_W^2 \sum_{k=0}^{Q-m} l[k] b_{k+m}, & 0 \leq m \leq Q \end{cases}$$

- For $m < 0$ we can exploit the symmetry of the ACF:

$$R_X[m] = R_X[-m]$$

- If we set $m=Q+1, Q+2, \dots, Q+P$ we obtain P linear equations in the P unknowns:

$$R_X[m] = \sum_{k=1}^P a_k R_X[m-k] \quad \text{for } m = Q+1, Q+2, \dots, Q+P$$

$$\begin{bmatrix} R_X[Q] & R_X[Q-1] & \cdots & R_X[Q-(P-1)] \\ R_X[Q+1] & R_X[Q] & \ddots & R_X[Q-(P-2)] \\ \vdots & \vdots & \ddots & \vdots \\ R_X[Q+(P-1)] & R_X[Q+(P-2)] & \cdots & R_X[Q] \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_P \end{bmatrix} = \begin{bmatrix} R_X[Q+1] \\ R_X[Q+2] \\ \vdots \\ R_X[Q+P] \end{bmatrix}$$

$$\mathbf{R}\mathbf{a} = \mathbf{r} \Rightarrow \mathbf{a} = \mathbf{R}^{-1}\mathbf{r}$$

- If we replace $\mathbf{R}$ and $\mathbf{r}$ with their estimates obtained by using the unbiased Sample Autocorrelation Function, we get an estimate of the AR coefficients: $\hat{\mathbf{a}} = \hat{\mathbf{R}}^{-1}\hat{\mathbf{r}}$

- By inserting the estimate of vector **a** in the previous equations for $m=0,1,2, \dots, Q$, we get $Q+1$ (nonlinear) equation in the $Q+1$ unknowns:

$$R_X[m] = \sum_{k=1}^P \hat{a}_k R_X[m-k] + \sigma_W^2 \sum_{k=0}^{Q-m} l[k; \hat{\mathbf{a}}, \mathbf{b}] b_{k+m}, \quad 0 \leq m \leq Q$$

- Since the above equations are non linear with respect to vector **b**, to estimate the MA coefficients is more computationally demanding than to estimate the AR coefficients.

- For **MA(Q) processes** $P=0$ and we have that $l[k]=b_k$, hence we get:



$$R_X[m] = \sum_{k=0}^Q b_k R_{XW}[m-k] = \begin{cases} 0, & m > Q \\ \sigma_W^2 \sum_{k=0}^{Q-m} l[k] b_{k+m}, & 0 \leq m \leq Q \end{cases}$$

$$= \begin{cases} 0, & m > Q \\ \sigma_W^2 \sum_{k=0}^{Q-m} b_k b_{k+m}, & 0 \leq m \leq Q \end{cases}$$

- This result tells us that: **(1)** the length of the ACF of an MA(Q) process is Q ; **(2)** for MA(Q) processes the relationship between the ACF values and the unknown parameters $\{b_k\}$ is **non-linear**, hence it is much more difficult to find the $\{b_k\}$ coefficients than the $\{a_k\}$ coefficients, given the ACF values.

- For **AR(P) processes** $Q=0$, taking into account that $l[0]=1$ and $b_0=1$, we get:

$$R_X[m] = \begin{cases} \sum_{k=1}^P a_k R_X[m-k], & m > 0 \\ \sum_{k=1}^P a_k R_X[m-k] + \sigma_w^2, & m = 0 \end{cases}$$


$$R_X[m] = \sum_{k=1}^P a_k R_X[m-k] + \sigma_w^2 \delta[m], \quad m \geq 0$$

- **Important:** for an **AR(P)** process there is a linear relationship between the ACF values and the unknown parameters (in this case: the power of the innovation process and the $\{a_k\}$ coefficients). The equations above are called Yule-Walker (YW) equations and can be expressed in matrix form by calculating it for $m=0,1,2,\dots,P$, being $P+1$ the number of unknowns.

$$R_X[m] - \sum_{k=1}^P a_k R_X[m-k] = \sigma_w^2 \delta[m], \quad m \geq 0$$

- Since we have $P+1$ unknowns, we need (at least) $P+1$ equations, so we calculate the YW equation for $m=0,1,2,\dots,P$:

$$m=0: \quad R_X[0] - a_1 R_X[-1] - a_2 R_X[-2] \cdots - a_P R_X[-P] = \sigma_w^2$$

$$m=1: \quad R_X[1] - a_1 R_X[0] - a_2 R_X[-1] \cdots - a_P R_X[-P+1] = 0$$

$$m=2: \quad R_X[2] - a_1 R_X[1] - a_2 R_X[0] \cdots - a_P R_X[-P+2] = 0$$

$$\vdots$$

$$m=P: \quad R_X[P] - a_1 R_X[P-1] - a_2 R_X[P-2] \cdots - a_P R_X[0] = 0$$

- This system of $P+1$ linear equations can be expressed in matrix form.

$$\begin{bmatrix} R_X[0] & R_X[-1] & R_X[-2] & \cdots & R_X[-P] \\ R_X[1] & R_X[0] & \ddots & \ddots & R_X[-(P-1)] \\ R_X[2] & R_X[1] & \ddots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & R_X[-1] \\ R_X[P] & R_X[P-1] & \cdots & R_X[1] & R_X[0] \end{bmatrix} \begin{bmatrix} 1 \\ -a_1 \\ -a_2 \\ \vdots \\ -a_P \end{bmatrix} = \begin{bmatrix} \sigma_w^2 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

- The **Yule-Walker** equations can be expressed in a more compact form as follows:

$$\mathbf{C}_X \triangleq \begin{bmatrix} R_X[0] & R_X[-1] & \cdots & R_X[-(P-1)] \\ R_X[1] & R_X[0] & \cdots & R_X[-(P-2)] \\ \vdots & \vdots & \ddots & \vdots \\ R_X[P-1] & R_X[P-2] & \cdots & R_X[0] \end{bmatrix}, \quad \mathbf{c}_X \triangleq \begin{bmatrix} R_X[1] \\ R_X[2] \\ \vdots \\ R_X[P] \end{bmatrix}, \quad \mathbf{a} \triangleq \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_P \end{bmatrix}$$

$$\begin{bmatrix} R_X[0] & \mathbf{c}_X^T \\ \mathbf{c}_X & \mathbf{C}_X \end{bmatrix} \begin{bmatrix} 1 \\ -\mathbf{a} \end{bmatrix} = \begin{bmatrix} \sigma_w^2 \\ \mathbf{0} \end{bmatrix} \Rightarrow \begin{cases} R_X[0] - \mathbf{c}_X^T \mathbf{a} = \sigma_w^2 \\ \mathbf{c}_X - \mathbf{C}_X \mathbf{a} = \mathbf{0} \end{cases}$$

- The second matrix equation can be easily solved w.r.t. $\mathbf{a}$ if the covariance matrix $\mathbf{C}_X$ is full rank (the covariance matrix of a regular process that satisfies the Paley-Wiener condition is always full-rank), and then the power of $W[n]$ is immediately derived:

$$\begin{cases} \mathbf{a} = \mathbf{C}_X^{-1} \mathbf{c}_X \\ \sigma_w^2 = R_X[0] - \mathbf{c}_X^T \mathbf{C}_X^{-1} \mathbf{c}_X \end{cases}$$

- Covariance matrix $\mathbf{C}_X$ is symmetric and it has the Toeplitz structure: these properties are exploited by the **Levinson-Durbin** and the **Schur** algorithms to solve the linear system of P equations with complexity of the order P^2 instead of P^3 .

- To identify an **AR(P) model**, i.e. derive the $\{a_k\}$ coefficients and the innovation process power, we have to solve a **linear** system of equations of order $P+1$.
- To identify a **MA(Q) model**, i.e. derive the $\{b_k\}$ coefficients and the power of the innovation process, we have to solve a **non-linear** system of equations of order $Q+1$; this is true also for an ARMA model. For this reason, AR models are much more frequently used than MA or ARMA models.

■ Identification of an AR(P) model:

- **Step 1:** estimate the ACF values for $m=0,1,2,\dots,P$, via the **sample estimates**:

$$\{X[n]\}_{n=0}^{N-1} \Rightarrow \hat{R}_X[m] = \frac{1}{N-m} \sum_{n=0}^{N-m-1} X[n]X[n+m], \quad m = 0, 1, 2, \dots, P$$

- where typically $P \ll N$, so we can use the unbiased sample estimator.

$$\hat{R}_X[m] = \frac{1}{N-m} \sum_{n=0}^{N-m-1} X[n]X[n+m], \quad m = 0, 1, 2, \dots, P$$



$$\begin{aligned} E\{\hat{R}_X[m]\} &= \frac{1}{N-m} \sum_{n=0}^{N-m-1} E\{X[n]X[n+m]\} \\ &= \frac{1}{N-m} \sum_{n=0}^{N-m-1} R_X[m] = R_X[m], \quad m = 0, 1, 2, \dots, P \end{aligned}$$

■ so the estimates are unbiased.

$$MSE\{\hat{R}_X[m]\} \cong \frac{\text{var}\{X[n]X[n+m]\}}{N-m}, \quad m = 0, 1, 2, \dots, P$$

- The **Sample Autocorrelation** estimator is **consistent**, i.e. the estimates of the ACF values for $m=0,1,2,\dots,P$, converge in probability and mean square sense to the true values when $N \rightarrow \infty$.

$$\hat{R}_X[m] = \frac{1}{N-m} \sum_{n=0}^{N-m-1} X[n]X[n+m], \quad m=0,1,2,\dots,P$$



$$\hat{\mathbf{C}}_X \triangleq \begin{bmatrix} \hat{R}_X[0] & \hat{R}_X[-1] & \cdots & \hat{R}_X[-(P-1)] \\ \hat{R}_X[1] & \hat{R}_X[0] & \cdots & \hat{R}_X[-(P-2)] \\ \vdots & \vdots & \ddots & \vdots \\ \hat{R}_X[P-1] & \hat{R}_X[P-2] & \cdots & \hat{R}_X[0] \end{bmatrix}, \quad \hat{\mathbf{c}}_X \triangleq \begin{bmatrix} \hat{R}_X[1] \\ \hat{R}_X[2] \\ \vdots \\ \hat{R}_X[P] \end{bmatrix}.$$

- **Step 2:** by applying the **Levinson-Durbin algorithm** or the **Schur algorithm**, solve efficiently the **Yule-Walker** linear system of equations and find the AR(P) model that better fits the estimated ACF sequence, i.e. the AR(P) model whose first $P+1$ values of the ACF are equal to the estimated ones:

$$\begin{cases} \hat{\mathbf{a}} = \hat{\mathbf{C}}_X^{-1} \hat{\mathbf{c}}_X \\ \hat{\sigma}_W^2 = \hat{R}_X[0] - \hat{\mathbf{c}}_X^T \hat{\mathbf{C}}_X^{-1} \hat{\mathbf{c}}_X \end{cases}$$

- **Step 3:** Once the AR(P) model parameters have been estimated, a (parametric) estimate of the **Power Spectral Density (PSD)** can be immediately obtained as follows:

$$\hat{S}_X(e^{j2\pi f}) = \hat{\sigma}_W^2 \left| \hat{L}(e^{j2\pi f}) \right|^2 = \frac{\hat{\sigma}_W^2}{\left| 1 - \sum_{k=1}^P \hat{a}_k e^{-j2\pi kf} \right|^2}$$

- The estimates of the ACF for $m > P$ can be derived from the recursive equation that we obtained for AR(P) processes (see at p. 76):

$$R_X[m] = \sum_{k=1}^P a_k R_X[m-k], \quad m > P$$

- By making use of the estimates of the ACF from $m=0$ to $m=P$, previously obtained by using the sample autocorrelation estimator, we can recursively derive an estimate of the ACF values for $m=P+1, P+2, P+3, \dots$, as follows:

$$\hat{R}_X[m] = \sum_{k=1}^P \hat{a}_k \hat{R}_X[m-k], \quad m = P+1, P+2, \dots$$

- The problem of model identification by using an **AR(P)** model consists in the estimation of the unknown parameters **$\mathbf{a}$** and the power of $W[n]$ of the AR(P) model whose first $P+1$ values of the ACF are equal to the $P+1$ ACF values obtained by the sample estimator, by processing N samples of $X[n]$:

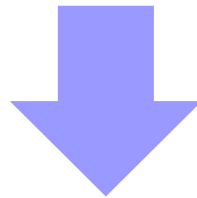
$$\{X[n]\}_{n=0}^{N-1} \Rightarrow \{\hat{R}_X[m]\}_{m=0}^P \Rightarrow \{\hat{\sigma}_W^2, \hat{\mathbf{a}}\} \Rightarrow \hat{S}_X(e^{j2\pi f})$$

- Model identification by using an AR(P) model requires the solution of a system of $P+1$ linear equations; once we have an estimate of **$\mathbf{a}$** , we derive an estimate of the power of the innovation process $W[n]$.
- Once the parameters of the AR(P) model have been estimated, a (parametric) estimate of the PSD is obtained as:

$$\hat{S}_X(e^{j2\pi f}) = \frac{\hat{\sigma}_W^2}{\left|1 - \sum_{k=1}^P \hat{a}_k e^{-j2\pi kf}\right|^2}$$

- **Example:** let us consider the following **broadband ARMA(4,4) process** whose innovation filter has system function:

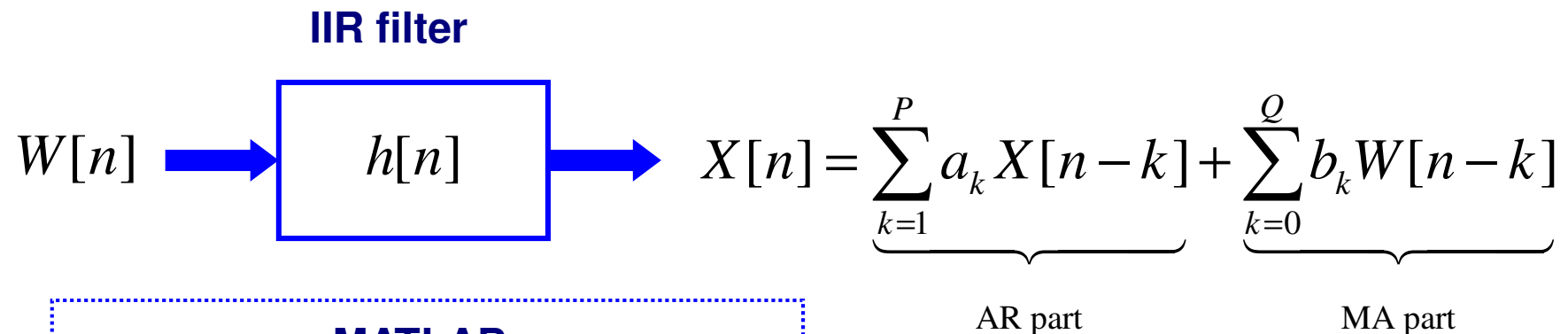
$$H(z) = \frac{B(z)}{A(z)} = \frac{1 - 1.3817z^{-1} + 1.5632z^{-2} - 0.8843z^{-3} + 0.4096z^{-4}}{1 - 0.3544z^{-1} + 0.3508z^{-2} + 0.1736z^{-3} + 0.2401z^{-4}}$$



$$X[n] = \underbrace{0.3544 \cdot X[n-1] - 0.3508 \cdot X[n-2] - 0.1736 \cdot X[n-3] - 0.2401 \cdot X[n-4]}_{\text{AR part}}$$

$$+ \underbrace{W[n] - 1.3817 \cdot W[n-1] + 1.5632 \cdot W[n-2] - 0.8843 \cdot W[n-3] + 0.4096 \cdot W[n-4]}_{\text{MA part}}$$

$$H(z) = \frac{B(z)}{A(z)} = \frac{1 - 1.3817z^{-1} + 1.5632z^{-2} - 0.8843z^{-3} + 0.4096z^{-4}}{1 - 0.3544z^{-1} + 0.3508z^{-2} + 0.1736z^{-3} + 0.2401z^{-4}}$$



MATLAB:

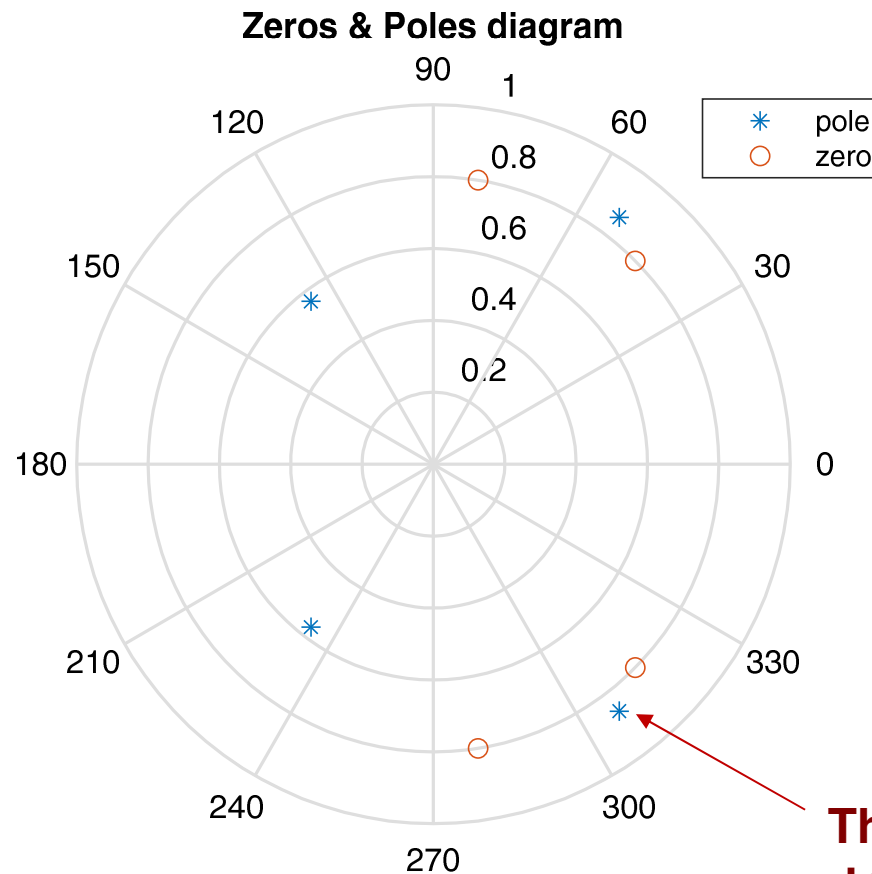
`w=randn(N,1), x=filter(b,[1 -aT]T,w)`

$$\sigma_w^2 = E\{W^2[n]\} = 1$$

$$\mathbf{b} = [1 \quad -1.3817 \quad 1.5632 \quad -0.8843 \quad 0.4096]^T$$

$$[1 - \mathbf{a}^T]^T = [1 \quad -0.3544 \quad 0.3508 \quad 0.1736 \quad 0.2401]^T$$

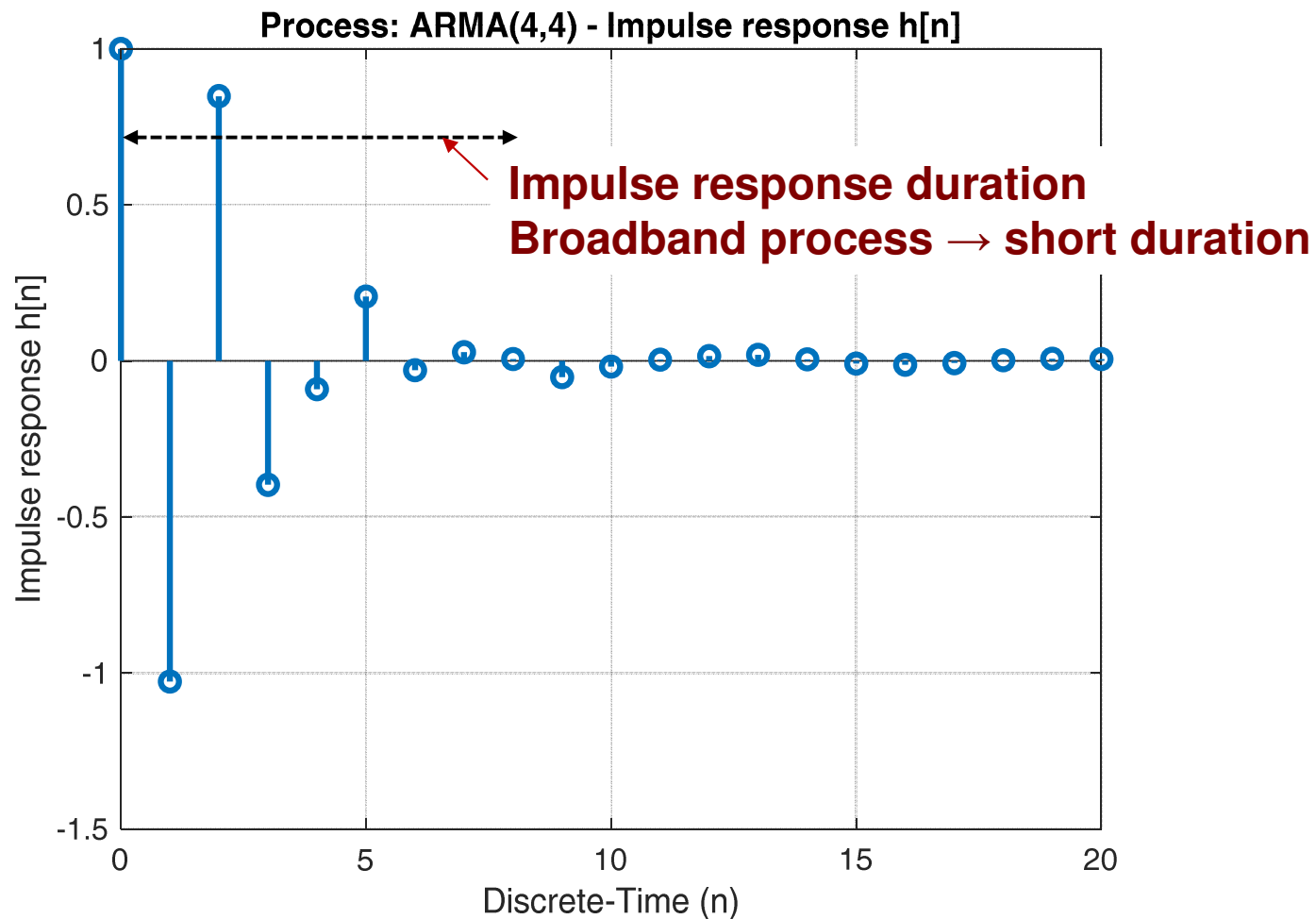
Poles-zeros diagram of the Innovation filter



$$z = \text{roots}(b), \quad p = \text{roots}([1 \ -a^T]^T)$$

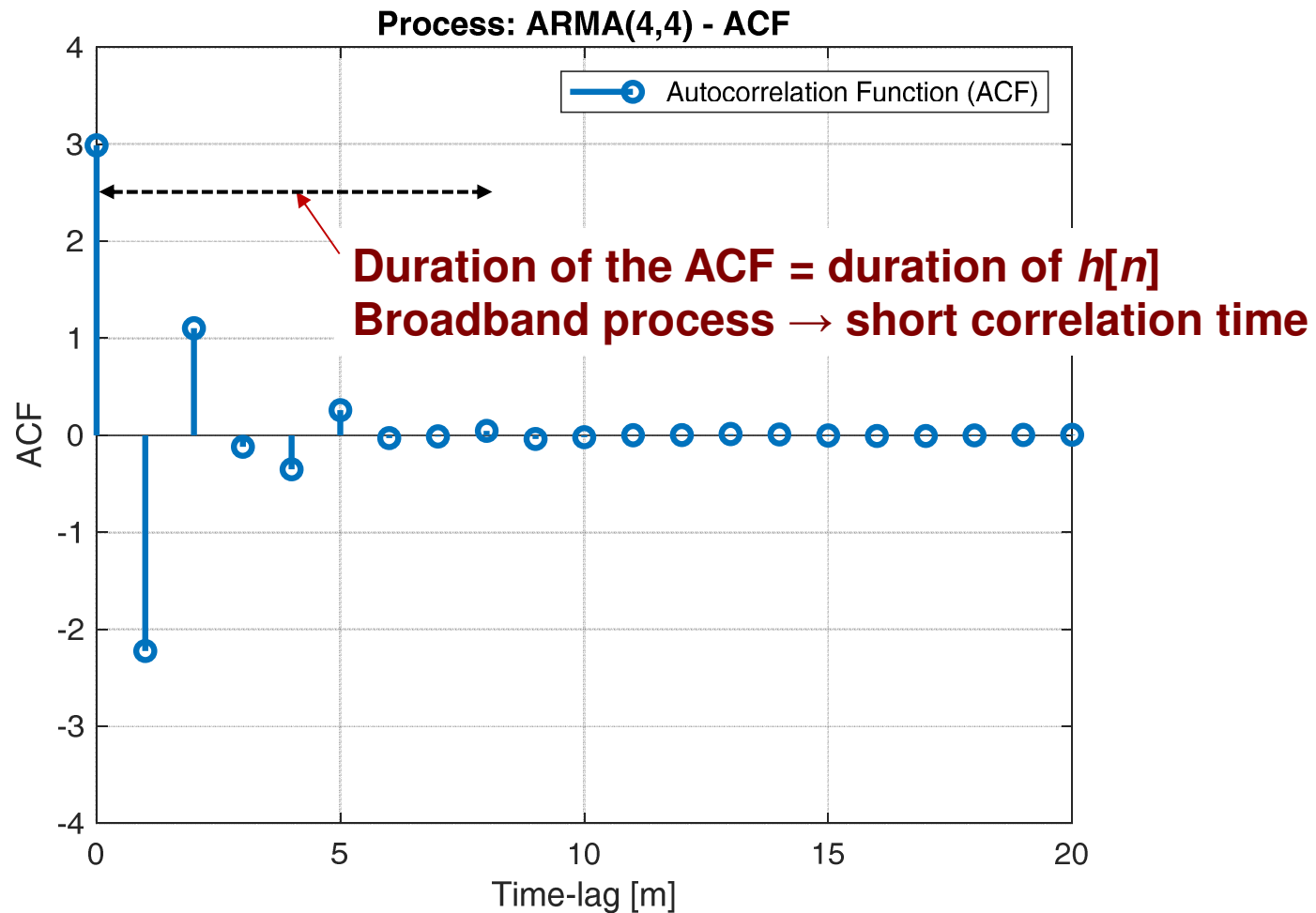
**The dominant poles are not too close to the unit circle
→ the process is broadband**

Impulse response of the innovation filter

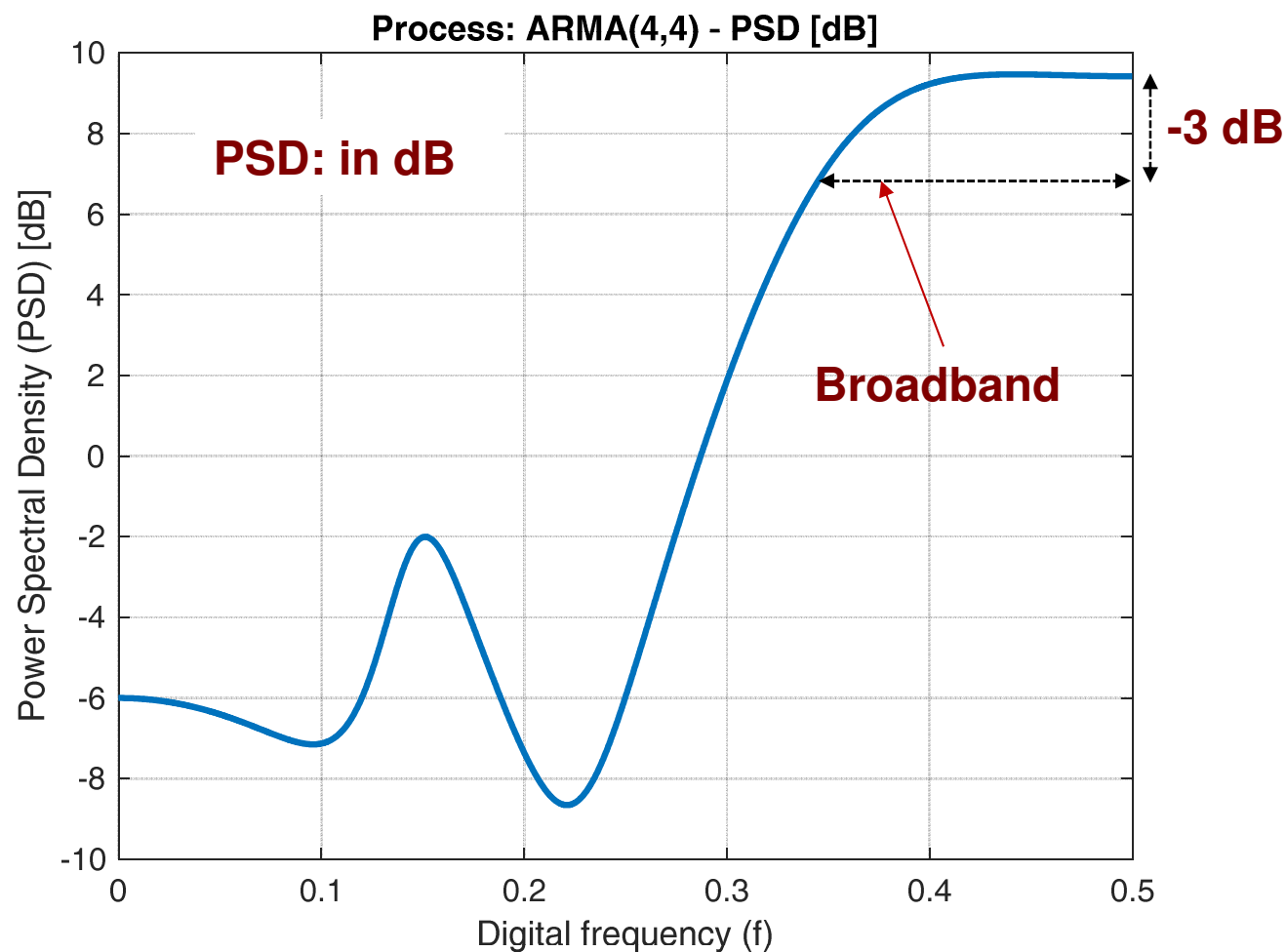


$$h = \text{dimpulse}(b^T, [1 \ -a^T])$$

Autocorrelation function (ACF)

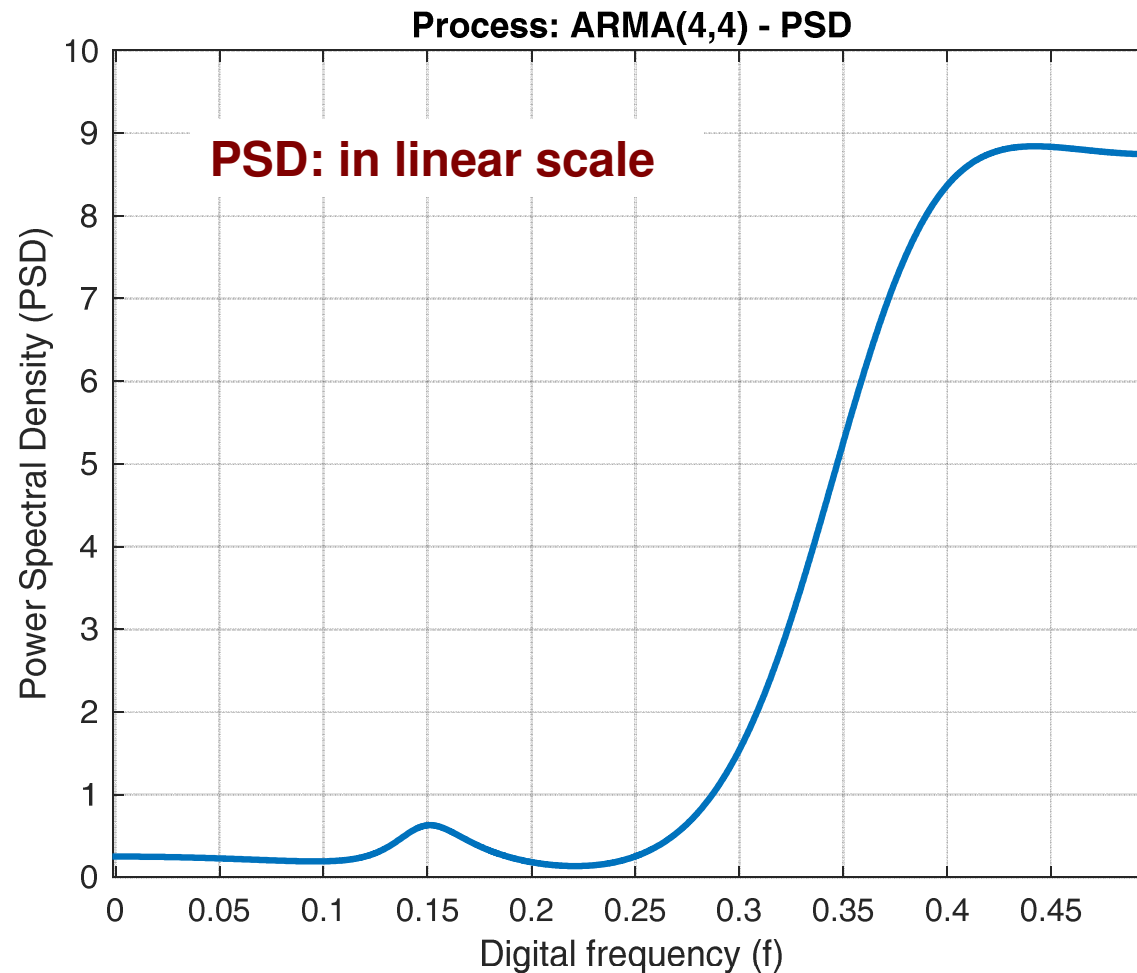


Power Spectral Density (PSD)





Power Spectral Density (PSD)



■ Definition of the PSD of a WSS discrete-time random process:

$$S_X(e^{j2\pi f}) = \lim_{N \rightarrow \infty} \frac{1}{N} E \left\{ \left| X_N(e^{j2\pi f}) \right|^2 \right\} = E \left\{ \lim_{N \rightarrow \infty} \frac{1}{N} \left| \sum_{n=0}^{N-1} x[n] e^{-j2\pi f n} \right|^2 \right\}$$

$$\text{where } X_N(e^{j2\pi f}) \triangleq DTFT\{X[n]\} = \sum_{n=0}^{N-1} x[n] e^{-j2\pi f n}$$

The method of the **Periodogram** (Schuster's Periodogram) derives directly from this definition of the PSD and it is a **non-parametric non-adaptive method**:

$$\hat{S}_X^{(p)}\left(e^{j2\pi \frac{k}{N}}\right) = \frac{1}{N} \left| \sum_{n=0}^{N-1} x[n] e^{-j2\pi \frac{k}{N} n} \right|^2, \quad 0 \leq k \leq N-1$$

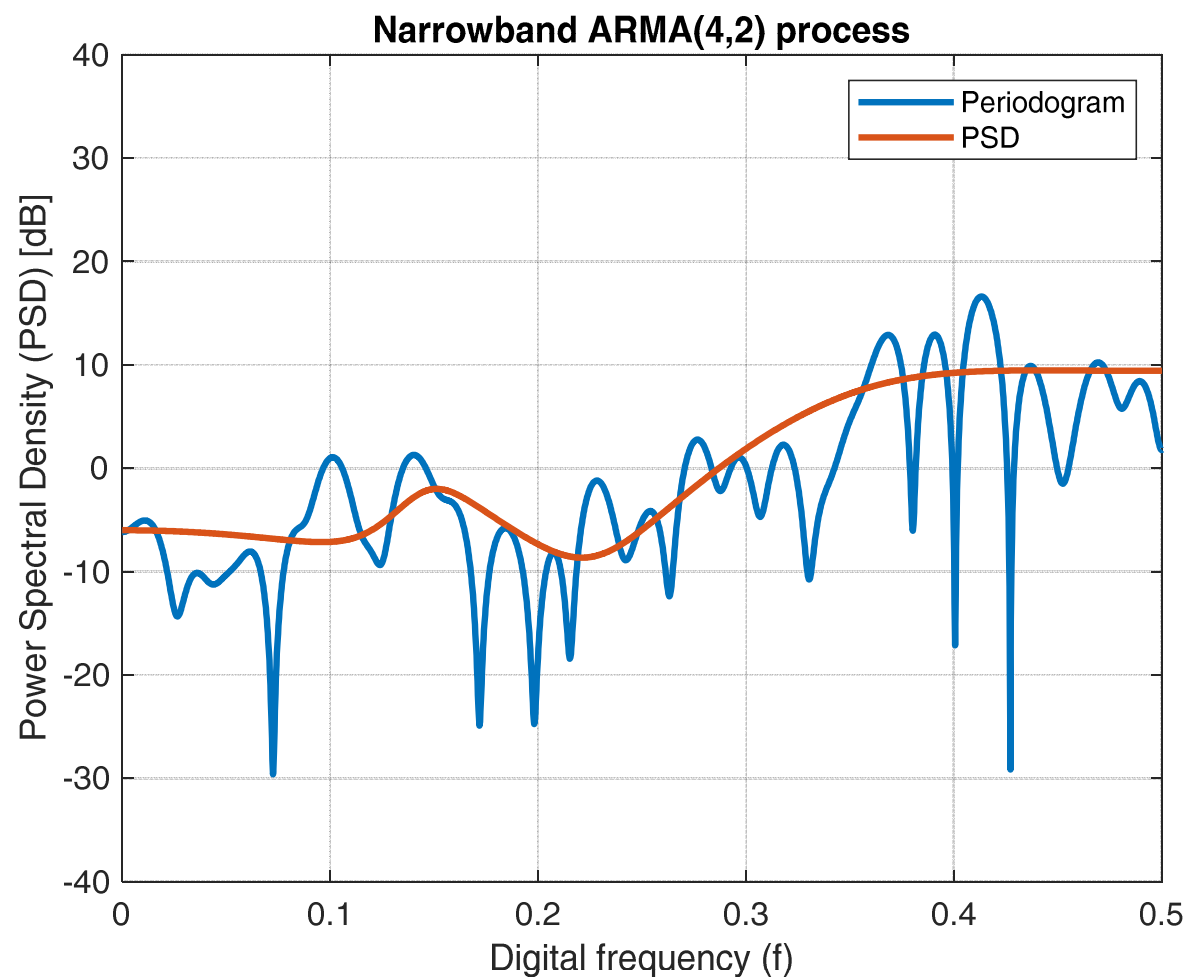
- **Periodogram**: when N goes to infinity, for regular processes, we have:

$$\lim_{N \rightarrow \infty} E \left\{ \hat{S}_X^{(p)} \left(e^{j2\pi \frac{k}{N}} \right) \right\} = S_X \left(e^{j2\pi \frac{k}{N}} \right)$$

$$\lim_{N \rightarrow \infty} \text{var} \left\{ \hat{S}_X^{(p)} \left(e^{j2\pi \frac{k}{N}} \right) \right\} = S_X^2 \left(e^{j2\pi \frac{k}{N}} \right)$$

In summary:

- Regarding **regular processes** (continuous PSD), the **Periodogram** is an asymptotically unbiased but not consistent estimator, since its variance does not tend to zero as N goes to infinity.
- The variance of the estimate is asymptotically proportional to the square of the value of the PSD, then the MSE is greater where the PSD is greater.
- For **predictable processes** (discrete PSD), the **Periodogram** is indeed an asymptotically unbiased and consistent estimator.

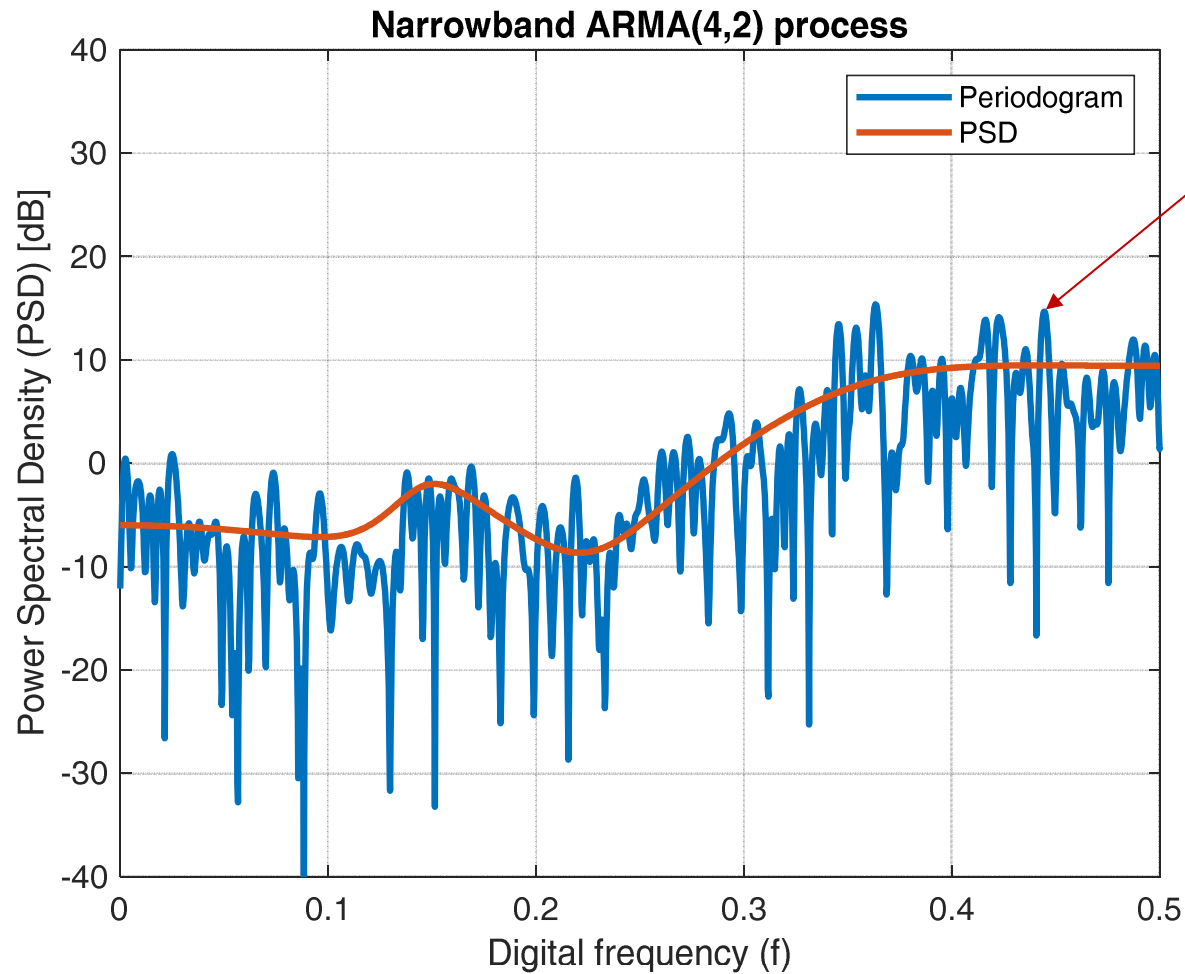


Sample size:
 $N=64$

Zero-padding:
 $N_z=4096$

Non-Parametric PSD estimation: Periodogram

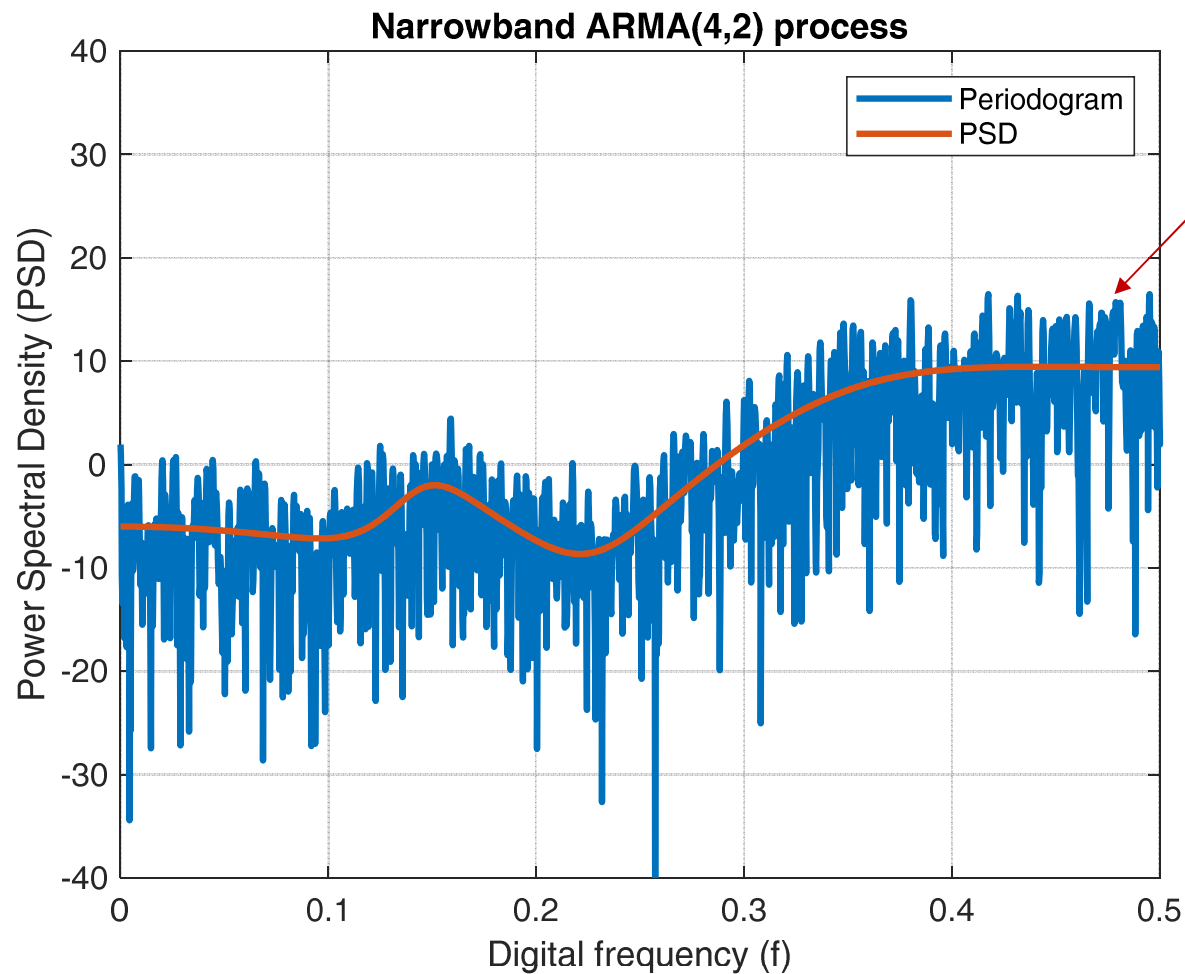
99



**The error variance
(and so the MSE)
does not decrease
when N increases!**

**Sample size:
 $N=256$**

**Zero-padding:
 $N_z=4096$**



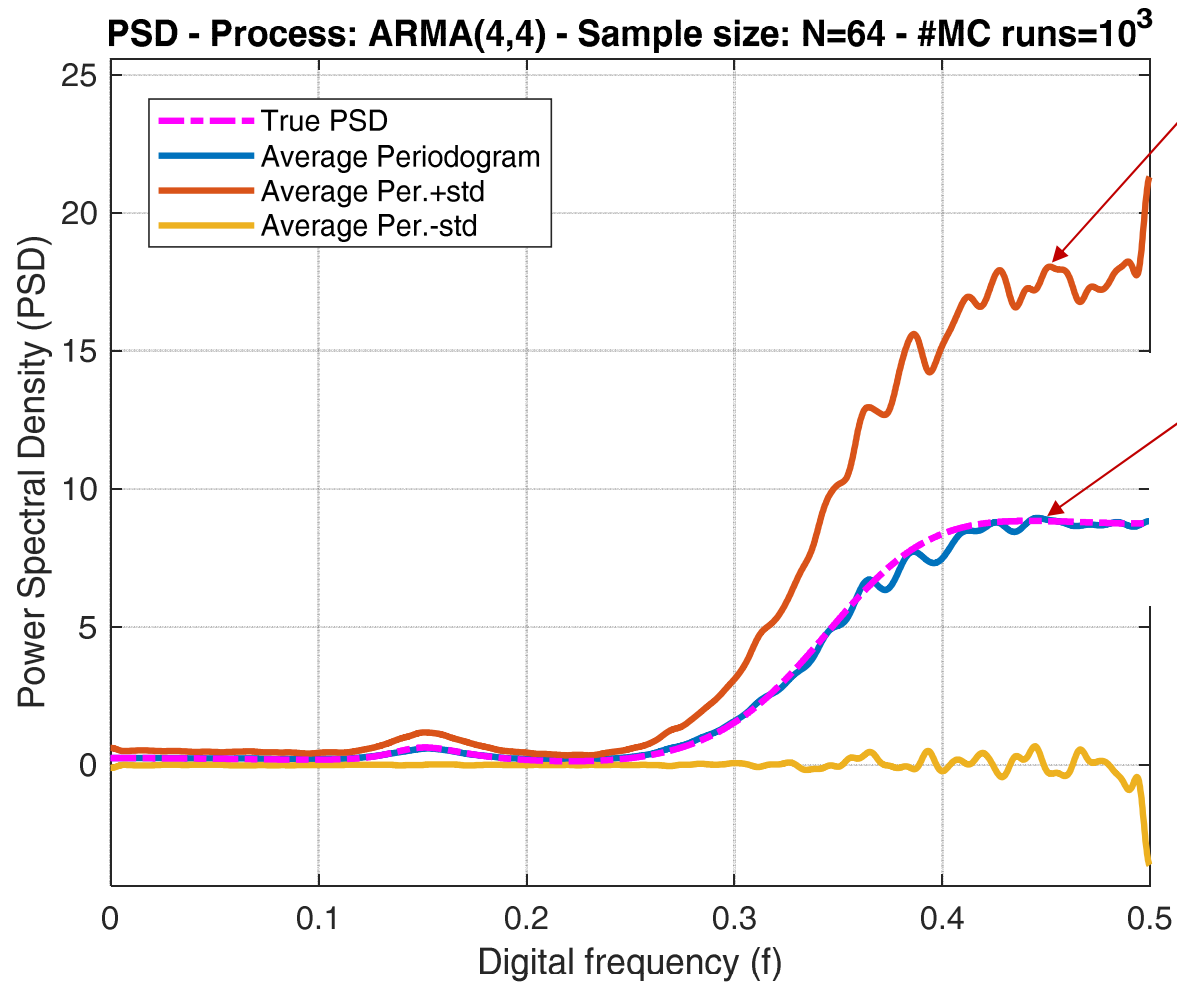
**The error variance
(and so the MSE)
does not decrease
when N increases!**

**Sample size:
 $N=1024$**

**Zero-padding:
 $N_z=4096$**

Non-Parametric PSD estimation: Periodogram

101

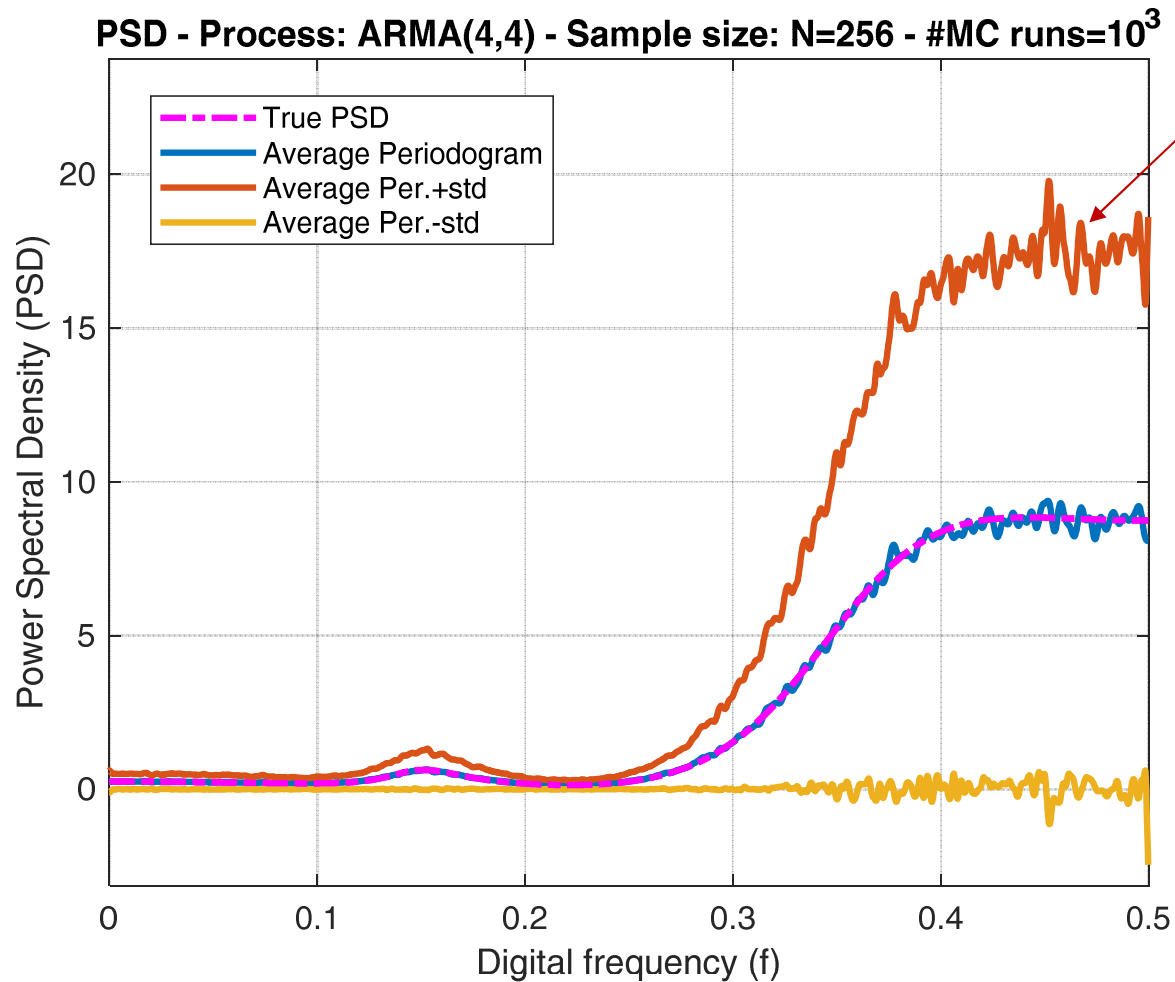


The error standard deviation is proportional to the true PSD

Periodogram averaged over $M=10^3$ realizations (Monte Carlo runs)

Sample size:
 $N=64$

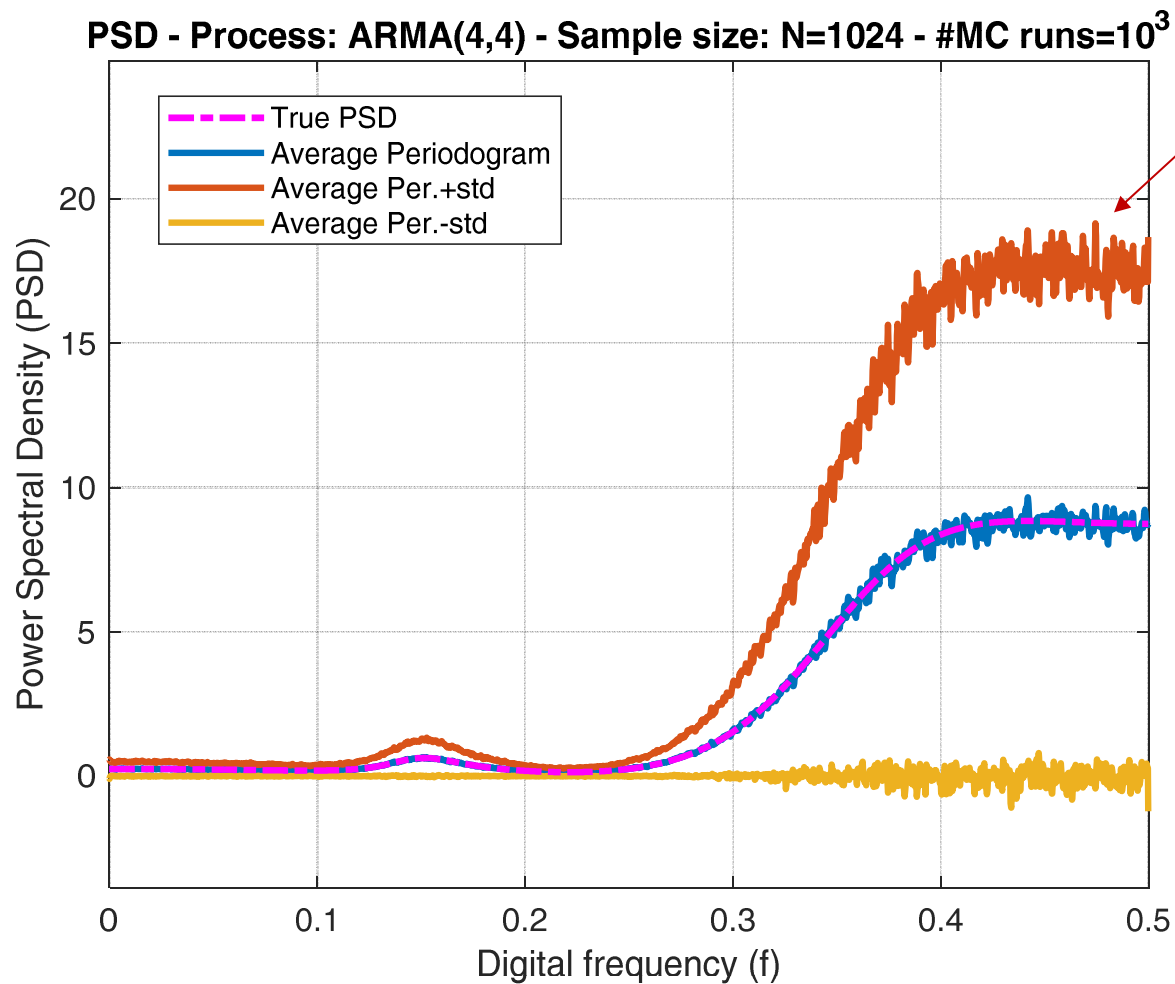
Zero-padding:
 $N_z=4096$



The error standard deviation does not decrease when N increases

Sample size:
 $N=256$

Zero-padding:
 $N_z=4096$



The error standard deviation does not decrease when N increases

Sample size:
 $N=1024$

Zero-padding:
 $N_z=4096$

- **Parametric estimation of the PSD:** we assume that the data can be modelled as an AR(P) process of order P properly selected.
- To identify the AR(P) model we first estimate the ACF values from lag $m=0$ to lag $m=P$ (with $P \ll N$) by using the unbiased sample estimator of the ACF, and then we solve the **Yule-Walker system** of P linear equations by using an efficient algorithm, e.g. the **Levinson-Durbin algorithm**.

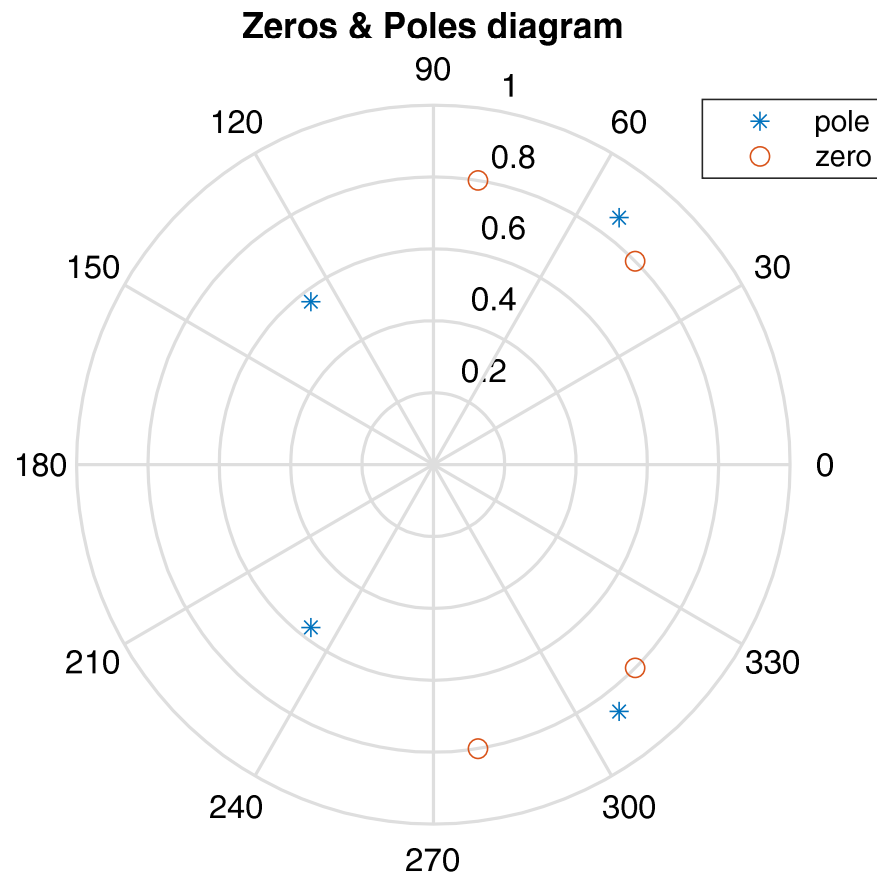
$$\{X[n]\}_{n=0}^{N-1} \Rightarrow \{\hat{R}_X[m]\}_{m=0}^P \Rightarrow \{\hat{\sigma}_W^2, \hat{\mathbf{a}}\}$$



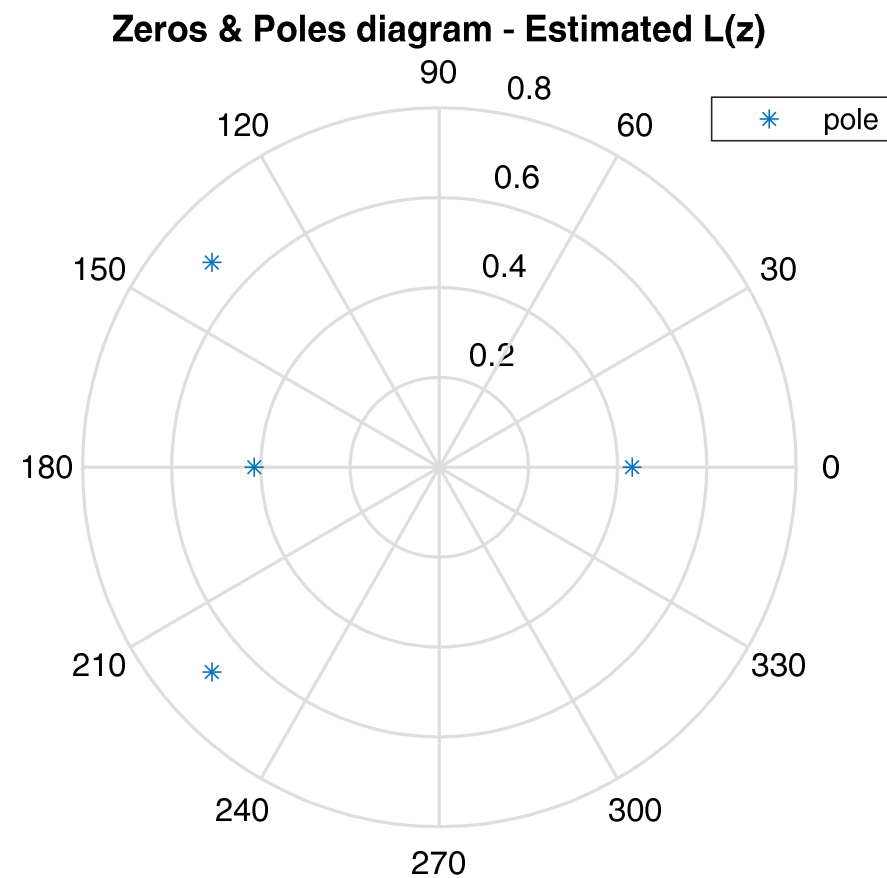
- Once we have estimated $\mathbf{a}$ and the power of $W[n]$, we get the parametric estimate of the PSD, under the assumption that it is an AR(P) process.

$$\hat{S}_X(e^{j2\pi f}) = \frac{\hat{\sigma}_W^2}{\left|1 - \sum_{k=1}^P \hat{a}_k e^{-j2\pi kf}\right|^2}$$

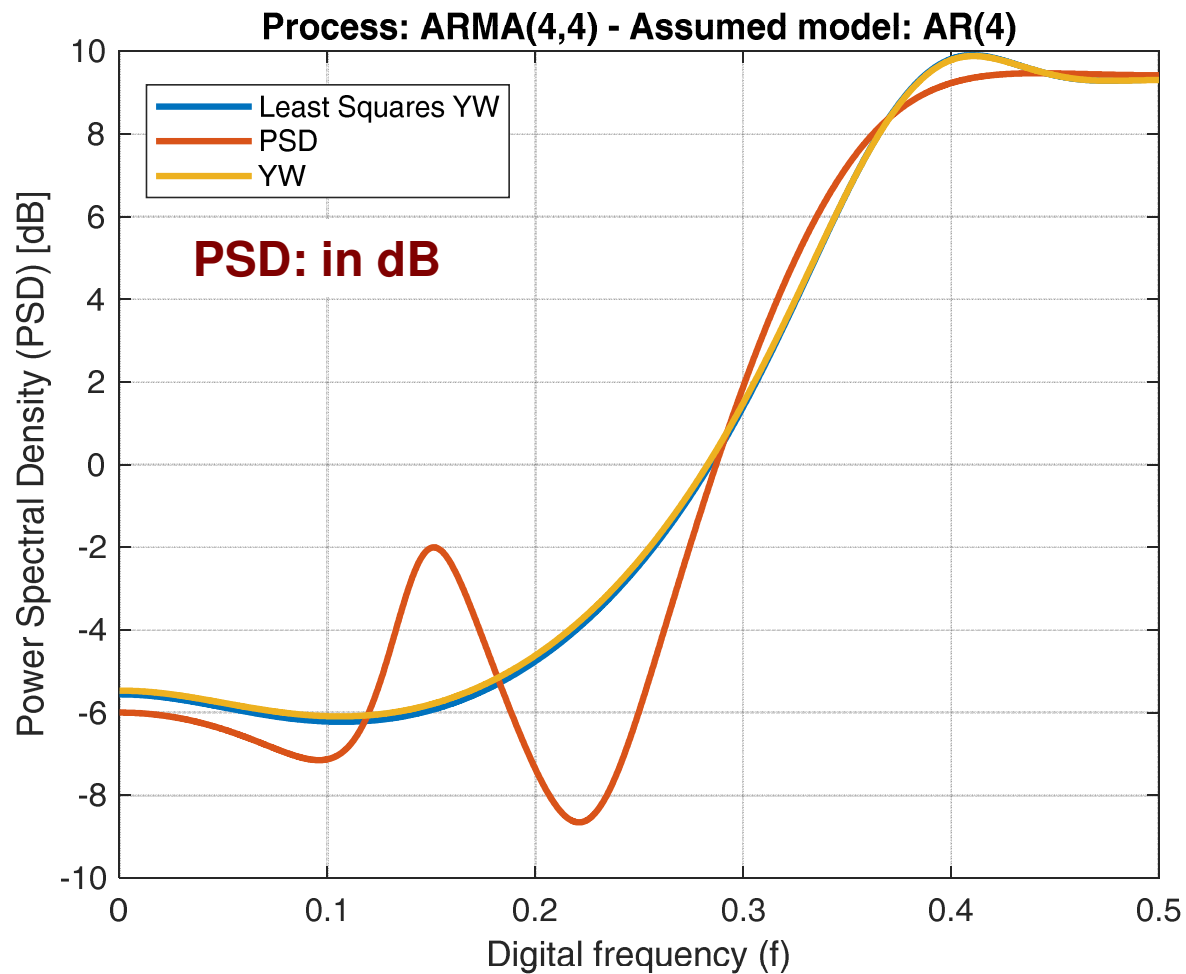
■ True model: ARMA(4,4)



■ Assumed model: AR(4)

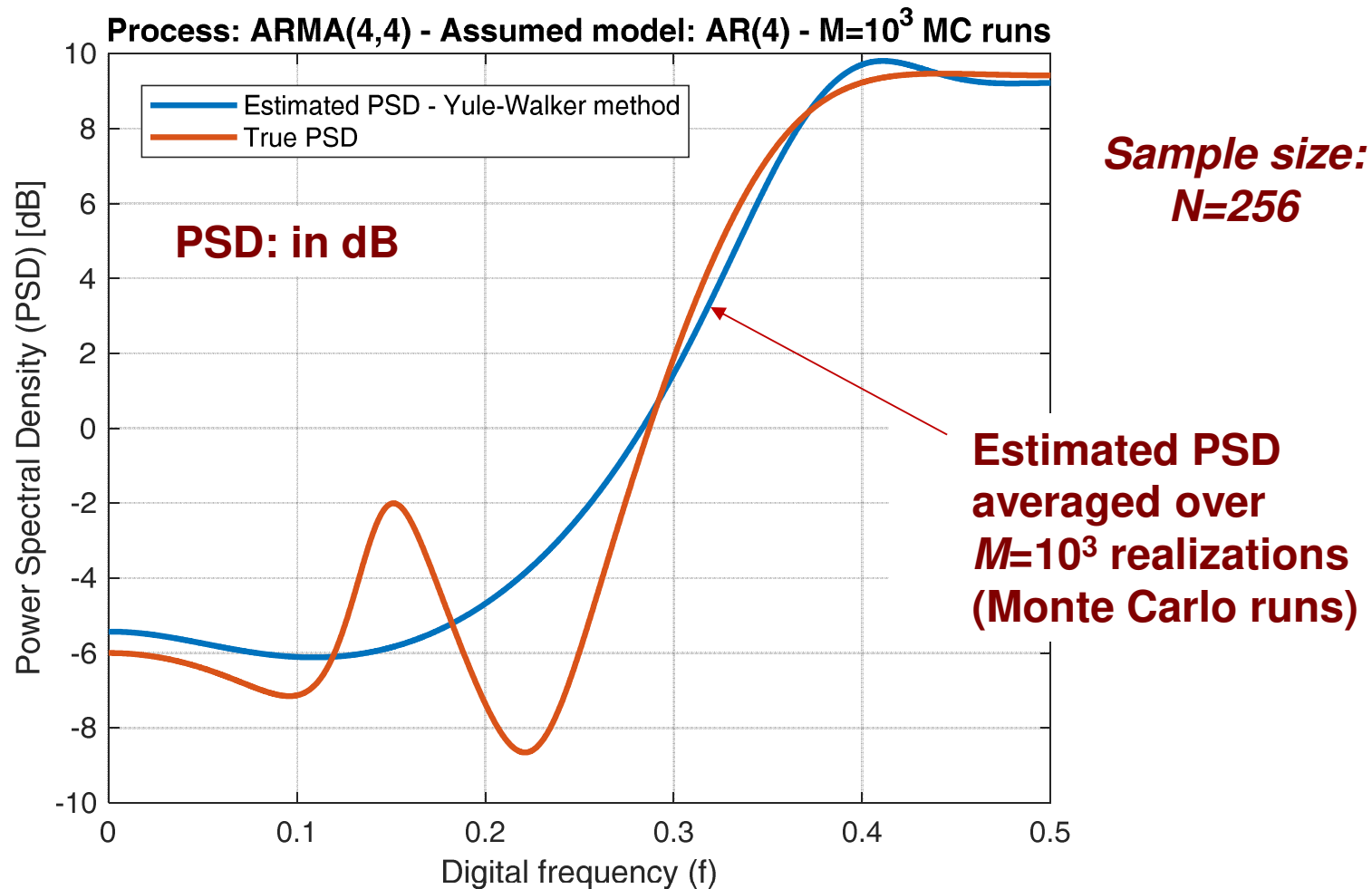


Assumed model: $AR(4)$

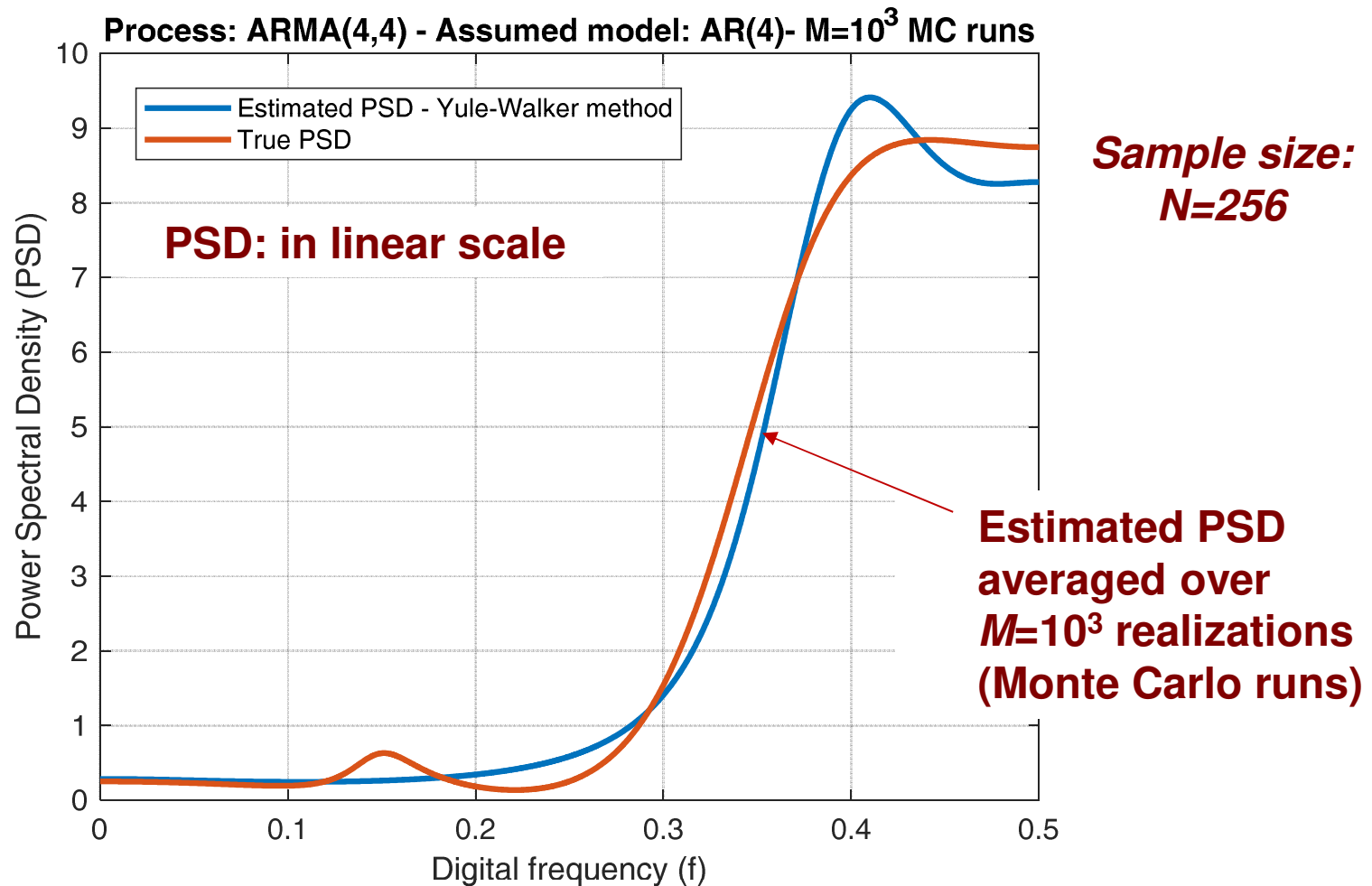


Sample size:
 $N=256$

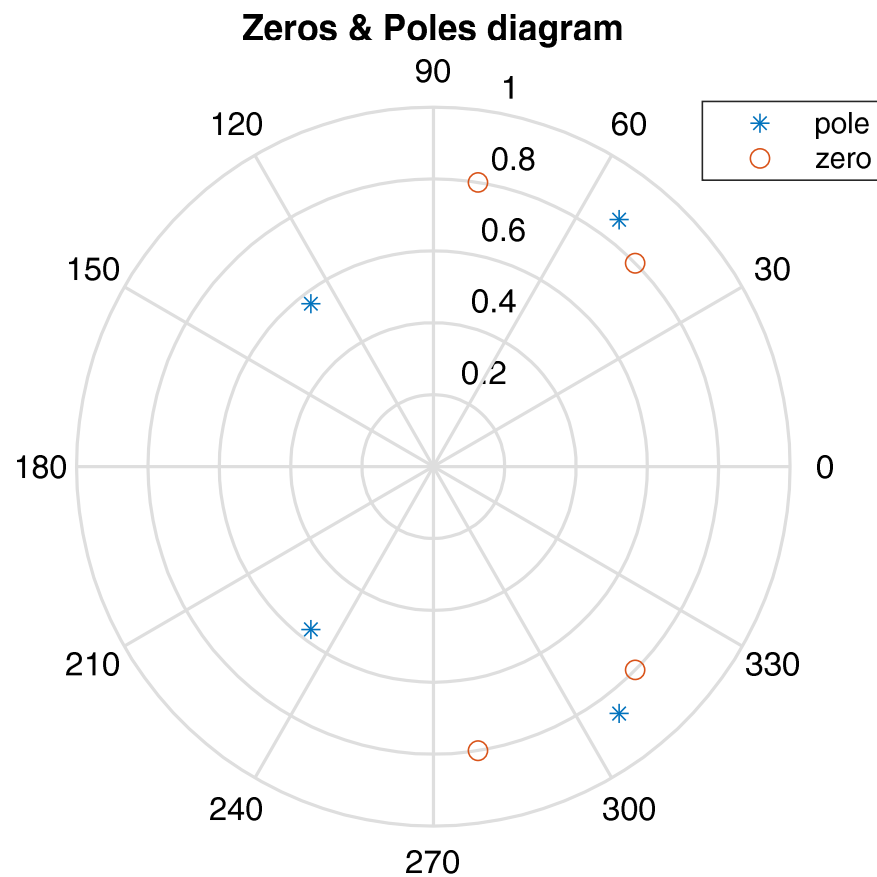
Assumed model: AR(4)



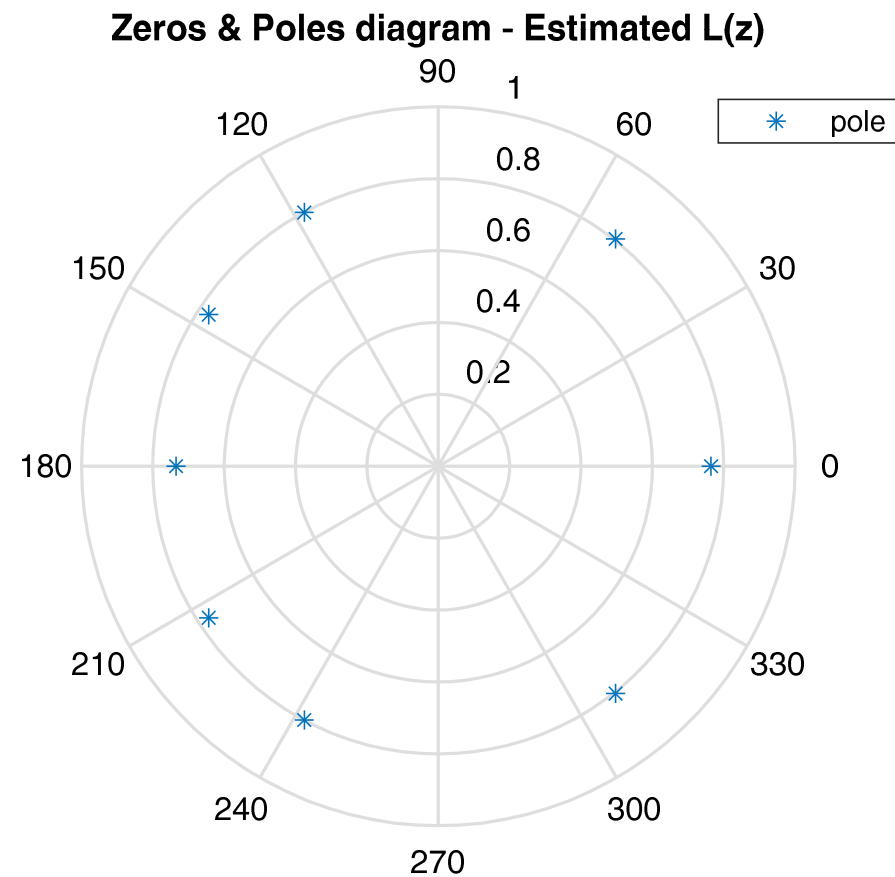
Assumed model: AR(4)



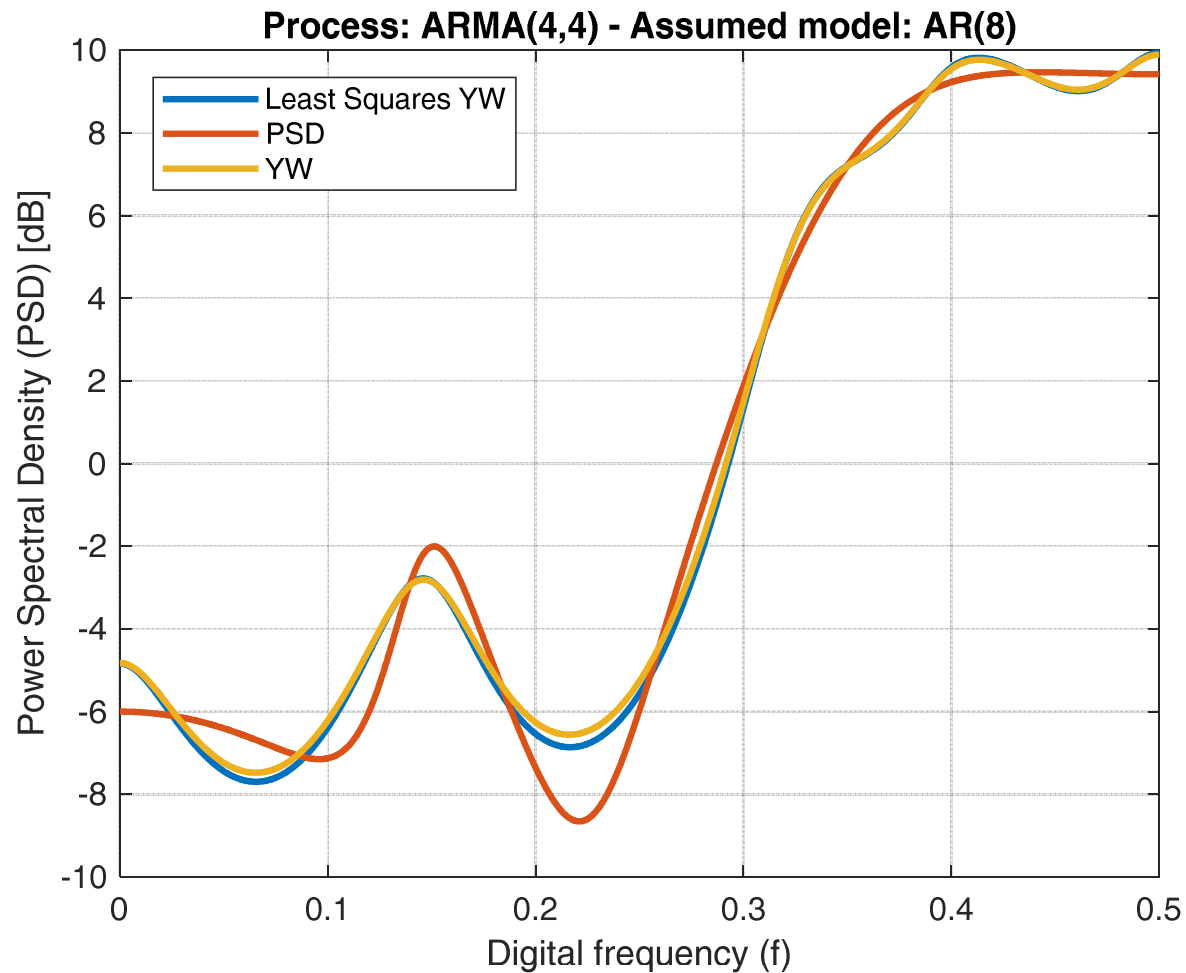
■ True model: ARMA(4,4)



■ Assumed model: AR(8)

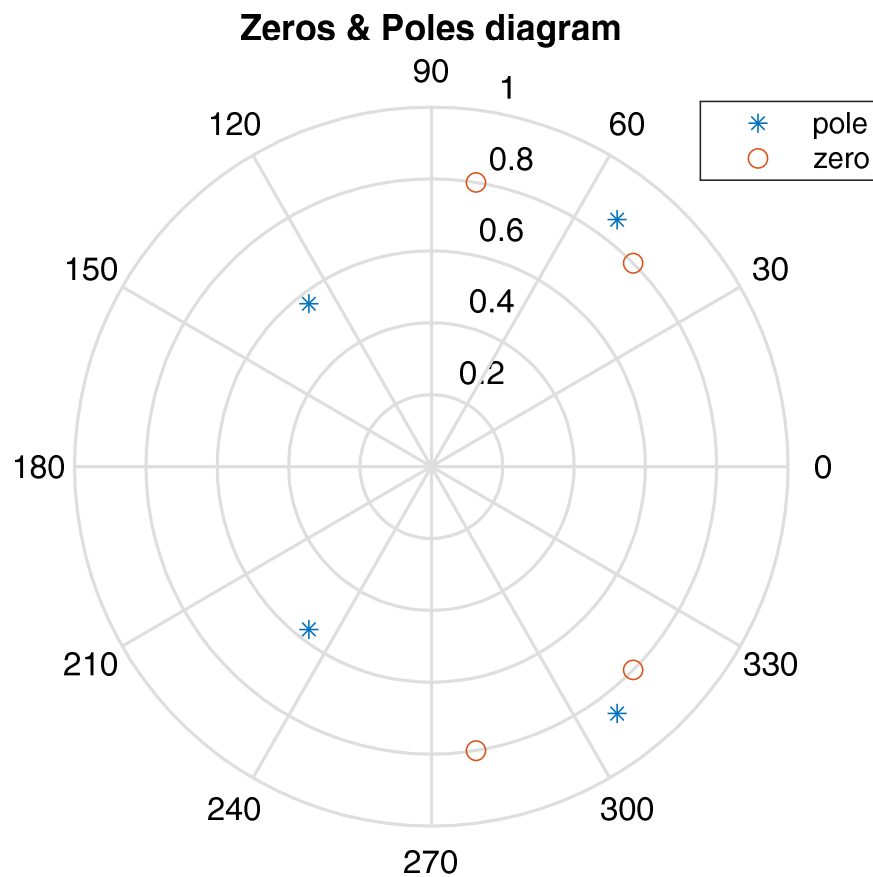


Assumed model: $AR(8)$

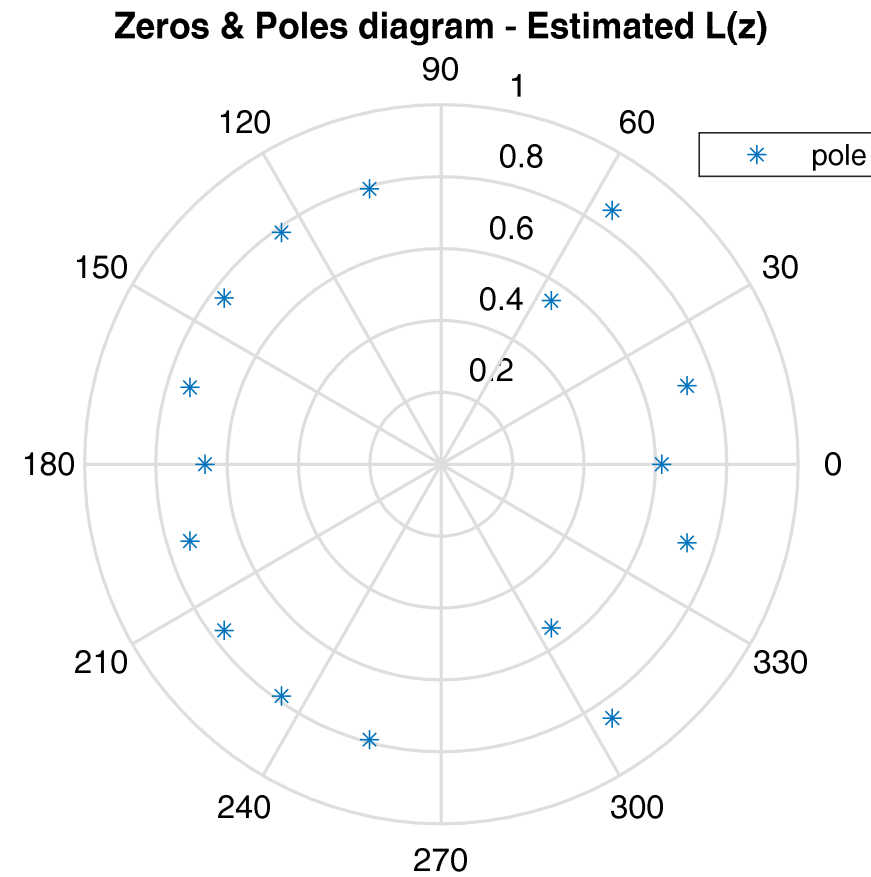


Sample size:
 $N=256$

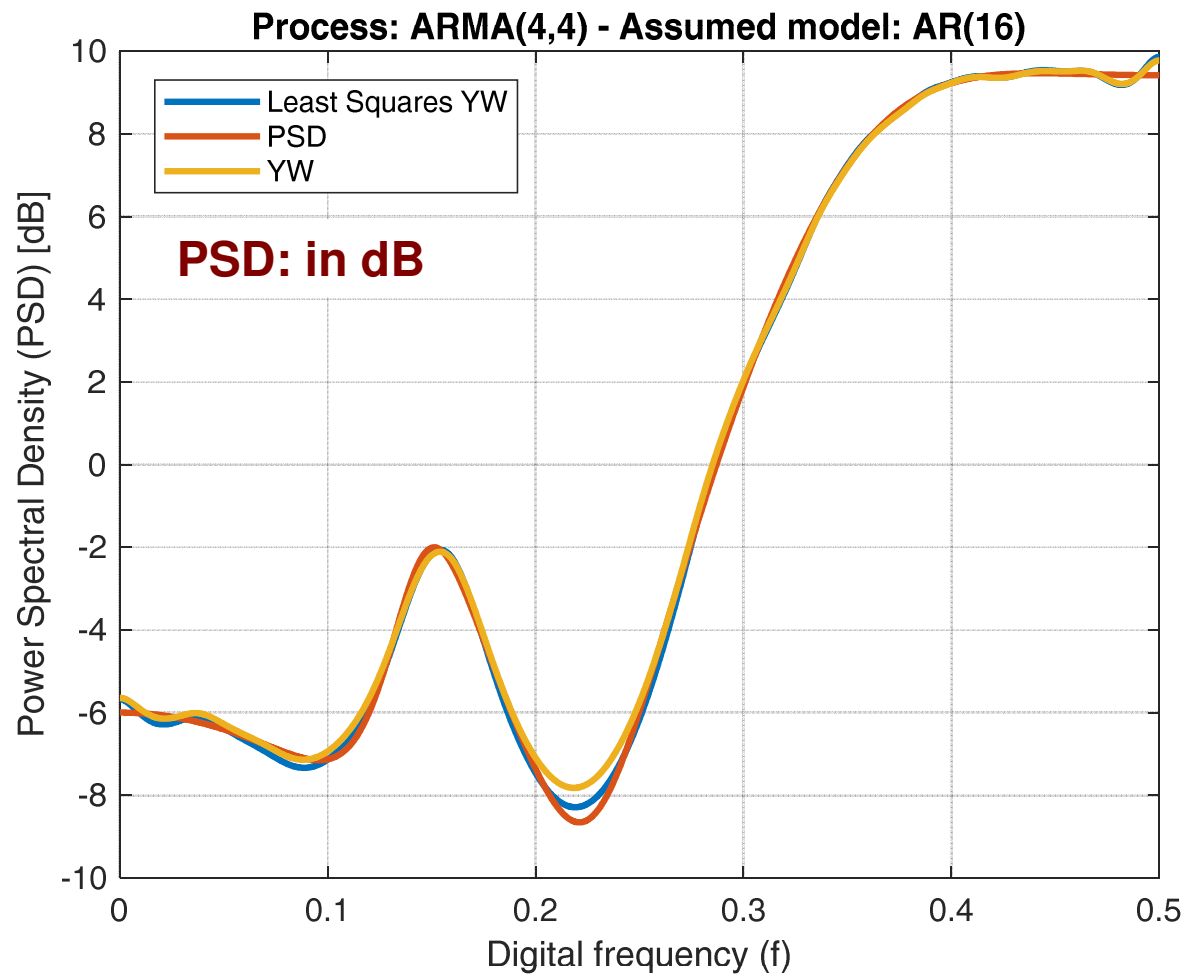
■ True model: ARMA(4,4)



■ Assumed model: AR(16)



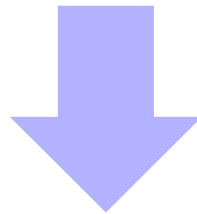
Assumed model: AR(16)



Sample size:
N=256

- **Example:** let us consider the following **narrowband ARMA(4,2) process** whose innovation filter has system function:

$$H(z) = \frac{B(z)}{A(z)} = \frac{1 + 1.5857z^{-1} + 0.9604z^{-2}}{1 - 1.6408z^{-1} + 2.2044z^{-2} - 1.4808z^{-3} + 0.8145z^{-4}}$$

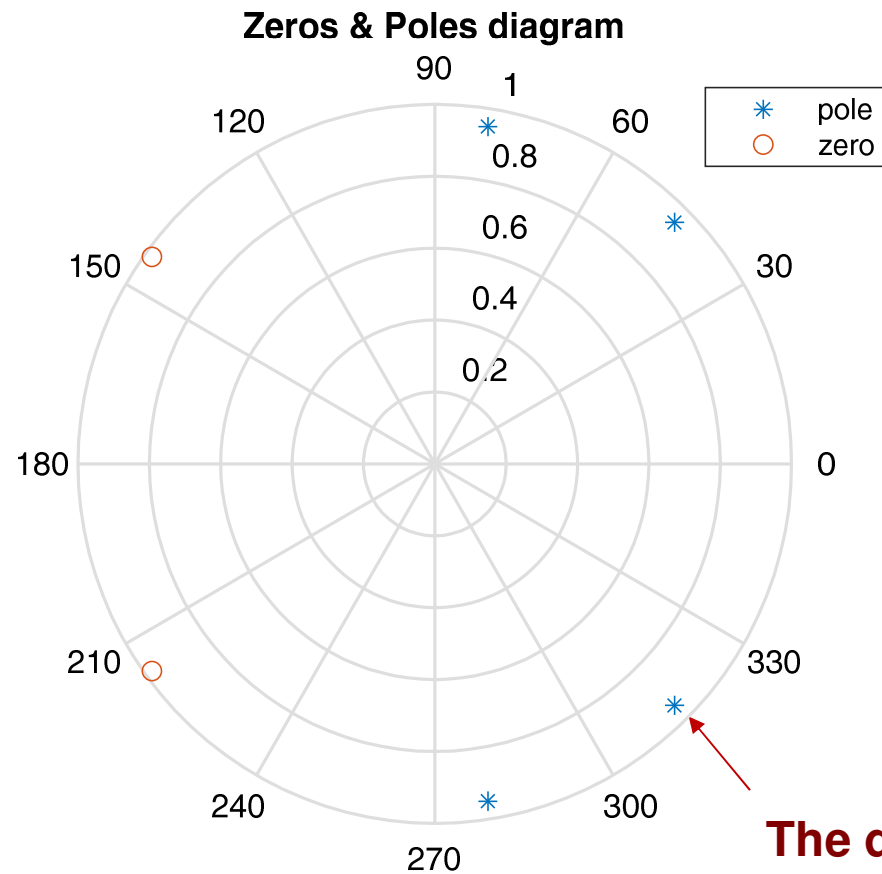


$$\mathbf{b} = [1 \quad 1.5857 \quad 0.9604]^T$$

$$[1 - \mathbf{a}^T]^T = [1 \quad -1.6408 \quad 2.2044 \quad -1.4808 \quad 0.8145]^T$$

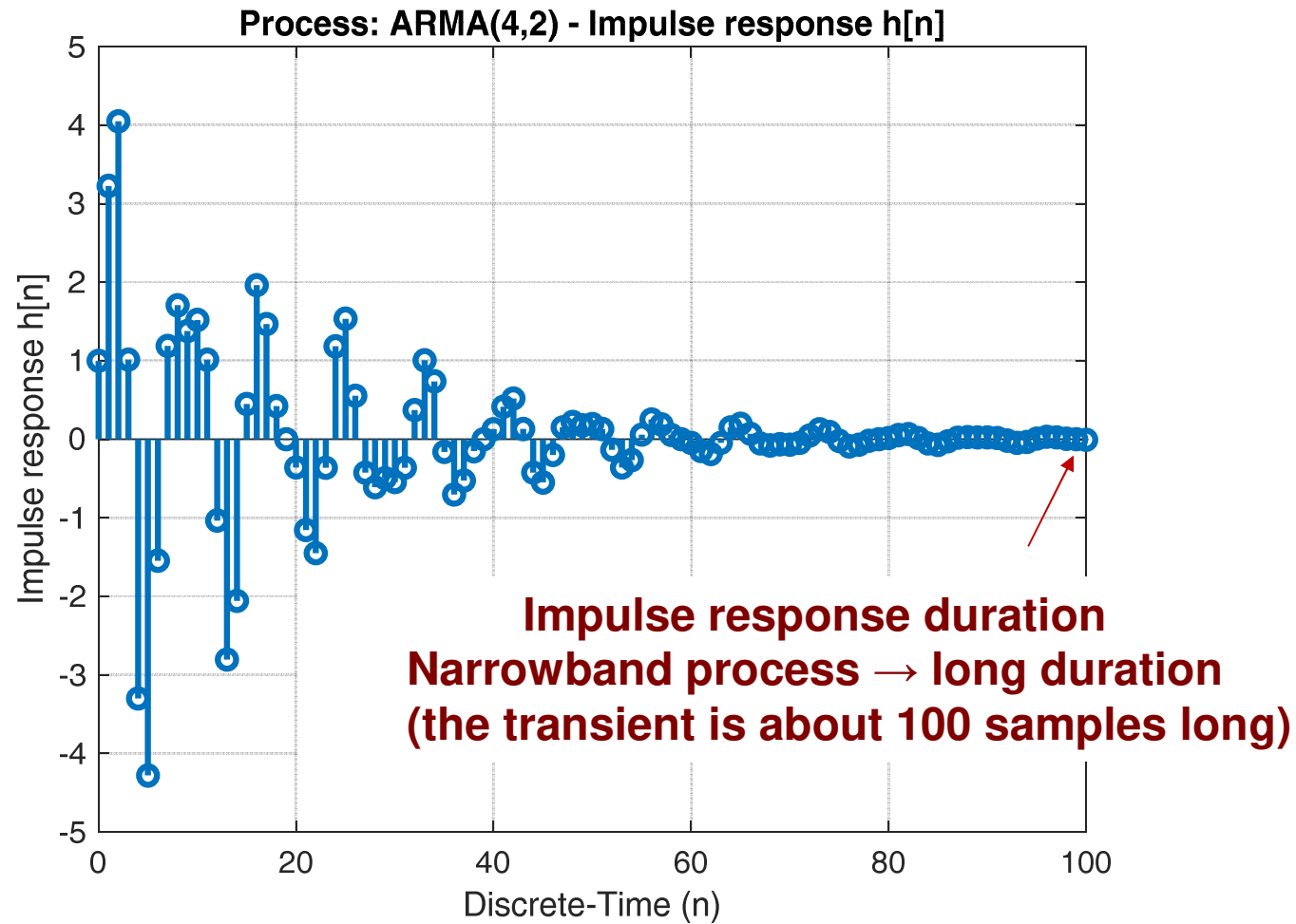
$$\sigma_w^2 = E\{W^2[n]\} = 1$$

Poles-zeros diagram of the innovation filter

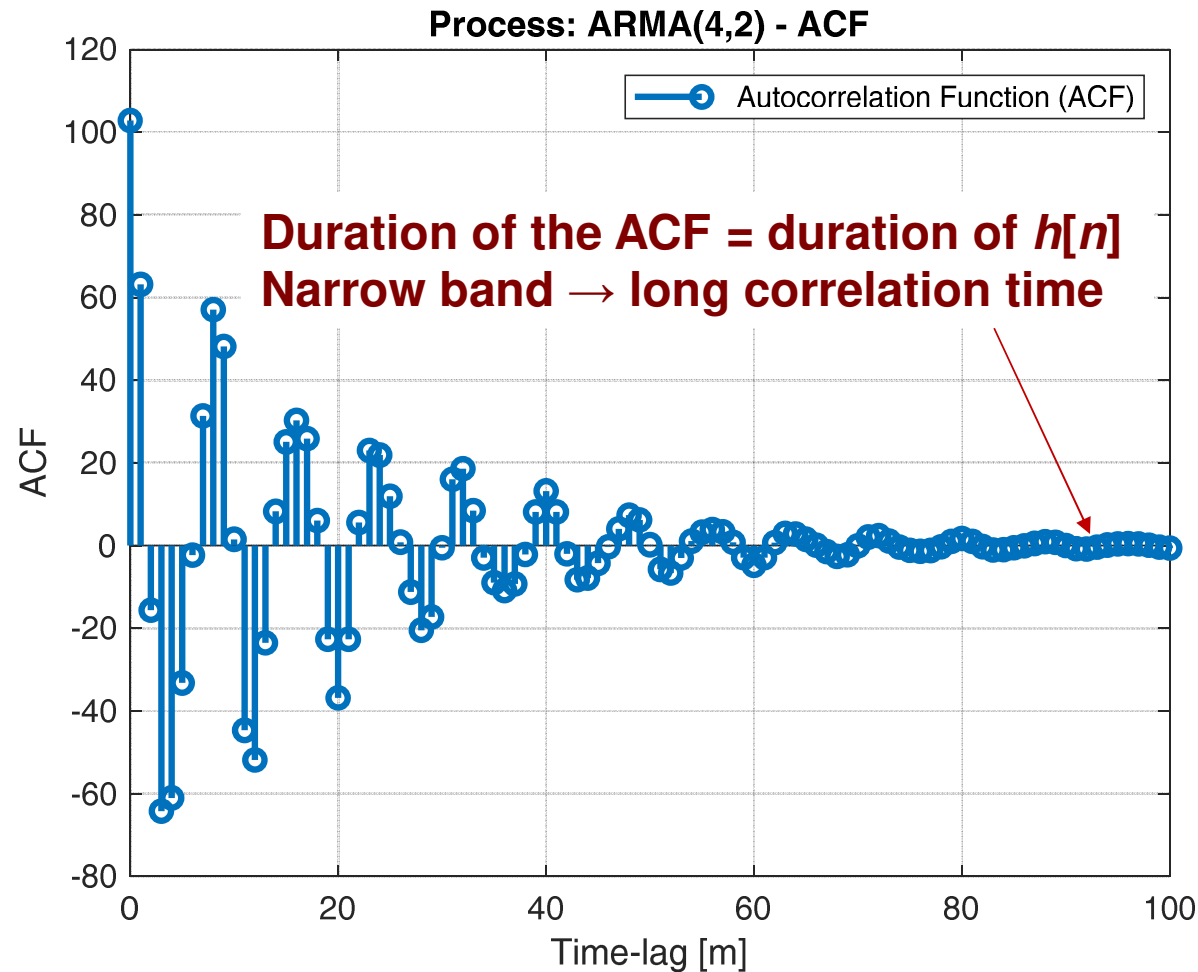


**The dominant poles are close to the unit circle
→ the process is narrowband**

Impulse response of the innovation filter

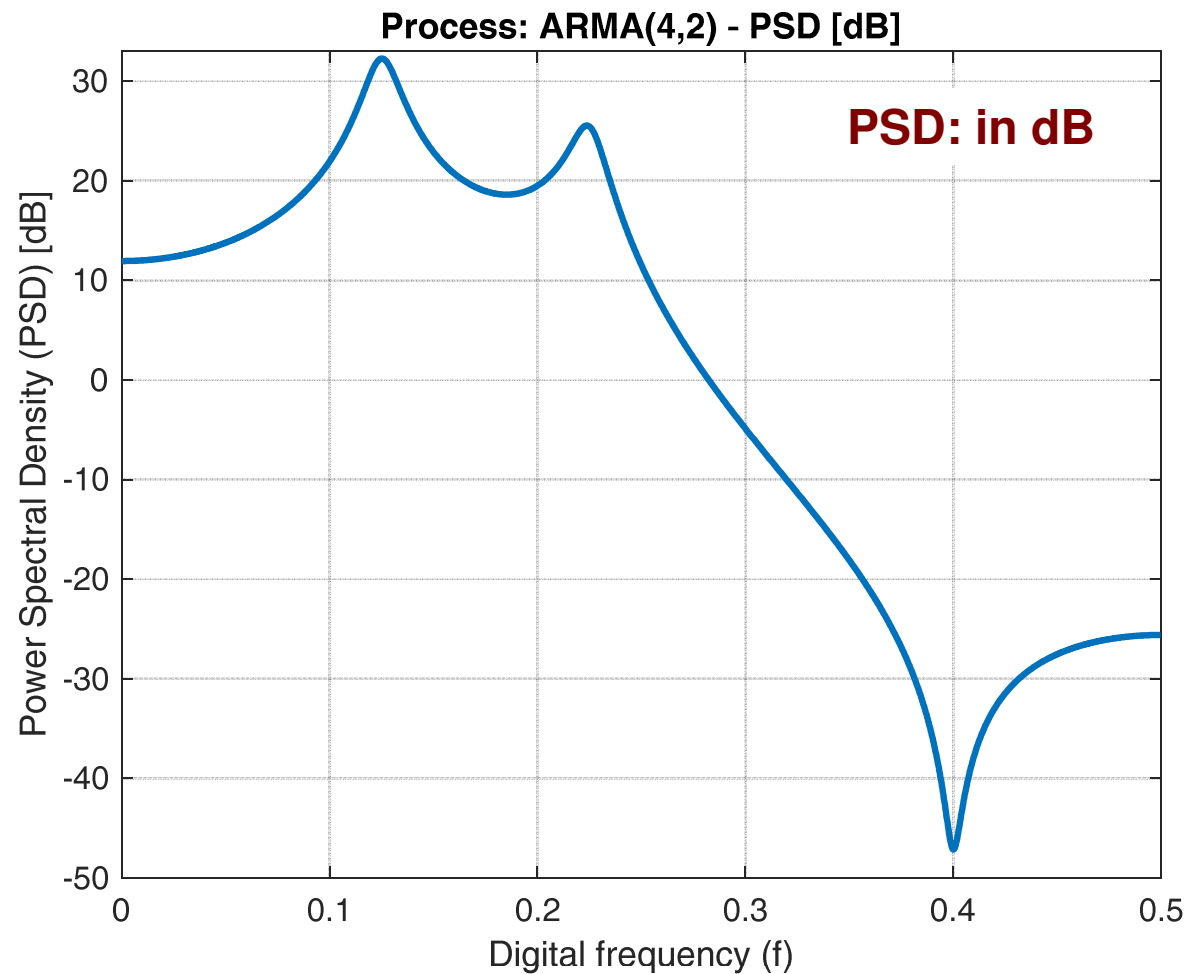


Autocorrelation function (ACF)



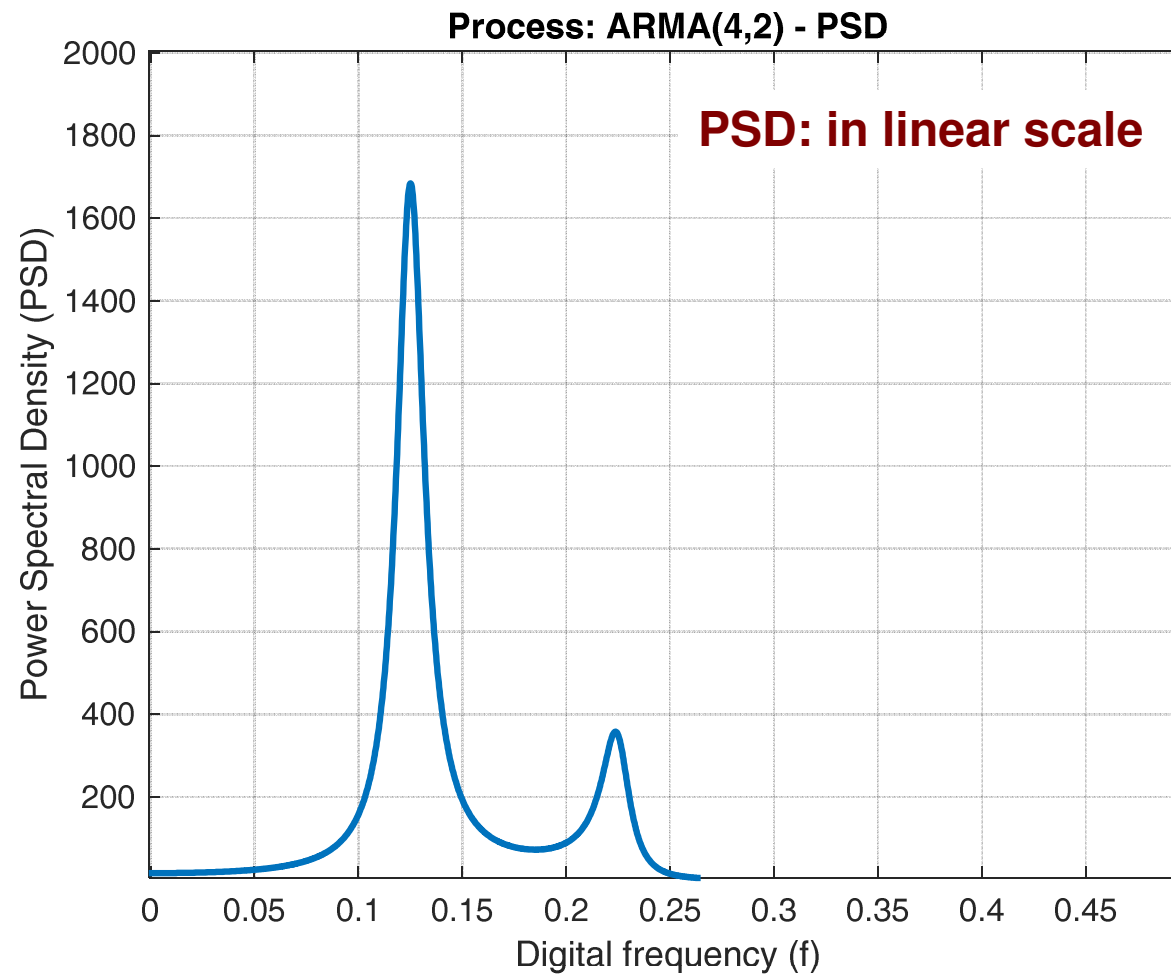


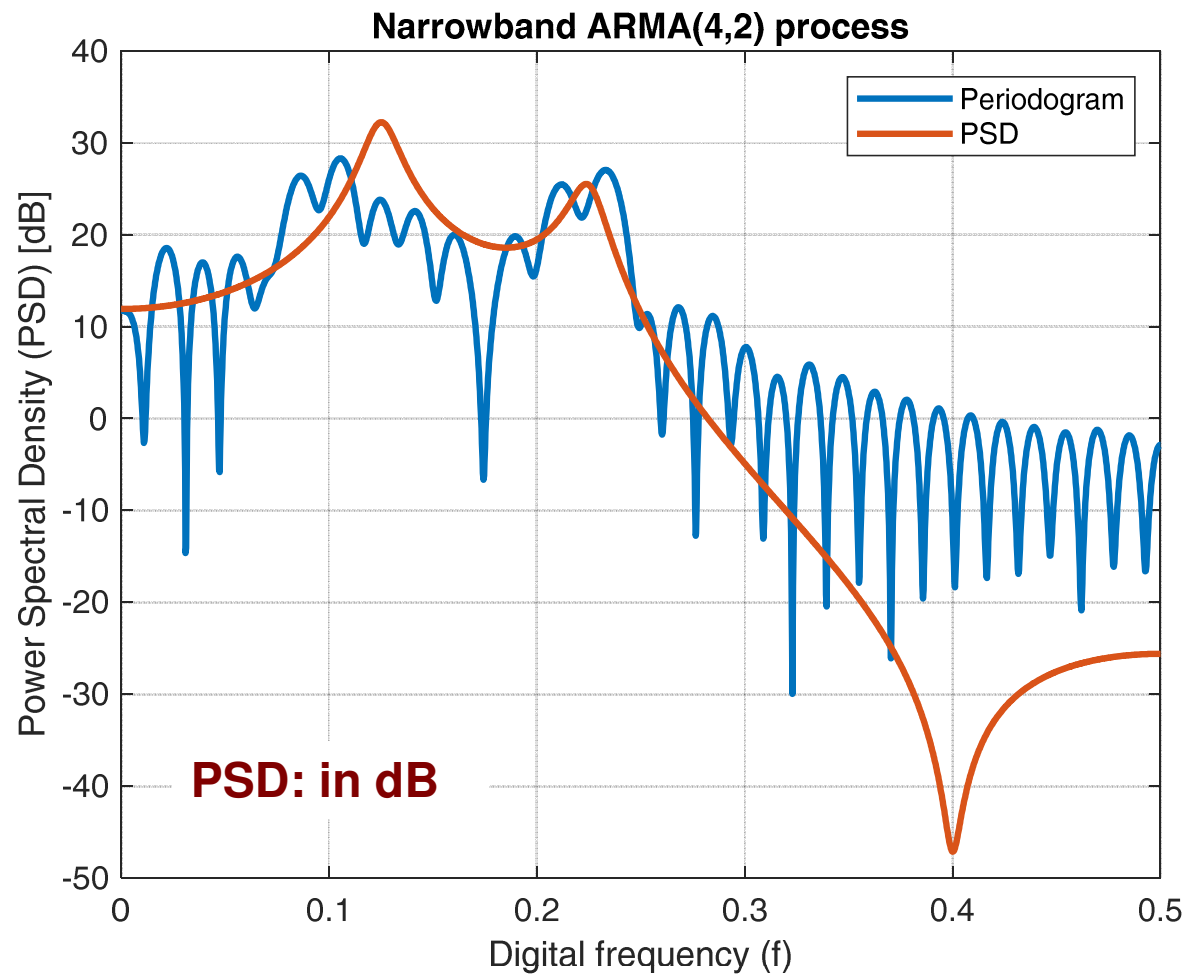
Power Spectral Density (PSD)





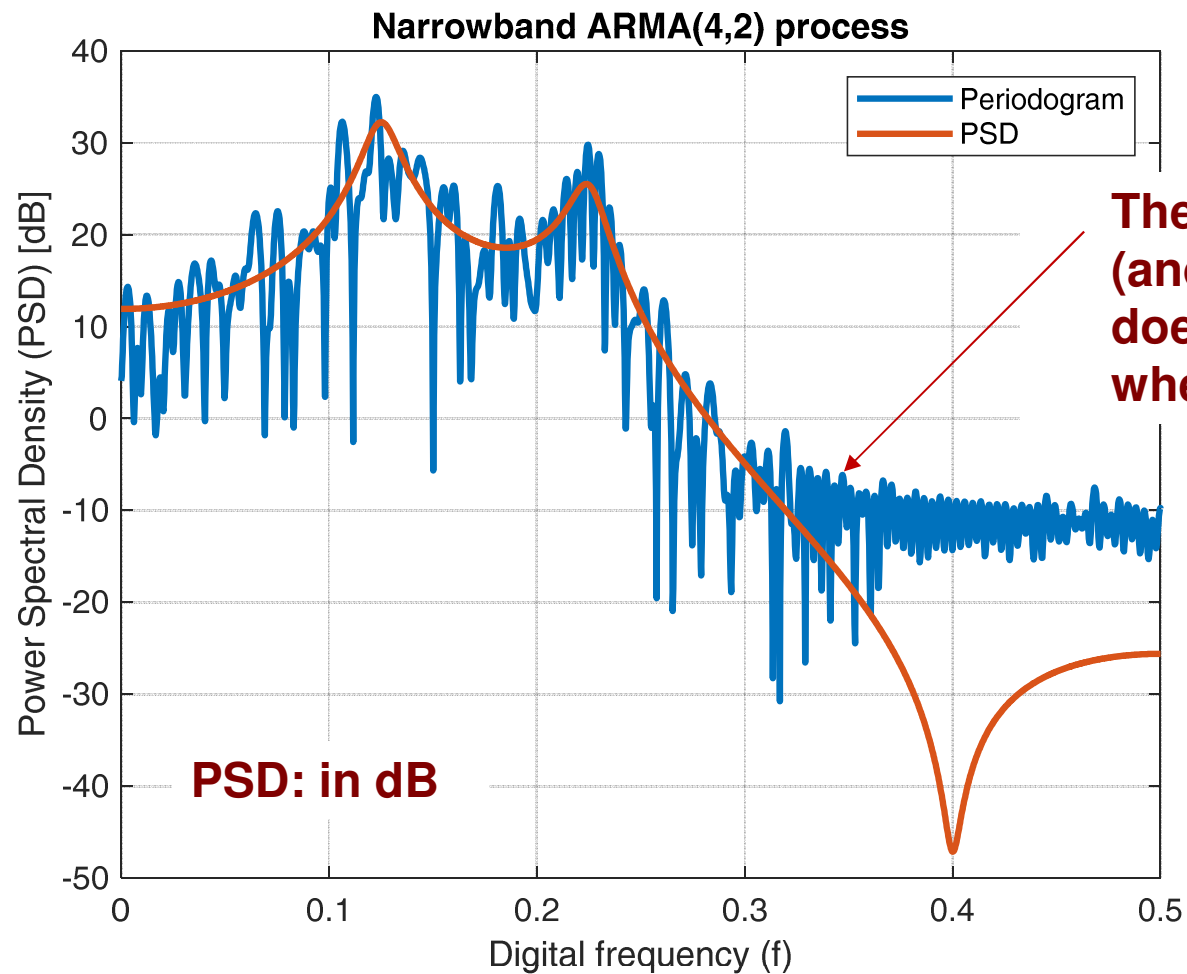
Power Spectral Density (PSD)





Sample size:
 $N=64$

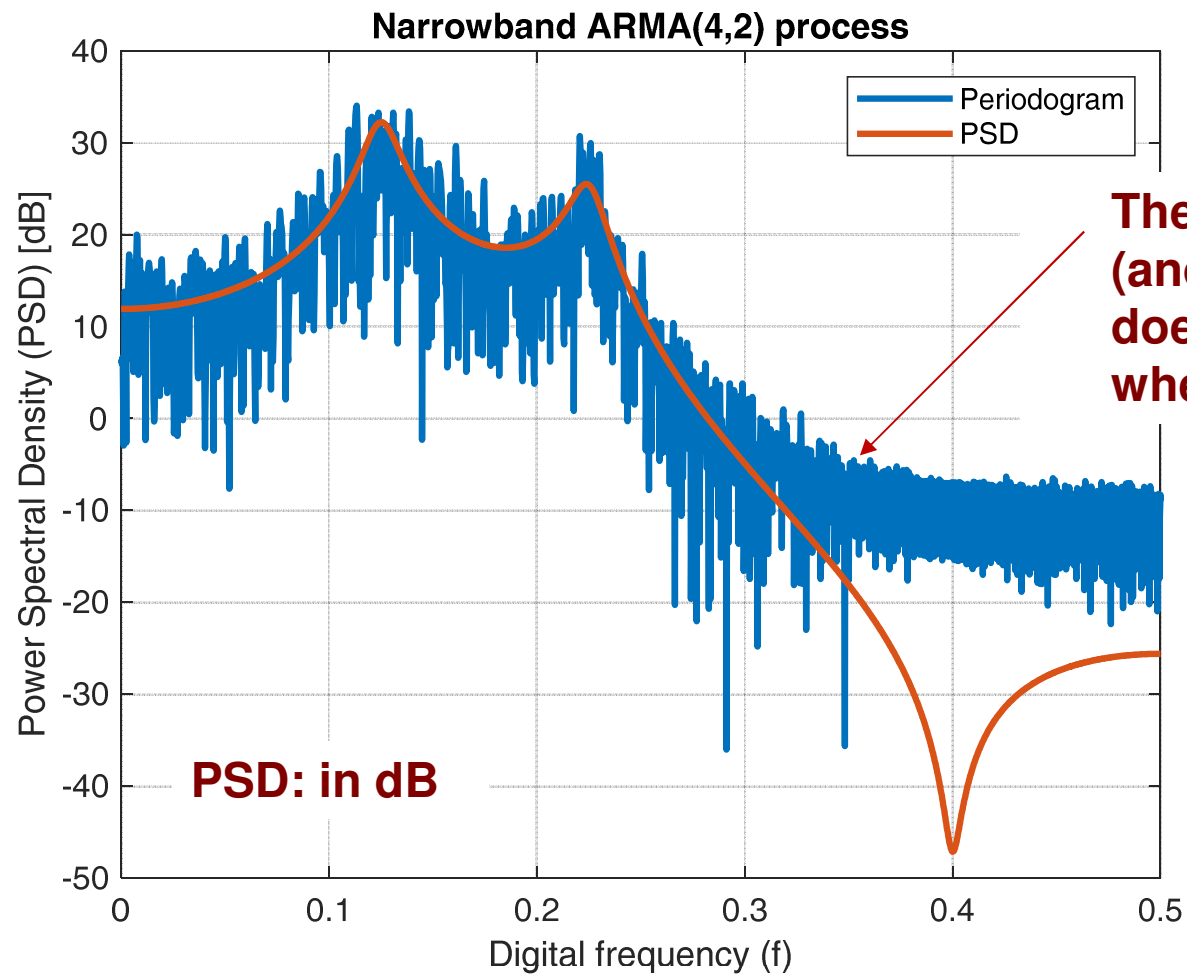
Zero-padding:
 $N_z=4096$



The error variance
(and so the MSE)
does not decrease
when N increases!

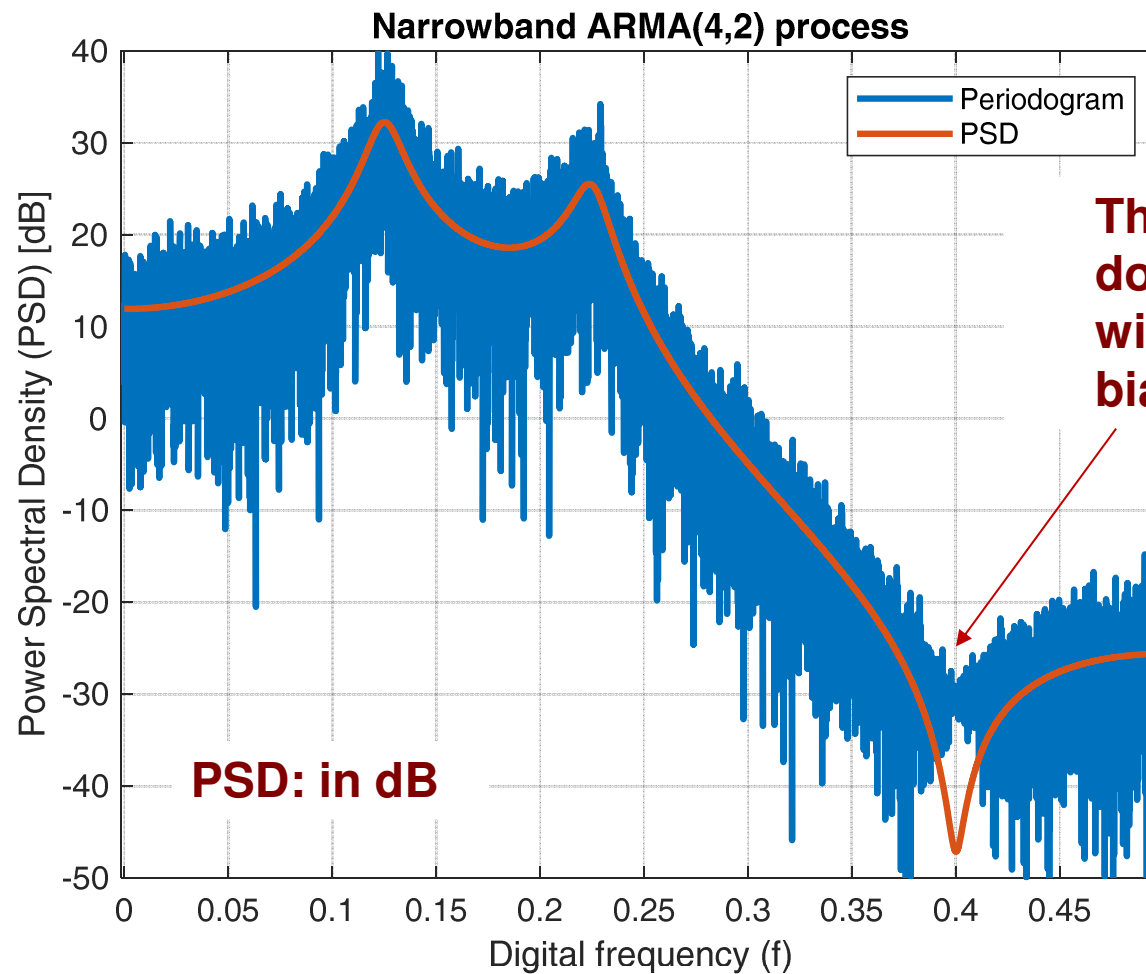
Sample size:
 $N=256$

Zero-padding:
 $N_z=4096$



Sample size:
 $N=1024$

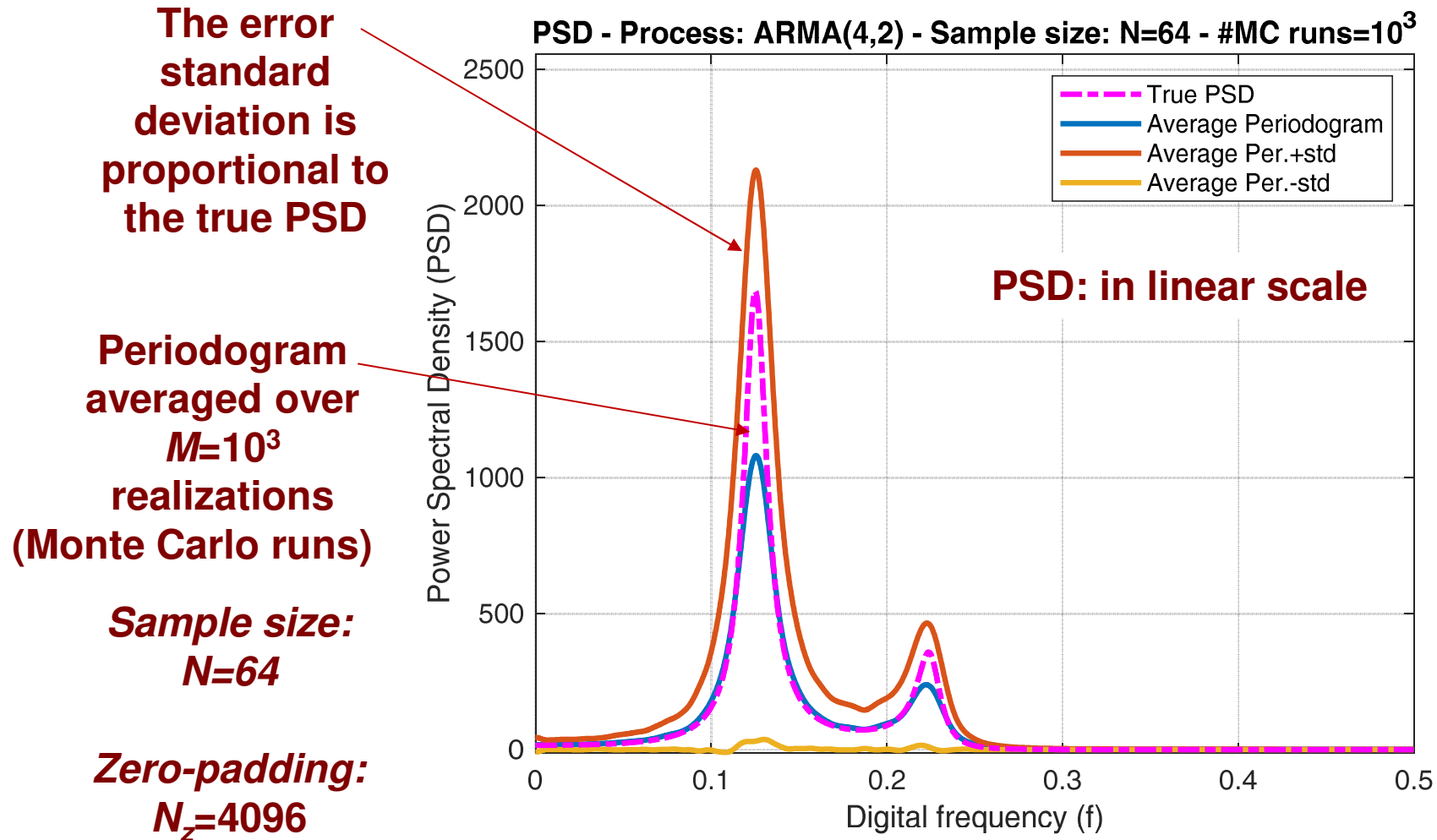
Zero-padding:
 $N_z=4096$



The error variance does not decrease with N but the bias tends to 0!

**Sample size:
 $N=16384$**

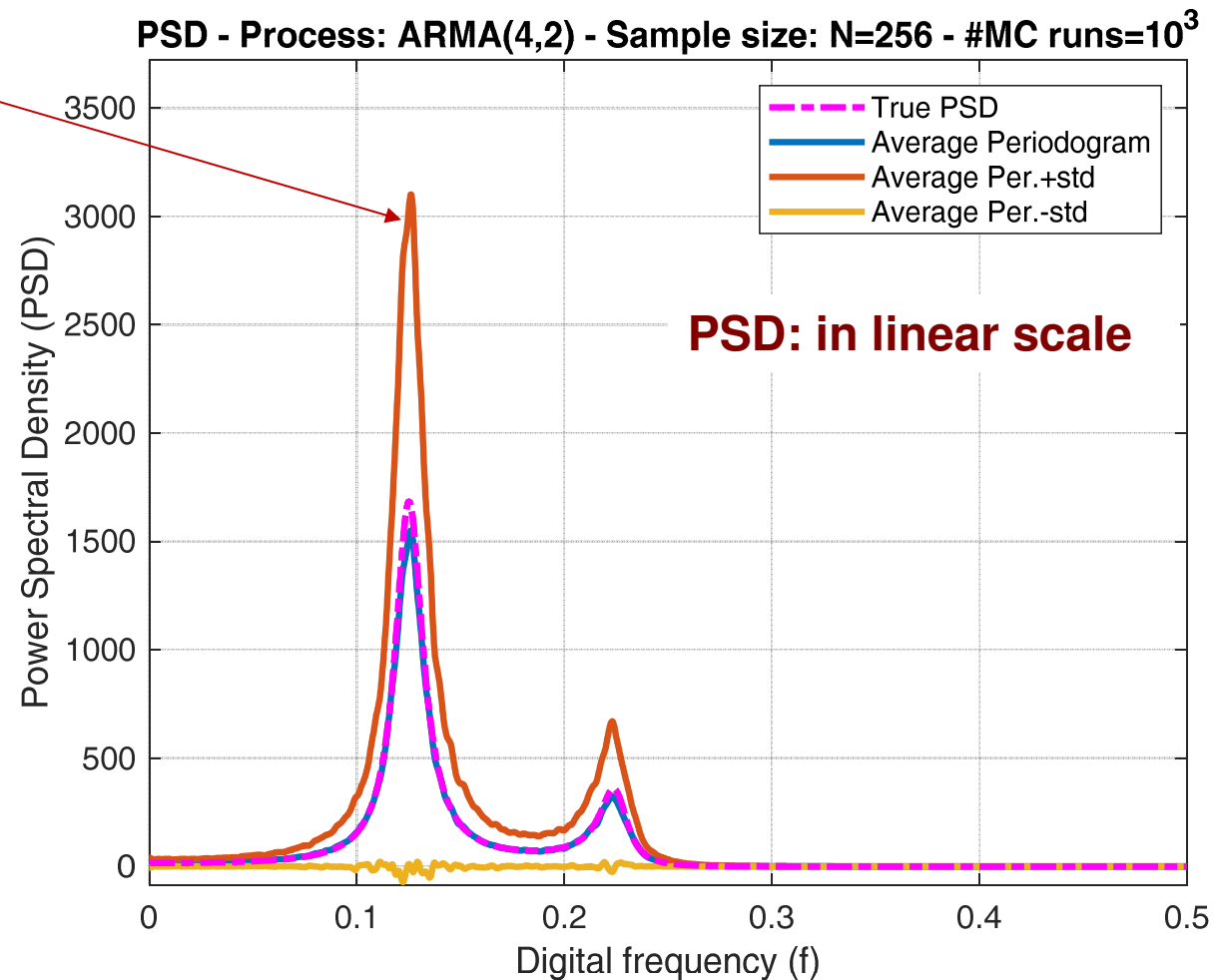
**Zero-padding:
 $N_z=4096$**



The error standard deviation does not decrease when N increases

**Sample size:
 $N=256$**

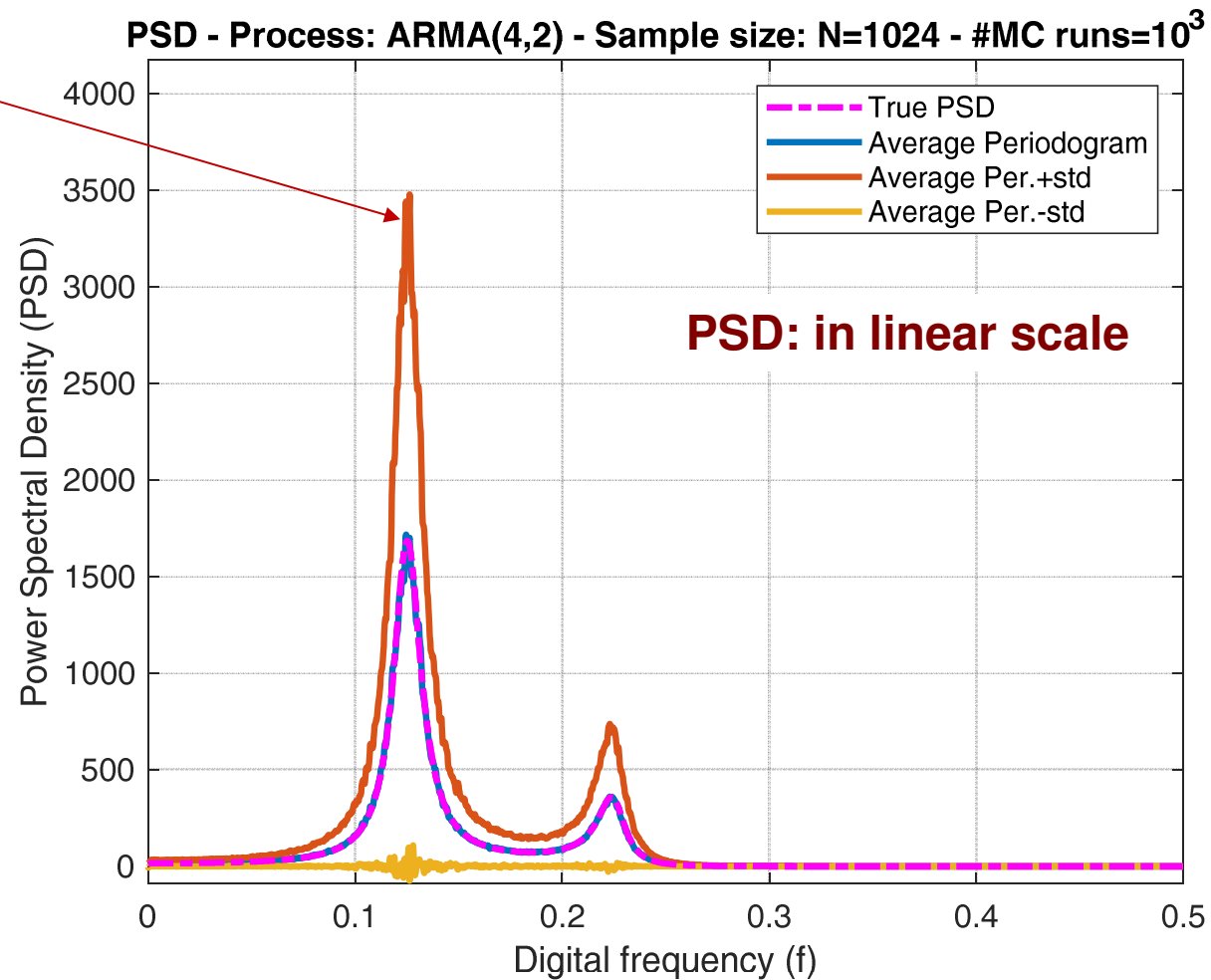
**Zero-padding:
 $N_z=4096$**



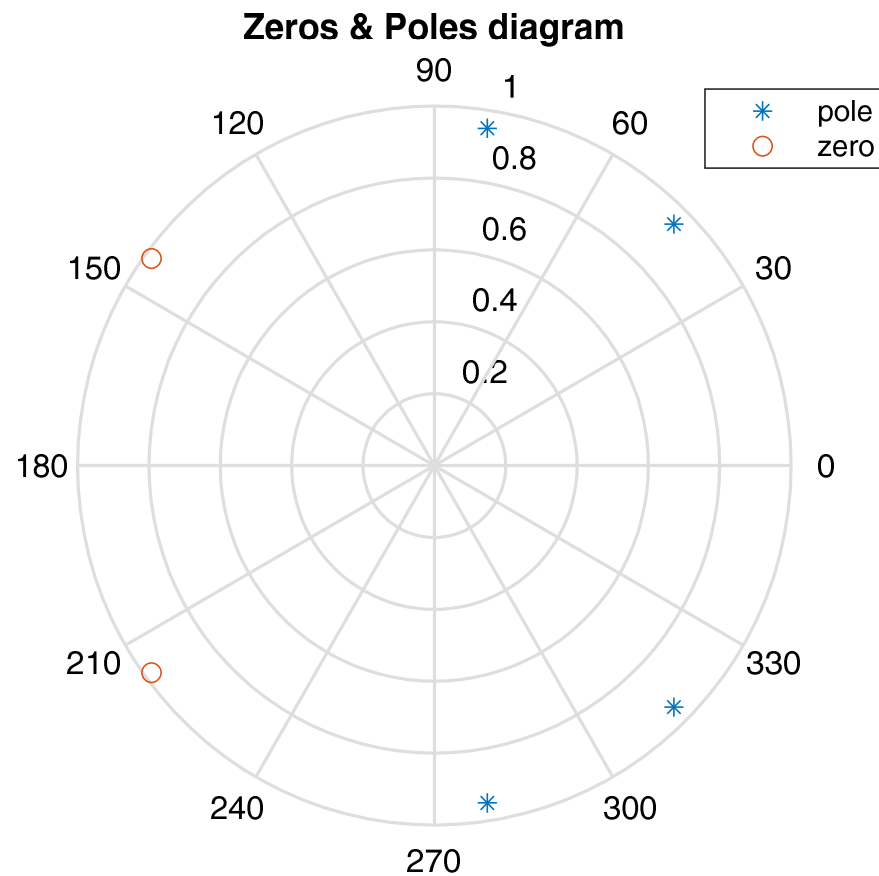
**The error
standard
deviation does
not decrease
when N
increases**

**Sample size:
 $N=1024$**

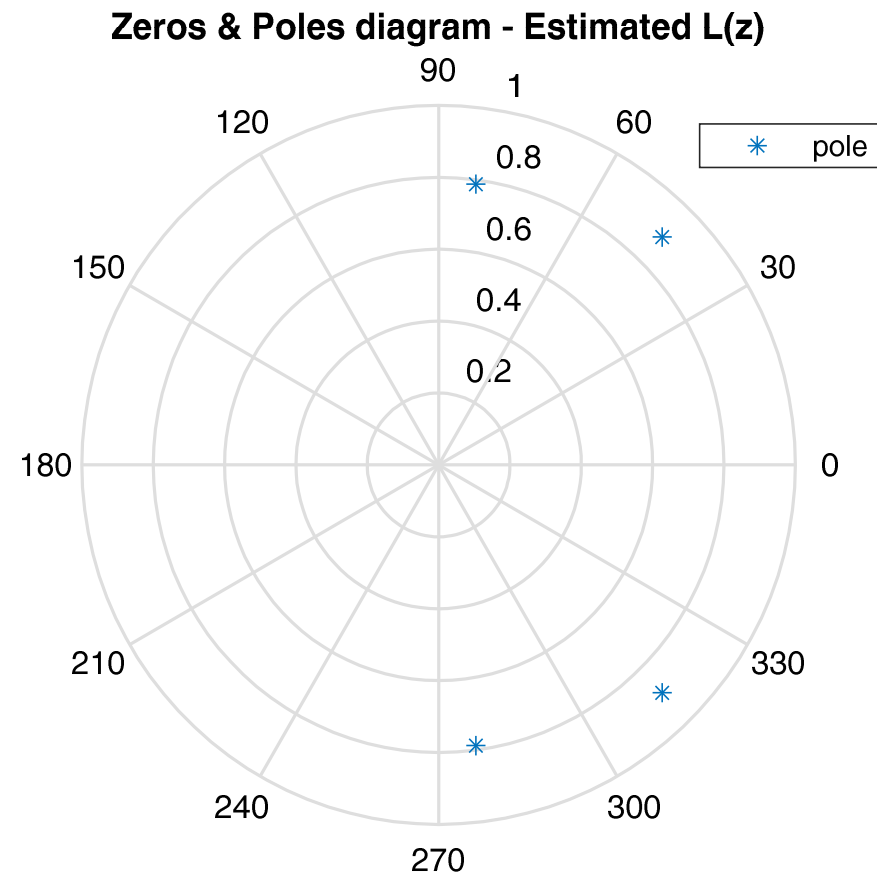
**Zero-padding:
 $N_z=4096$**



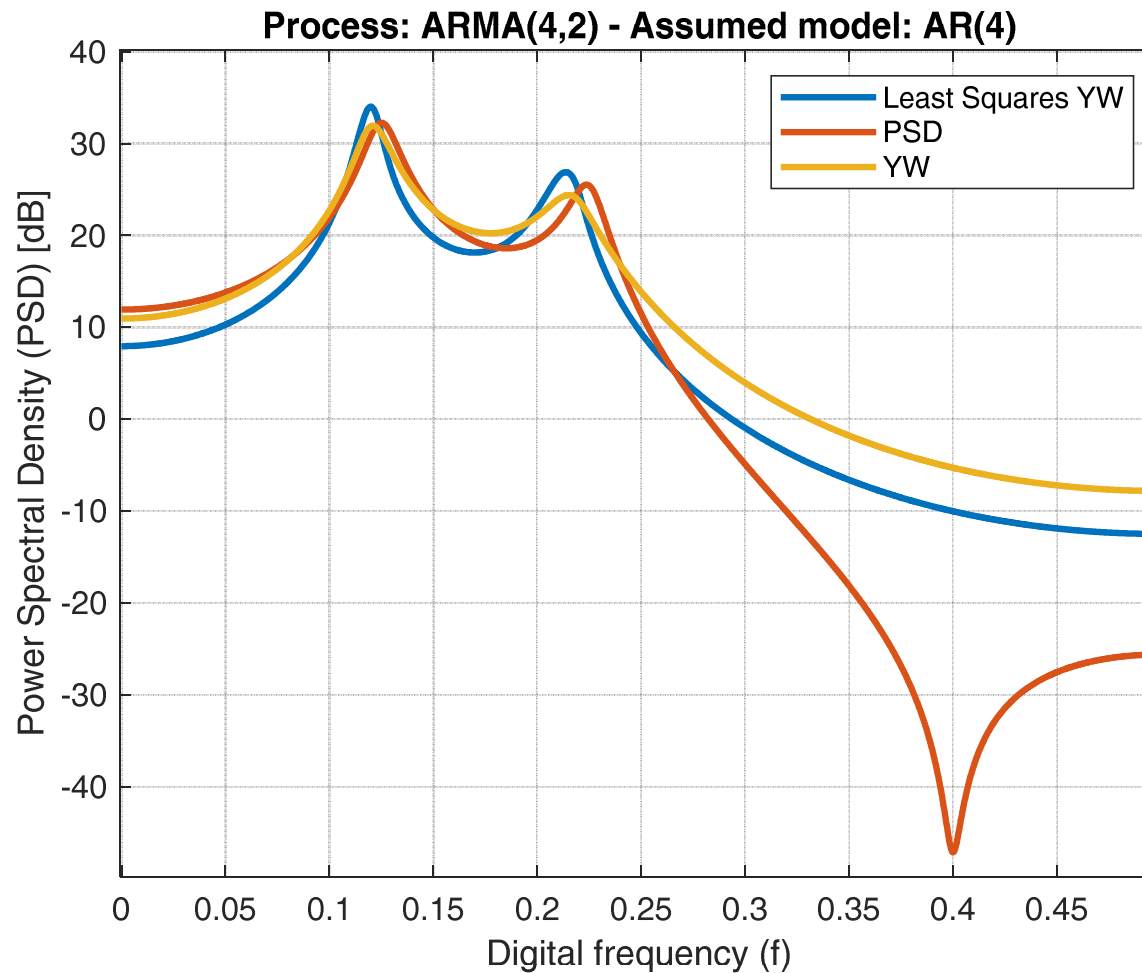
■ True model: ARMA(4,2)



■ Assumed model: AR(4)

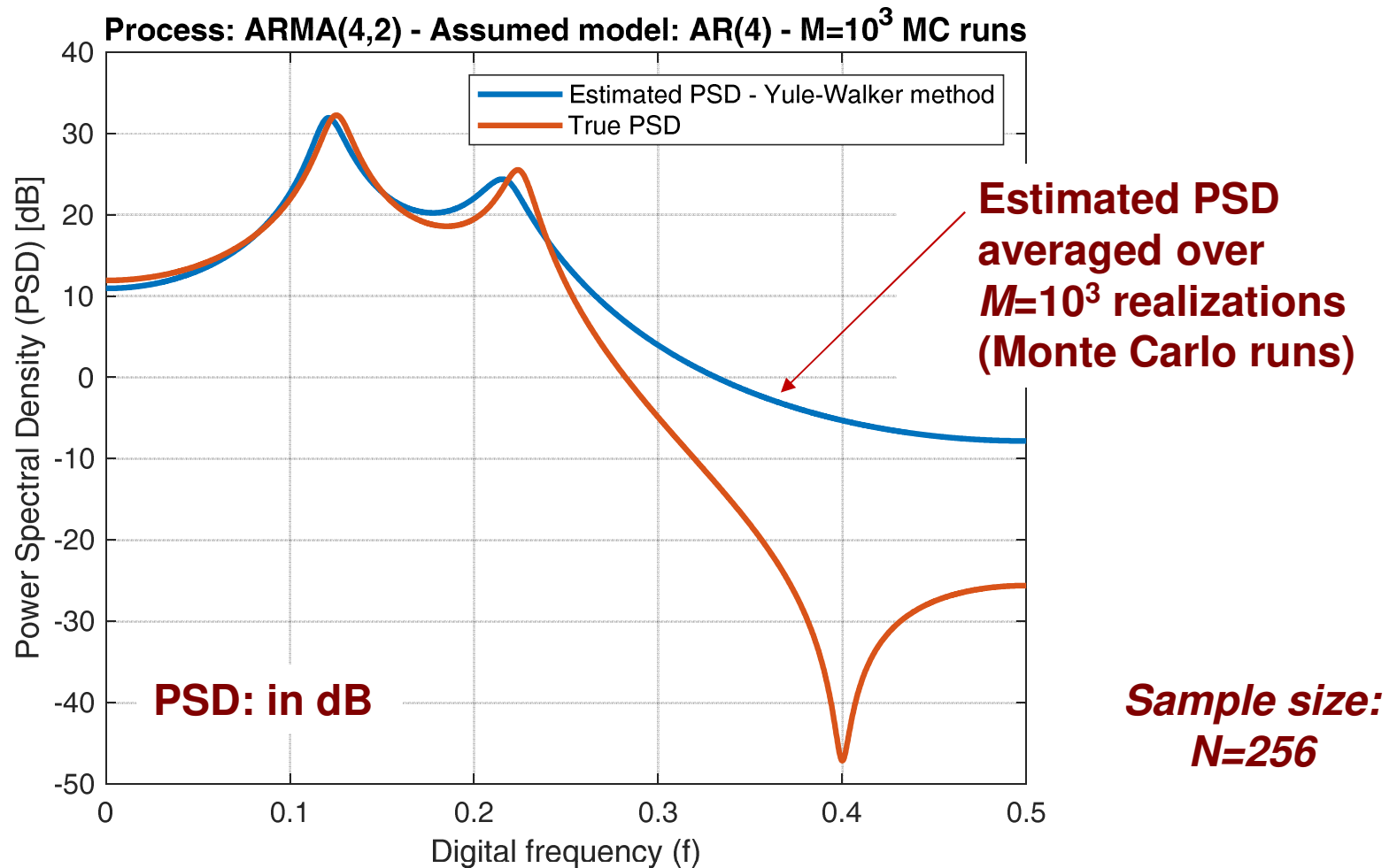


Assumed model: $AR(4)$

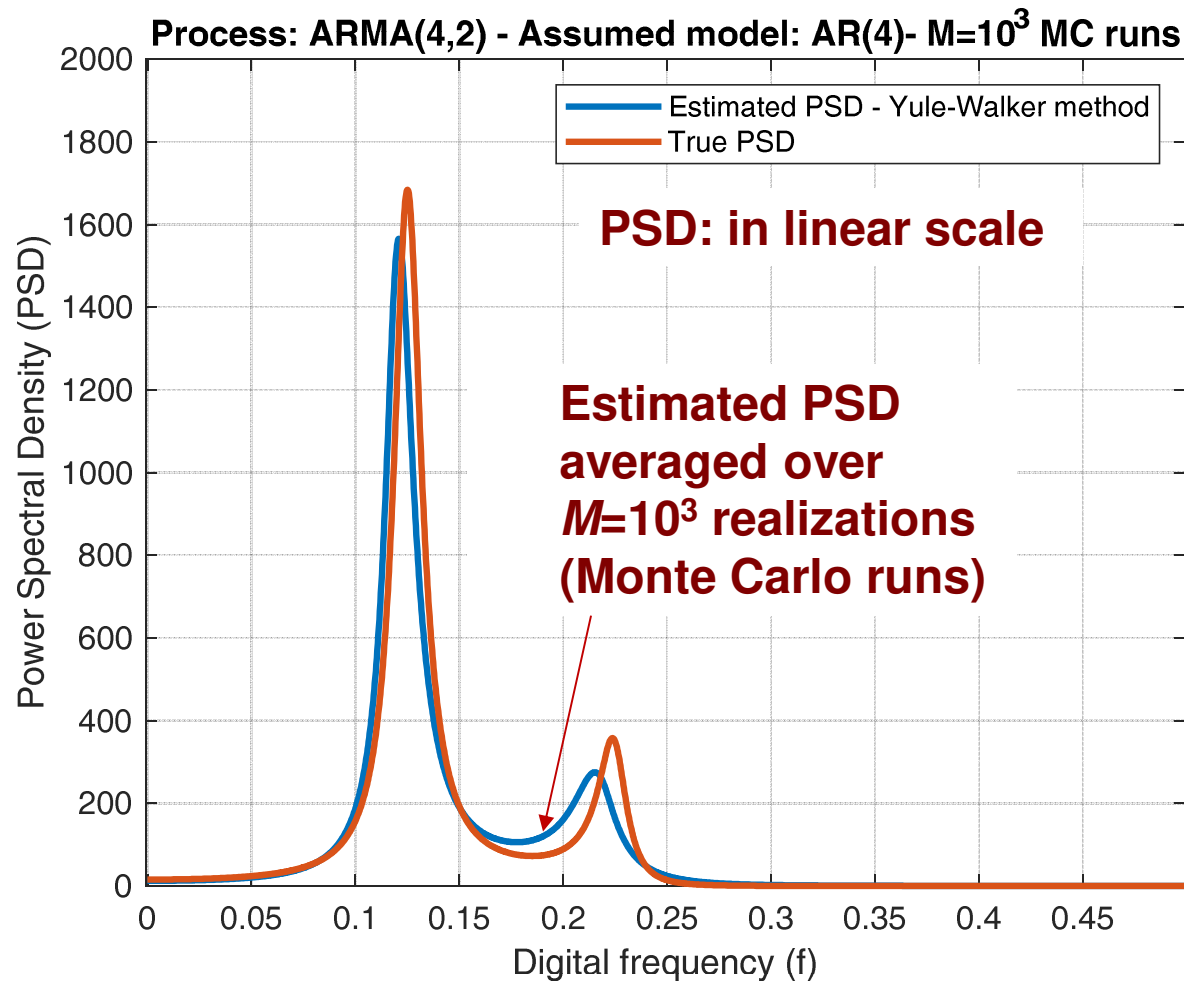


Sample size:
 $N=256$

Assumed model: AR(4)



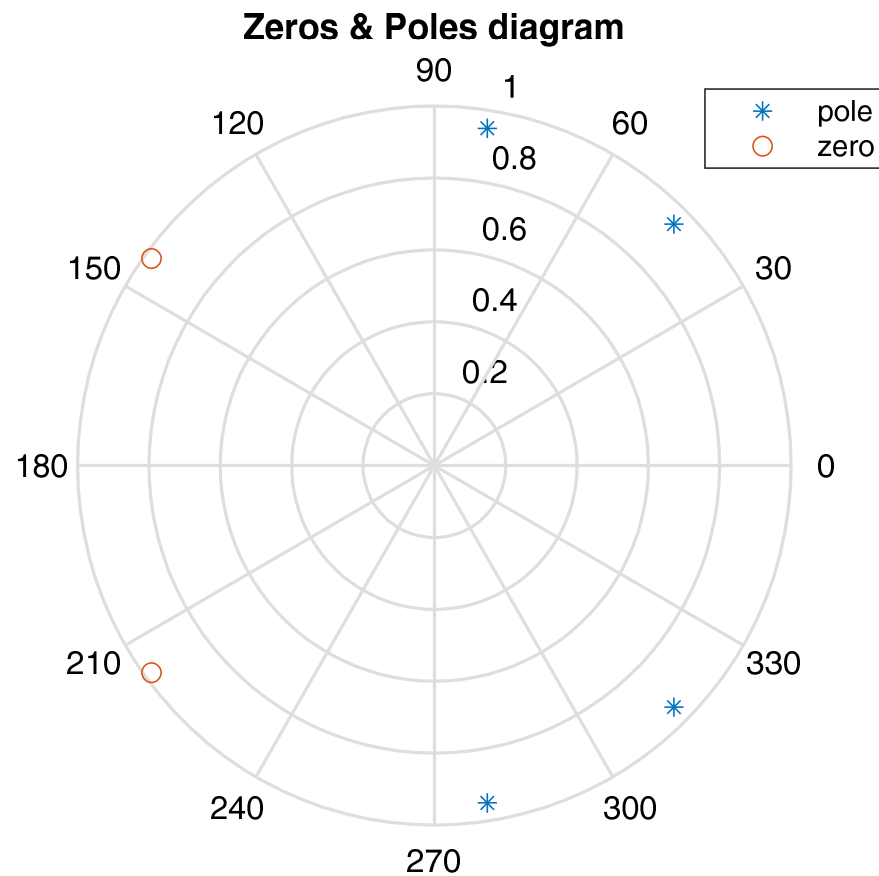
Assumed model: $AR(4)$



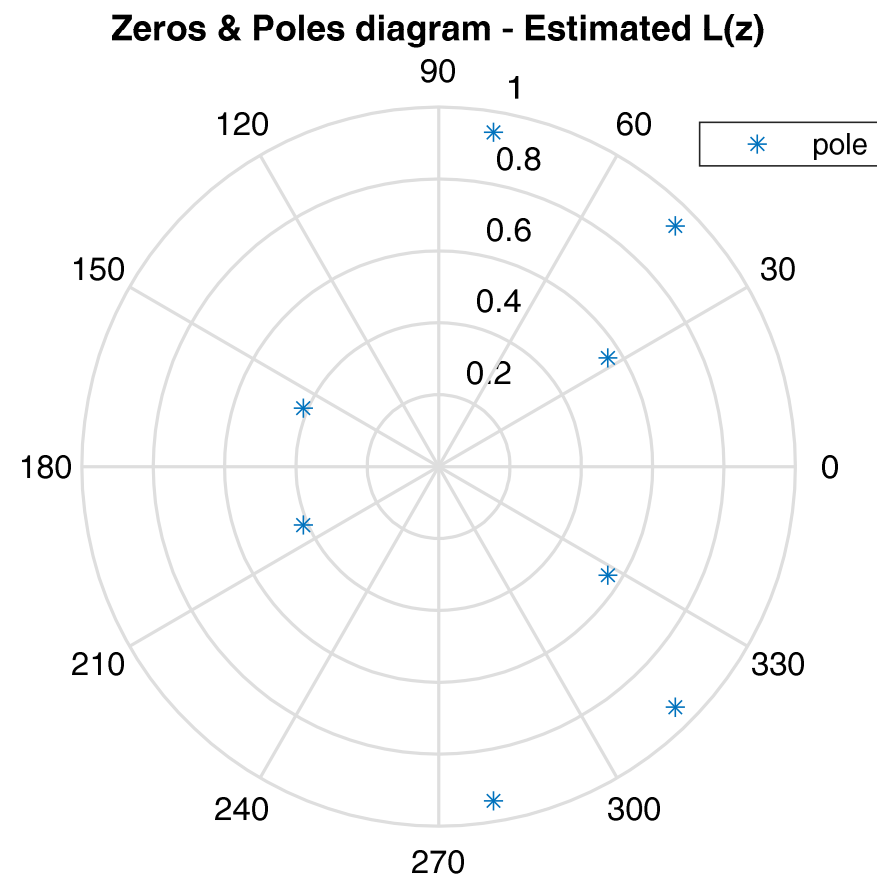
We can see the difference also in linear scale

Sample size:
 $N=256$

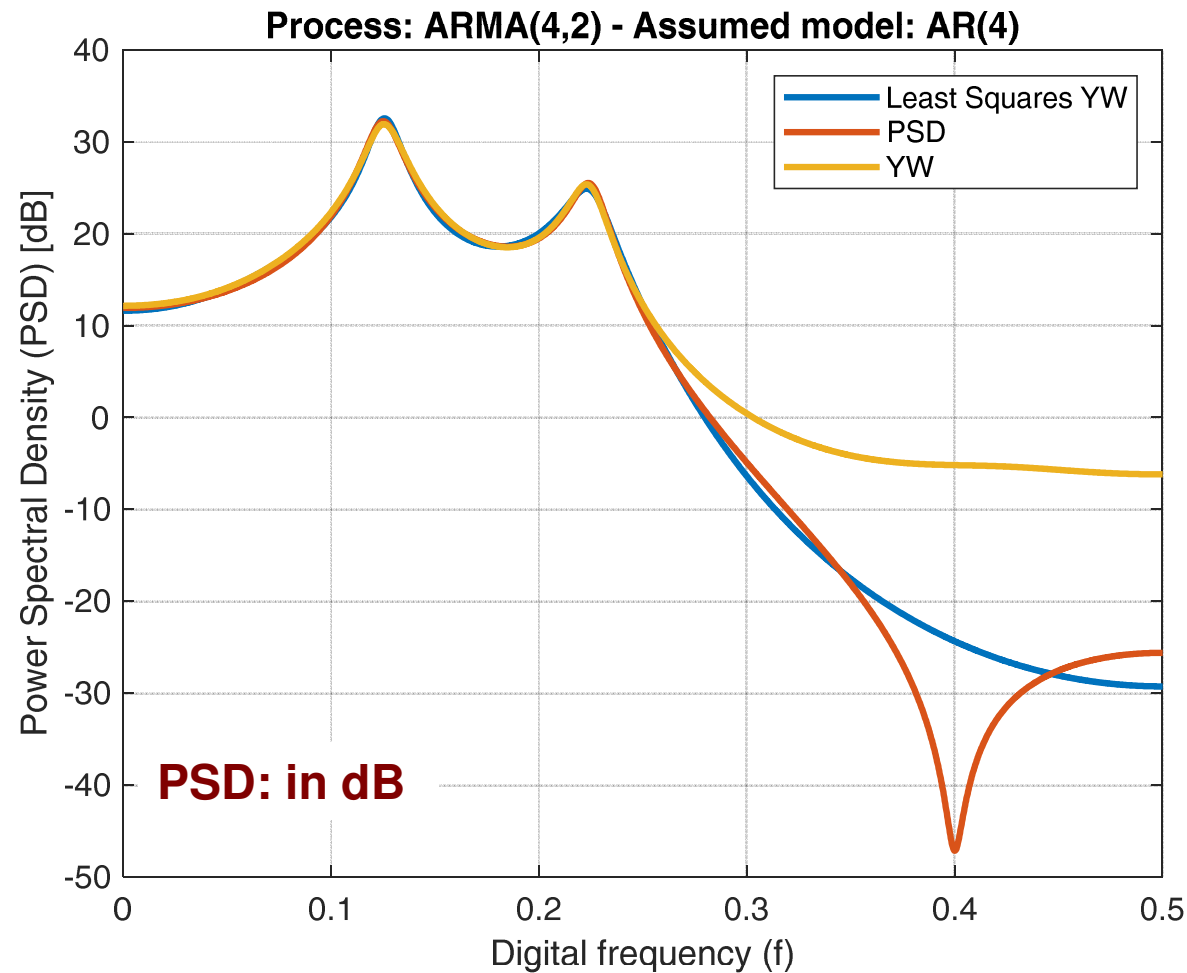
■ True model: ARMA(4,2)



■ Assumed model: AR(8)

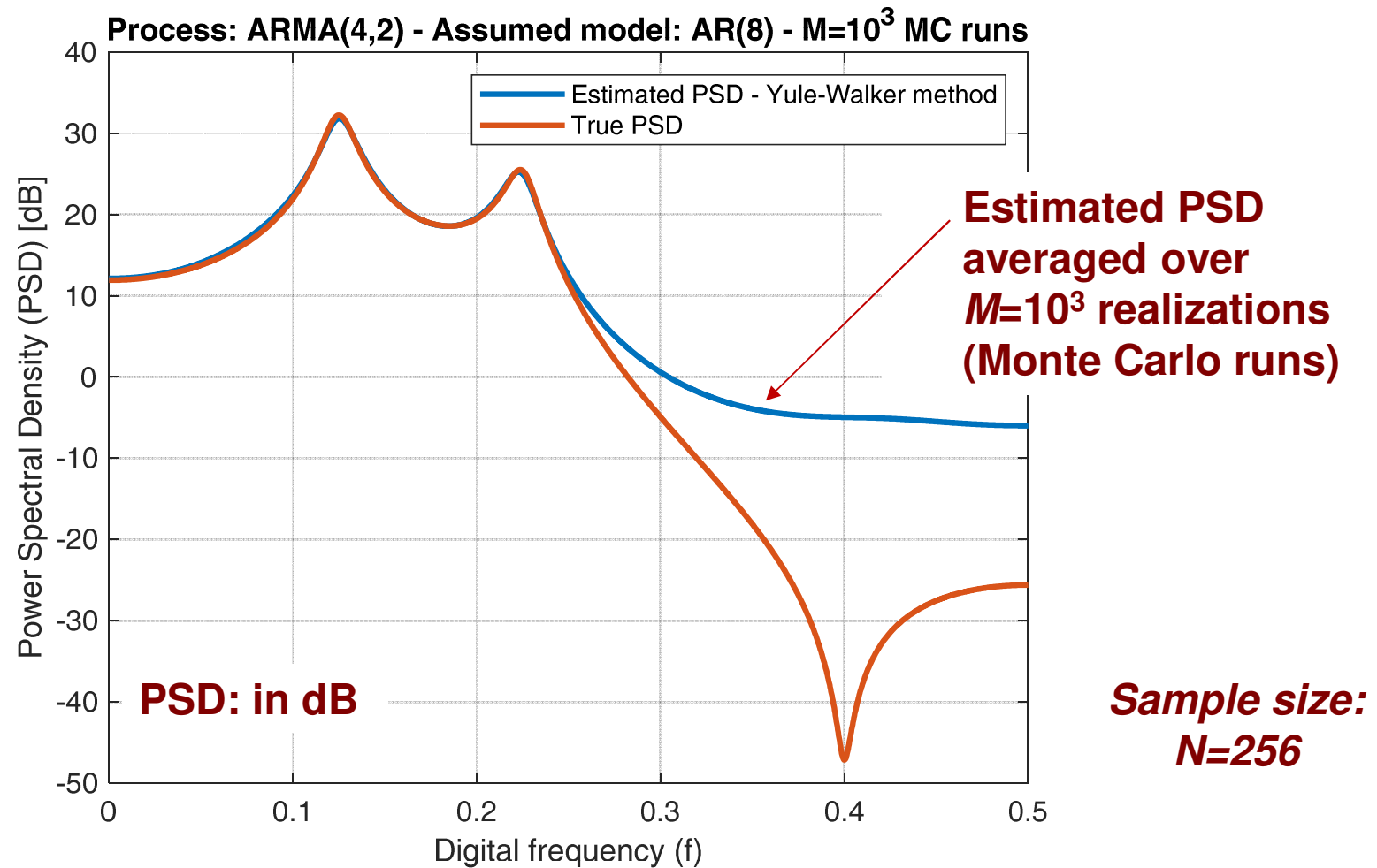


Assumed model: $AR(8)$

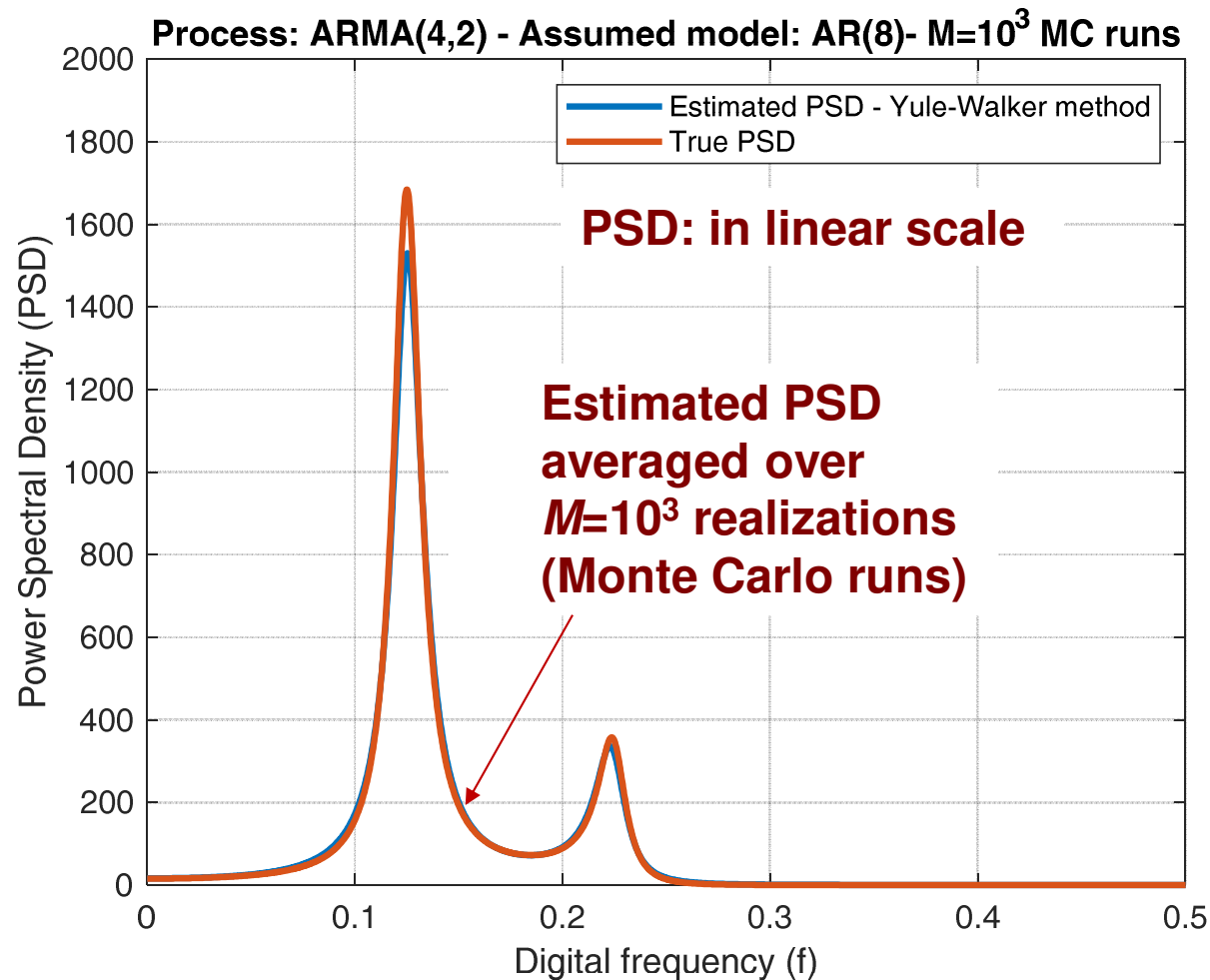


Sample size:
 $N=256$

Assumed model: AR(8)



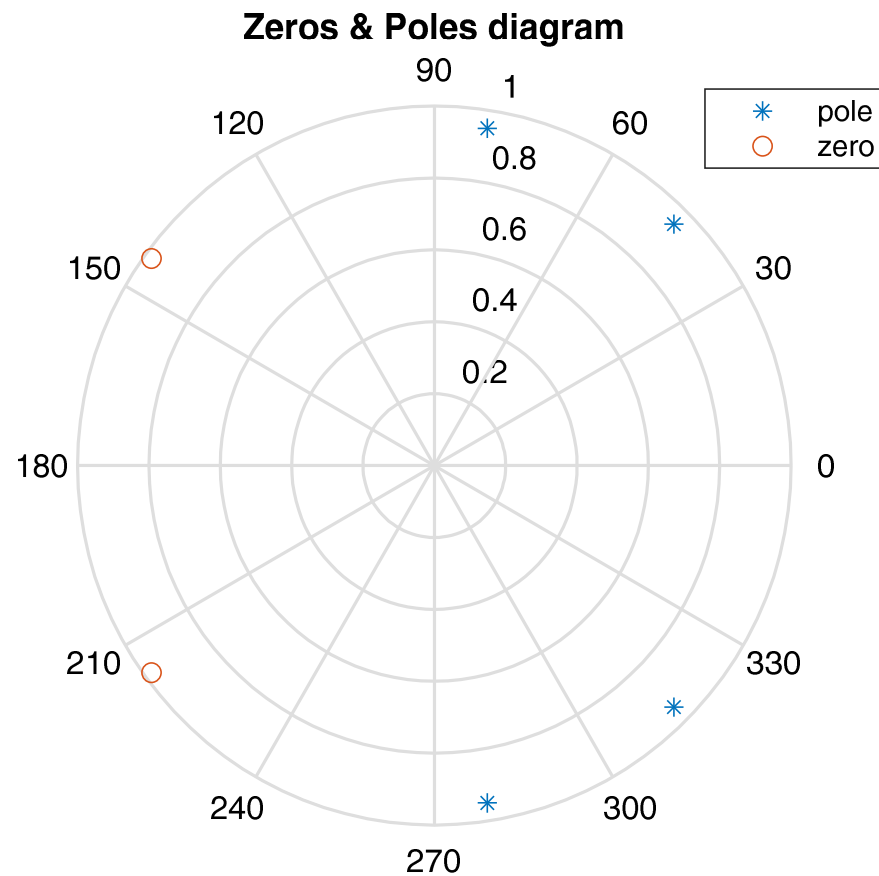
Assumed model: $AR(8)$



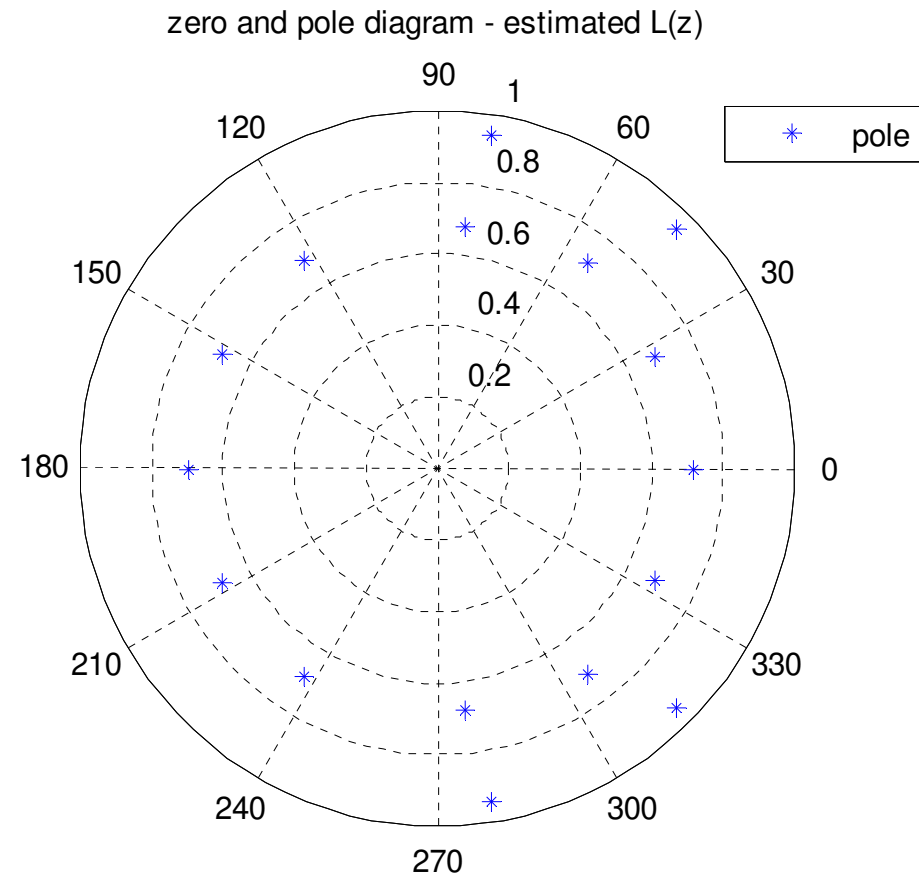
We can't see
the difference
in linear scale

Sample size:
 $N=256$

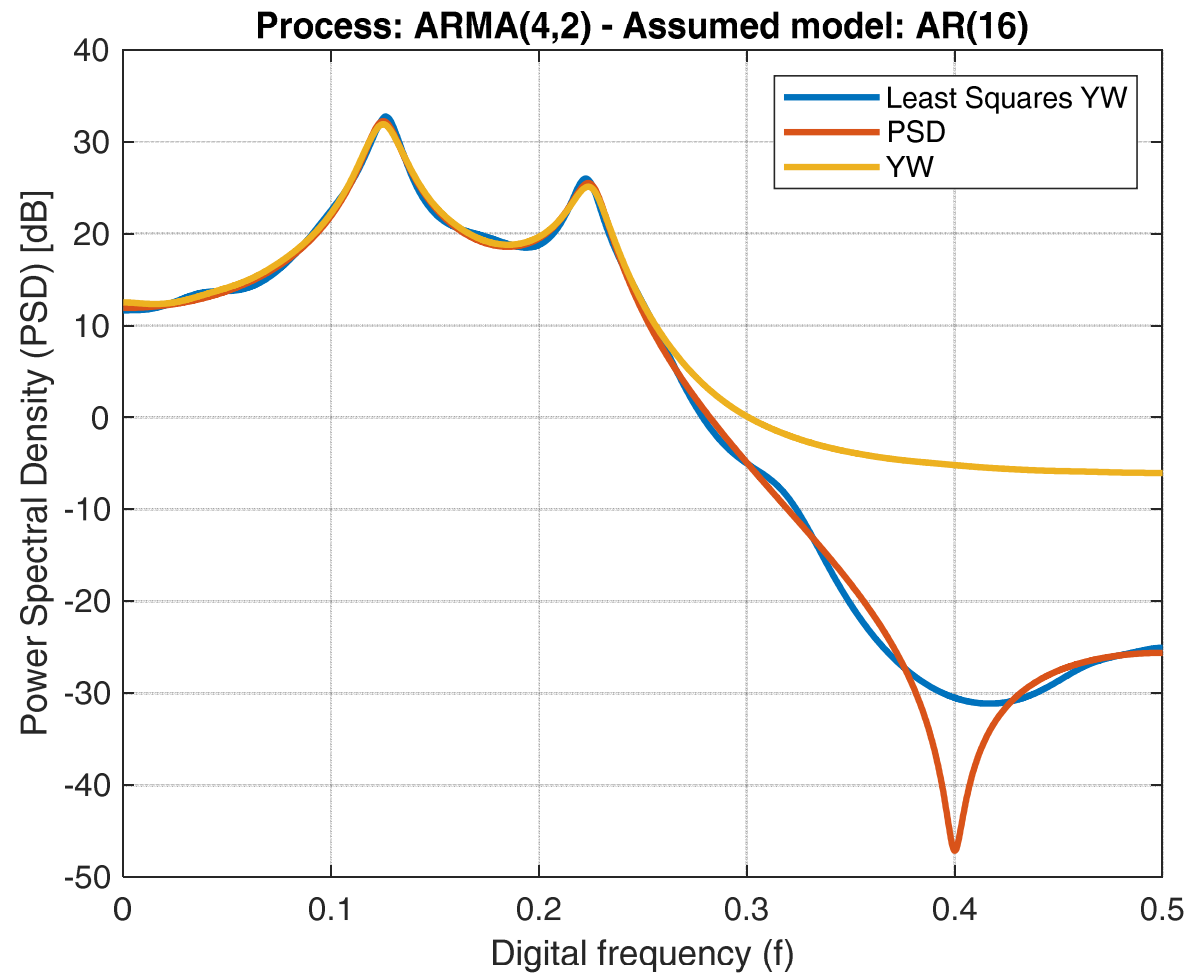
■ True model: ARMA(4,2)



■ Assumed model: AR(16)

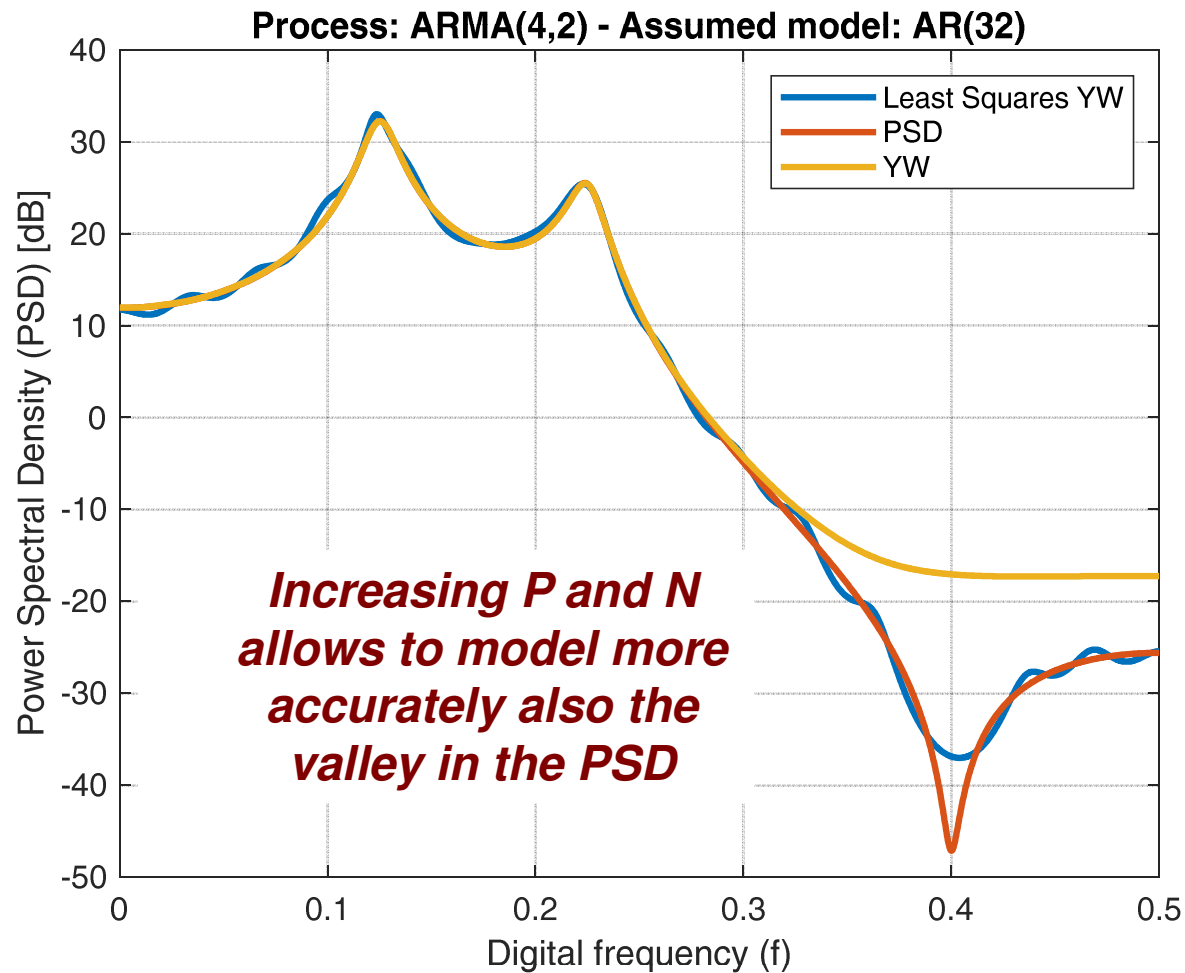


Assumed model: $AR(16)$



Sample size:
 $N=256$

Assumed model: AR(32)



*Sample size:
 $N=4096$*

■ Multicomponent sinusoidal signal in AWGN:

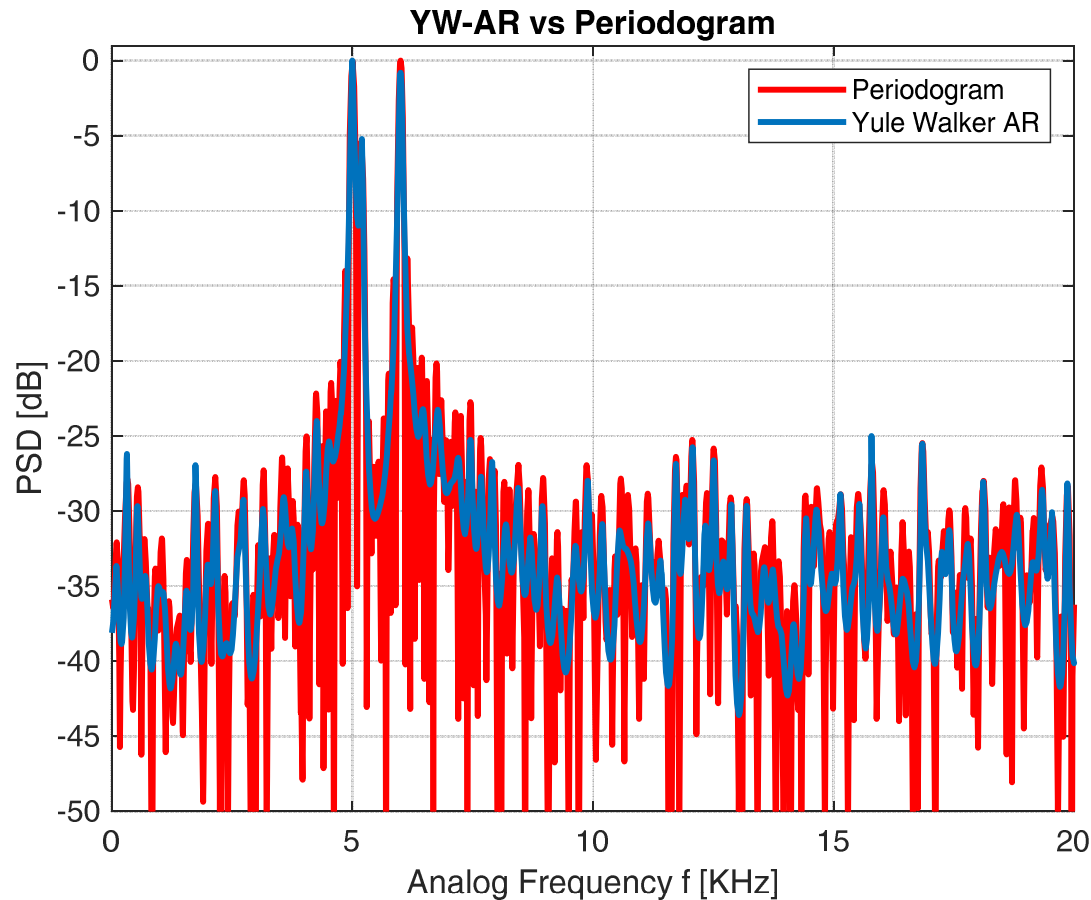
$$X(t) = s(t) + W(t), \quad t \in [0, T)$$

$$s(t) = \sum_{k=1}^K A_k \cos(2\pi f_k t + \theta_k), \quad \text{with } f_k T \gg 1, \quad W(t) \text{ AWGN: } S_W(f) = \frac{N_0}{2}$$

■ Since the poles of the $AR(P)$ model determine the peaks in the spectrum, we can estimate the frequencies of a **multicomponent sinusoidal signal** in AWGN by estimating the locations of the K highest peaks in the (estimated) PSD. The estimates of the K frequencies are obtained from the phases of the K poles (with positive angle) closest to the unit circle (**YW method**):

$$\hat{f}_k = \frac{1}{2\pi} \angle p_k, \quad k = 1, 2, \dots, K$$

■ Once we have estimated the frequencies, amplitudes and phases can be estimated in the usual way from the modulo and phase of the data vector Fourier Transform at the estimated frequencies.



$$B = 20 \text{ KHz}$$

$$T = 10 \text{ msec}$$

$$A_1 = 10, A_2 = 5, A_3 = 10$$

$$\theta_1 = 0, \theta_2 = 2\pi/3, \theta_3 = 2\pi/5$$

$$f_1 = 5 \text{ KHz},$$

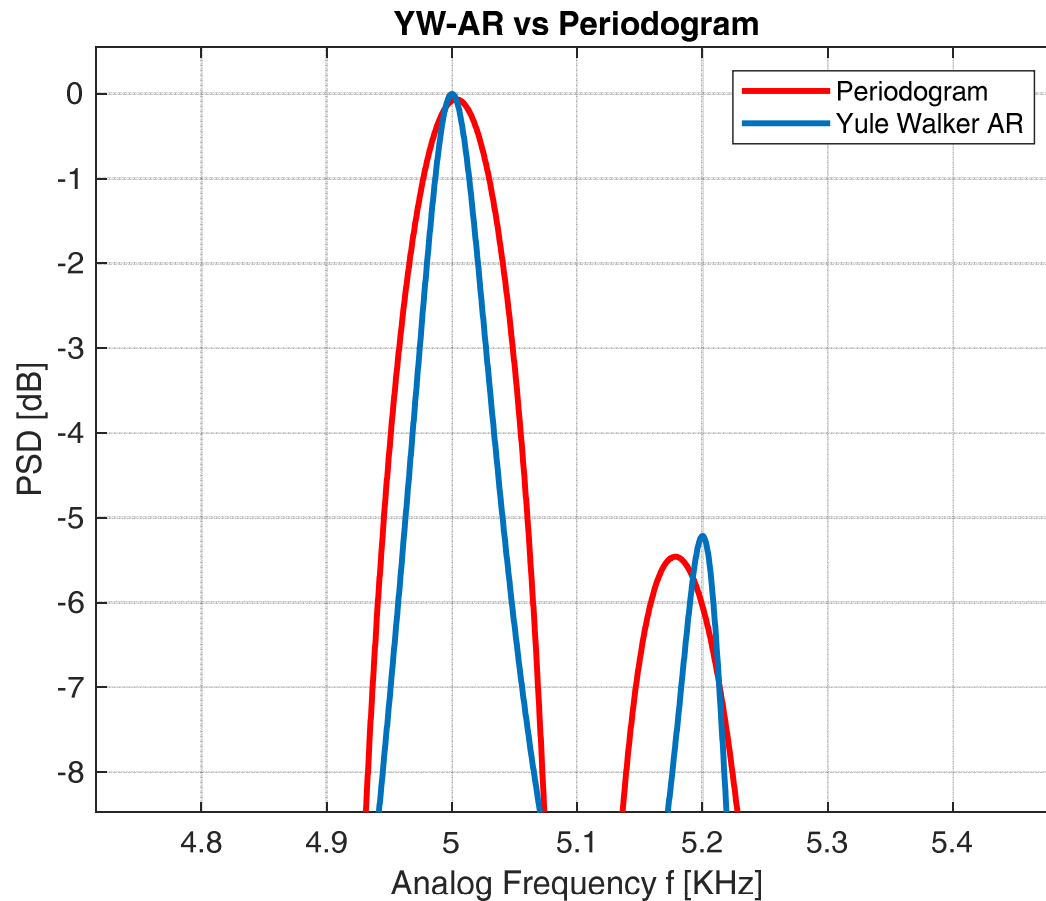
$$f_2 = 5 \text{ KHz} + \frac{2}{T} = 5.2 \text{ KHz}$$

$$f_3 = 6 \text{ KHz}$$

$$SNR_{in} = 10 \text{ dB}$$

$$N = 400$$

$$N_{zp} = 2^{19}$$



$$B = 20 \text{ KHz}$$

$$T = 10 \text{ msec}$$

$$A_1 = 10, A_2 = 5, A_3 = 10$$

$$\theta_1 = 0, \theta_2 = 2\pi/3, \theta_3 = 2\pi/5$$

$$f_1 = 5 \text{ KHz},$$

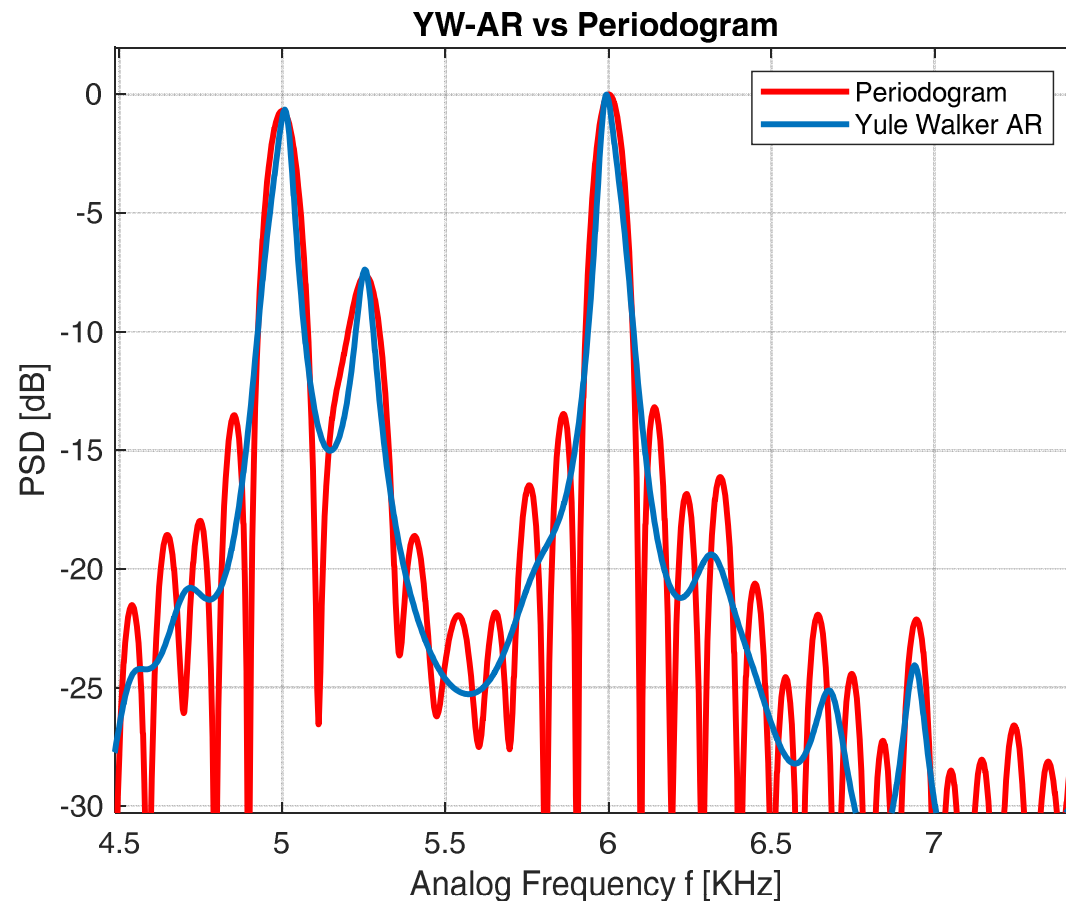
$$f_2 = 5 \text{ KHz} + \frac{2}{T} = 5.2 \text{ KHz}$$

$$f_3 = 6 \text{ KHz}$$

$$SNR_{in} = 10 \text{ dB}$$

$$N = 400$$

$$N_{zp} = 2^{19}$$



$$B = 20 \text{ KHz}$$

$$T = 10 \text{ msec}$$

$$A_1 = 10, A_2 = 5, A_3 = 10$$

$$\theta_1 = 0, \theta_2 = 2\pi/3, \theta_3 = 2\pi/5$$

$$f_1 = 5 \text{ KHz},$$

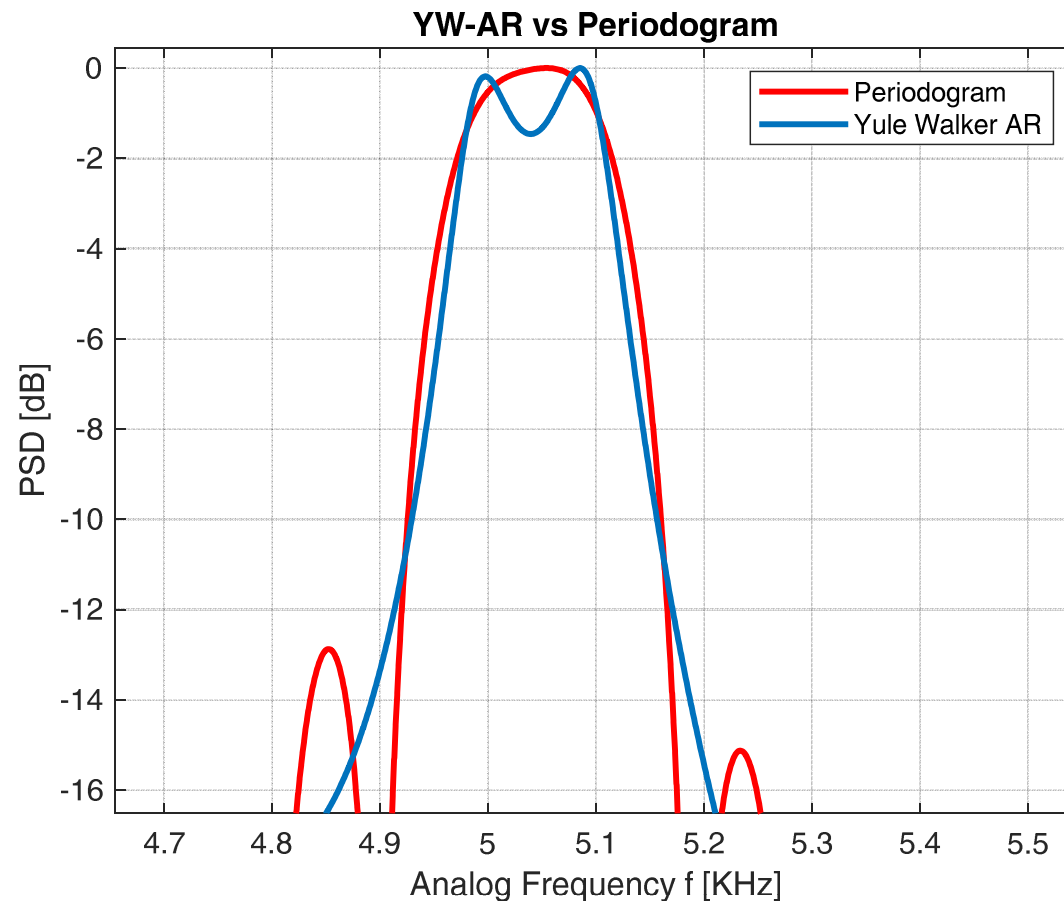
$$f_2 = 5 \text{ KHz} + \frac{2.5}{T} = 5.25 \text{ KHz}$$

$$f_3 = 6 \text{ KHz}$$

$$SNR_{in} = 10 \text{ dB}$$

$$N = 400$$

$$N_{zp} = 2^{19}$$



$$B = 20 \text{ KHz}$$

$$T = 10 \text{ msec}$$

$$A_1 = 10, A_2 = 10, A_3 = 10$$

$$\theta_1 = 0, \theta_2 = 2\pi/3, \theta_3 = 2\pi/5$$

$$f_1 = 5 \text{ KHz},$$

$$f_2 = 5 \text{ KHz} + \frac{0.9}{T} = 5.09 \text{ KHz}$$

$$f_3 = 6 \text{ KHz}$$

$$SNR_{in} = 10 \text{ dB}$$

$$N = 400$$

$$N_{zp} = 2^{19}$$

■ YW-AR is a superresolution method to estimate the frequencies of a multicomponent sinusoidal signal in AWGN.

- In summary, the **ARMA (P,Q) model identification** problem consists in the estimation of the unknown parameters **a**, **b**, and the power of the innovation process $W[n]$ of the ARMA model that better fits the estimated ACF sequence, having available N samples of a realization of the process $X[n]$:

$$\{X[n]\}_{n=0}^{N-1} \Rightarrow \{\hat{R}_X[m]\}_{m=0}^{Q+P} \Rightarrow \{\hat{\sigma}_W^2, \hat{\mathbf{a}}, \hat{\mathbf{b}}\} \Rightarrow \hat{S}_X(e^{j2\pi f})$$

- For **AR(P) processes**, **model identification** requires the solution of a system of $P+1$ linear equations, that can be solved efficiently by using the Levinson-Durbin algorithm or the Shur algorithm.
- The system of equations to be solved to estimate the vector **b** for MA(Q) or ARMA(P,Q) processes is instead non-linear.
- The **Model Order Selection (MOS)** problem is not treated here.

- The most important methods to estimate the parameters of an ARMA (P, Q) process, given P and Q , are:

- **Modified Yule-Walker Method (Matlab: mywarma.m)**
- **Two-Stage Least-Squares Method (Matlab: lsarma.m)**

- We will not describe in details these two methods (*See Stoica-Moses book*).

Method	Computational Burden	Accuracy	$\hat{S}_x(e^{j2\pi f}) > 0?$	Use for
AR: YW or LS	low	medium	Yes	Spectra with (narrow) peaks but no valley
MA: BT	low	low-medium	No	Broadband spectra possibly with valleys but no peaks
ARMA: MYW	low-medium	medium	No	Spectra with both peaks and (not too deep) valleys
ARMA: 2-Stage LS	medium-high	medium-high	Yes	As above